CH2　應力

2-1. 正向應力與剪應力

一、正向力之張應力與壓應力

如圖　所示，我們可由兩端承受軸向負載P的等剖面桿來說明，當我們施加一拉力作用於桿件時，桿件在外力的作用下，桿件內部各部分之間的因外力作用而引起的附加相互作用力，此種在桿件內部產生之抗力稱爲內力，而此力的強度，也就是單位面積的正向力，稱之爲應力（stress），且由於此應力作用方向和剖面垂直，故稱爲正向應力（normal stress)，通常以希臘字母σ（sigma）表示，$\sigma > 0$ 稱之爲張應力（Tensile Stress），$\sigma < 0$ 稱之爲壓應力（Compressive Stress），其中該應力是均勻分佈整個截面，其合力等於應力σ乘上截面積A，又物體呈平衡狀態，內力之合力將等於拉力P的大小，但方向相反其分別說明如下：

(1) 如圖1.1當作用於構件的外力，合力的作用線與構件的軸線重合，構件將產生軸向拉伸或壓縮變形，橫截面上的內力稱爲軸力，軸力用N表示，方向與軸線重合。

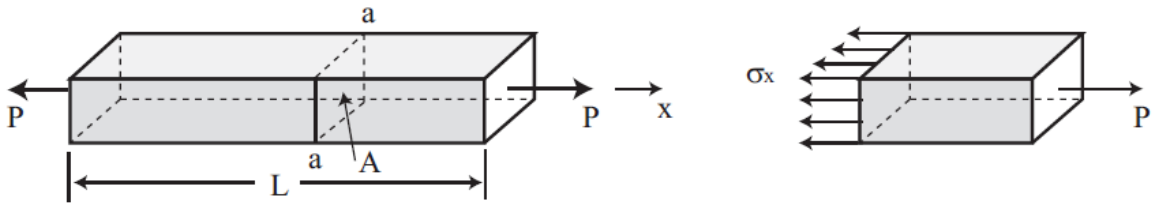
(2) 欲求某一截面處a-a處的內力時，可利用截面法沿該截面假想地把構件切開使其分爲兩部分，此時橫截面所受到的應力強度可表示爲 $\sigma_x = \frac{P}{A}$，其中A可表示橫截面積，一般拉伸應力視爲正（拉應力），壓縮應力（壓應力）視爲負。

(3) 桿件應滿足下列假設：

A. 桿件須爲直桿，且爲均質材料（即桿件內部各部位置之性質均相同）。

B. 軸向拉力之作用線需通過桿件橫斷面之形心。

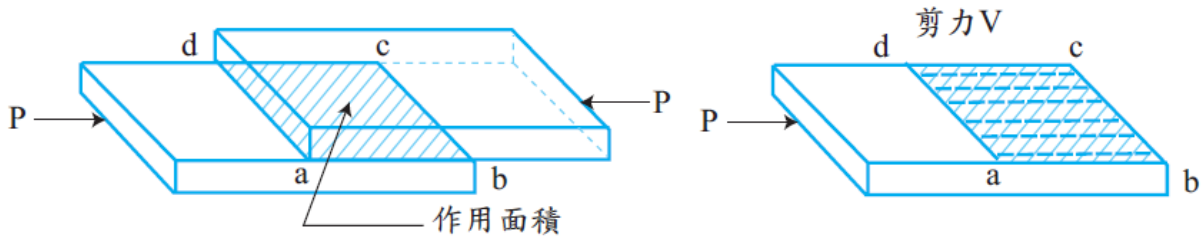
C. 所考慮之斷面需遠離桿端之施力點，或遠離斷面積有突然變化之處。



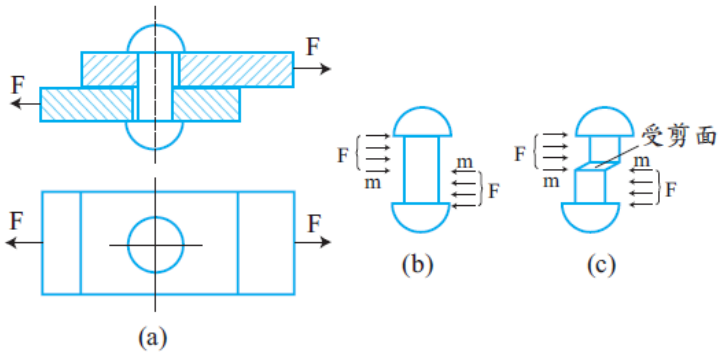
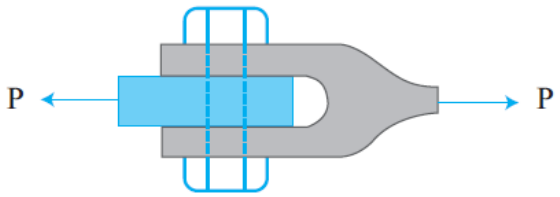
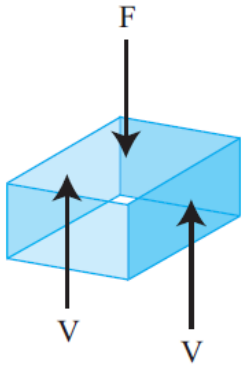
二、剪力之剪應力

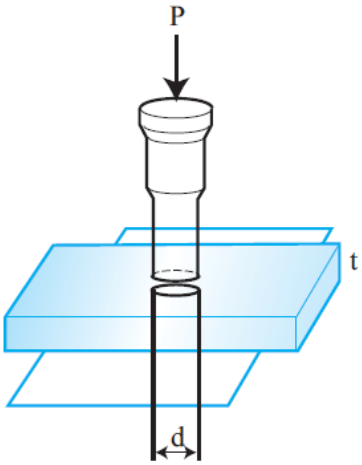
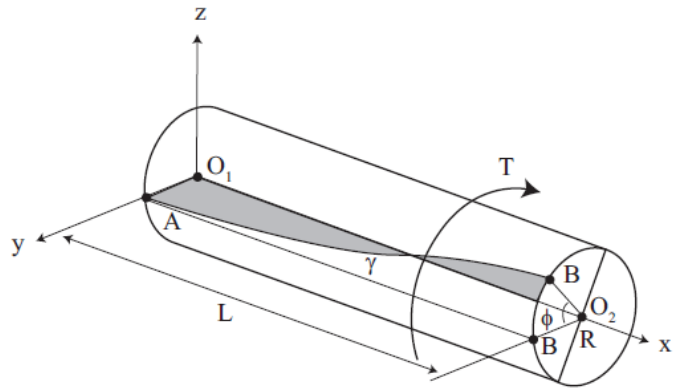
1.剪力及剪應力

剪力及剪應力定義：**一物體受作用力作用時，使該物體其中一部份與另一部份產生沿外力作用方向，發生相對滑動或剪斷的趨勢時，我們稱此種作用力為剪力**(shear force)，分佈於與剪力平行之平面上的應力稱爲剪應力，亦即是單位面積內所受的剪力，通常以希臘符號 τ 表示，兩個力之間的截面沿作用面積相對錯動，可能被剪斷的截面稱爲剪切面，如圖　所示，該物體受到P作用力，則其作用面積所受的剪力爲V，其中V＝P，則**剪應力為** $\tau = \frac{V}{A}$ (A：作用面積)。



(2) 剪力的形式

受力型式	破壞模式
	單剪：$\tau = \dfrac{F}{n\left(\dfrac{\pi \cdot d^2}{4}\right)}$ n：總鉚釘數量 d：鉚釘直徑
	雙剪：$\tau = \dfrac{P}{2n\left(\dfrac{\pi \cdot d^2}{4}\right)}$ n：總鉚釘數目 d：鉚釘直徑
	$F = 2V$ $\Rightarrow$剪應力 $\tau = \dfrac{V}{A} = \dfrac{F}{2A}$ A：作用面積

受力型式	破壞模式
	1. 剪床剪切時的撕裂剪應力 剪應力：$\tau=\frac{P}{\pi dt}$ d：衝頭直徑 t：板厚度 2. 衝頭所受的壓應力 壓應力：$\sigma=\frac{P}{\frac{\pi}{4}\times d^2}$
	1. 當r＝R時有最大剪應力 $\tau_{max}=\frac{TR}{J}$ 2. $\tau=G\gamma=Gr\frac{\varphi}{\ell}\Rightarrow$扭轉角度$\varphi=\frac{T\ell}{GJ}$(其中$\ell$表圓軸桿長) 3. 實心圓軸的極慣性矩$J=\frac{\pi D^4}{32}$，空心圓軸$J=\frac{\pi}{32}(D^4-d^4)$

三、應力及其單位

應力及其單位

(1) 應力：材料在單位面積上所承受的內力。

$$\sigma=\frac{P}{A}$$

(2) 單位：　A.重力單位：kgf/cm^2

B.絕對單位：$N/m^2(Pa)$、MPa、GPa

C.換算單位：$1MPa=10^6Pa=10^6N/m^2=1N/mm^2$

$1GPa=10^9Pa=10^3N/mm^2=1kN/mm^2$

$1Pa=1.02\times10^{-5}kg/cm^2$

$1MPa=10.2kg/cm^2$

四、安全係數

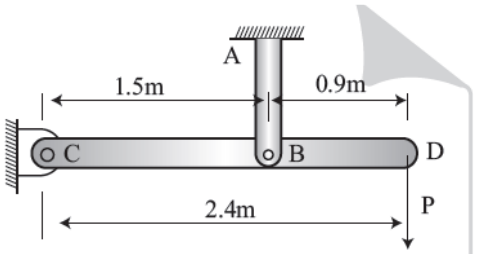
在工程設計上爲確保結構物的安全，並滿足強度、剛度及穩定度之要求，故結構實際所能支持的負載必須大於操作時所需承受的負載，在設計結構物時，使材料在安全範圍內所能承受的最大工作應力，稱爲工作容許應力(allowable stress)，當材料受外力而至發生降伏或斷裂的情形，此時材料所受的應力稱爲破壞應力，將結構實際可承受之破壞應力與容許應力之比值稱爲安全係數或安全因數，通常以n表示：

$$n = \frac{\text{破壞應力}}{\text{容許應力}}$$

爲避免結構的損壞，安全因數必須大於1，且安全因數愈大結構愈安全，通常在1～10之間，一般如果是延性材料的話，材料維持在線彈性區域內，避免負載卸除時產生永久變形的破壞，因此是以材料的降伏應力爲破壞應力，脆性材料的話，材料在破壞時並沒有明顯的降伏情況，則以材料的極限應力爲破壞應力。

範例 1

如圖所示，在水平桿CBD的端點D處，承受一負荷P，如垂直桿AB之截面積為500mm^2，其所生之應力為20MPa，試求此負荷P為若干N？



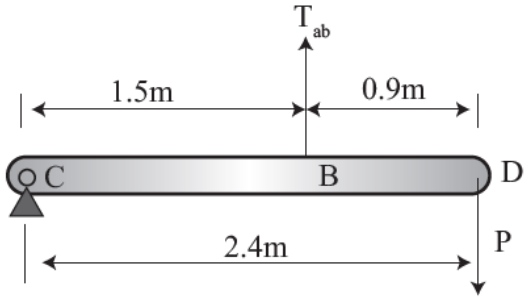
【解析】

$$\sigma_{ab}=\frac{T_{ab}}{A}$$

$$\therefore T_{ab}=\sigma_{ab}\times A=(2\times10^{7})(5\times10^{-4})$$

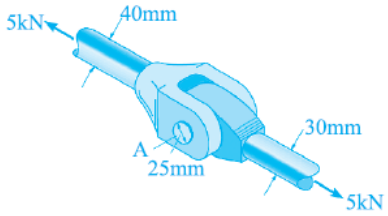
$$=10000\text{N}\quad \Sigma M_C=10000\times1.5-P\times2.4=0$$

$$\therefore P=6250\text{N}$$



範例 2

軛桿連結用來支撐一張力5kN。求每根桿的平均正向應力及構件間之銷A的平均剪應力。



【解析】

(1)桿40mm受正向應力 $\sigma_{40}=\dfrac{P}{A}=\dfrac{5(10^3)}{\frac{\pi}{4}(40)^2}=3.98\text{MPa}$

(2)桿30mm受正向應力 $\sigma_{30}=\dfrac{V}{A}=\dfrac{5(10^3)}{\frac{\pi}{4}(30)^2}=7.07\text{MPa}$

(3)銷A受剪應力 $\tau_{avg}=\dfrac{P}{A}=\dfrac{2.5(10^3)}{\frac{\pi}{4}(25)^2}=5.09\text{MPa}$

範例 3

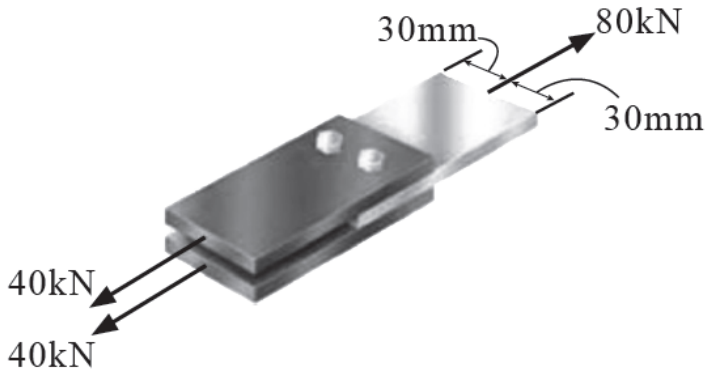
如圖所示之接頭由兩支螺栓固定，假設板厚皆為12mm，材料之剪力強度為350MPa，抗拉、抗壓強度皆為600MPa，並可忽略摩擦之影響。在設計安全係數(factor of safety)要求為2.5，且在下列各設計考量下，試求螺栓所需之直徑：

(一) 螺栓剪壞。

(二) 板孔承壓(bearing)破壞。

(三) 板拉壞。

(四) 整體(overall)接頭設計安全。



【103 地特四等】

【解析】

(一) $\tau = \frac{V}{As} \le \frac{Ss}{Fs}$

$$\frac{40\times10^3}{\frac{\pi}{4}d^2\times2} \le \frac{350}{2.5} \Rightarrow d \ge 13.48(mm) \quad 取 d = 13.5(mm)$$

(二) $\sigma_b = \frac{P}{Ac} \le \frac{Sc}{Fs}$

$$\frac{80\times10^3}{d\times12\times2} \le \frac{600}{2.5} \Rightarrow \sigma \ge 13.88(mm) \quad 取\ d = 14(mm)$$

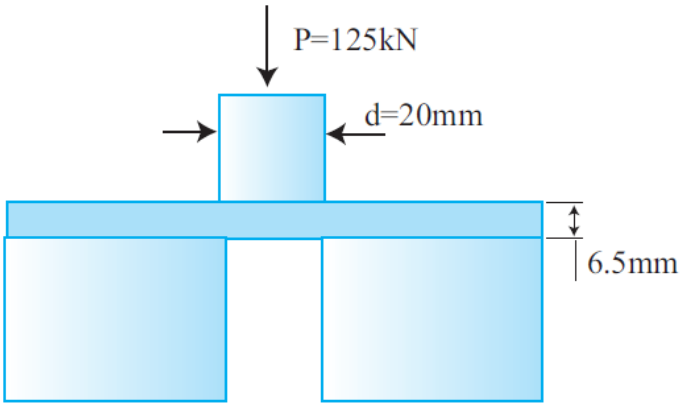
(三) $\sigma_t = \frac{P}{At} \le \frac{St}{Fs}$

$$\frac{80\times10^3}{(60-2d)12} \le \frac{600}{2.5} \quad d \le 16.11 \quad 取 d = 16(mm)$$

(四) 由上計算可知，取d＝16(mm)

範例 4

下圖所示有一沖頭用來在鋼板上沖製圓孔，該沖頭之直徑為20mm，鋼板之厚度為6.5mm。假設沖頭之出力為P＝125kN，請求出鋼板所受之平均剪應力及沖頭所受之平均壓應力（不考慮該沖頭之衝擊影響）。



【台科機械】

【解析】

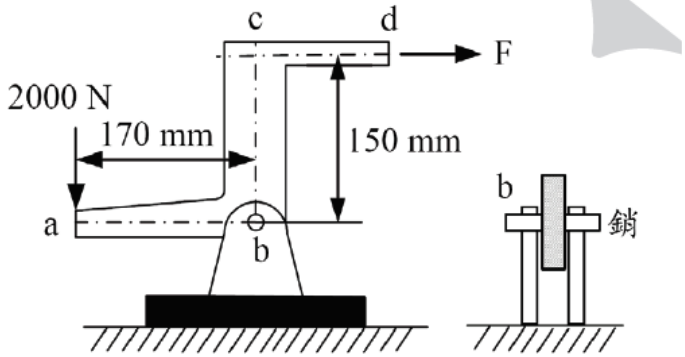
(1)鋼板所受之平均剪應力：

$$\tau = \frac{P}{A} = \frac{125\times10^3}{\pi\times(0.02)\times(0.0065)} = 306067198.25\,\text{Pa} = 306.07(\text{MPa})$$

(2)平均壓應力：$\sigma = \dfrac{P}{A} = \dfrac{125\times10^3}{\frac{\pi}{4}\times(0.02)^2} = 397887357.73\,\text{Pa} = 397.89(\text{MPa})$

範例 5

如圖所示，ab長度為170mm，bc長度為150mm，在a端施加一垂直向下之力量2000N，在d端則有一水平力F保持平衡，b銷裝置如圖之右邊小圖，b銷直徑為6mm，試求b銷所受之平均剪應力為多少？(101關四)



【解析】

$$\sum M_b = 0$$

$$F \times 150 = 2000 \times 170 \Rightarrow F = 2266.67(N)$$

$$\sum F_x = 0 \Rightarrow b_x = 2266.67(N)$$

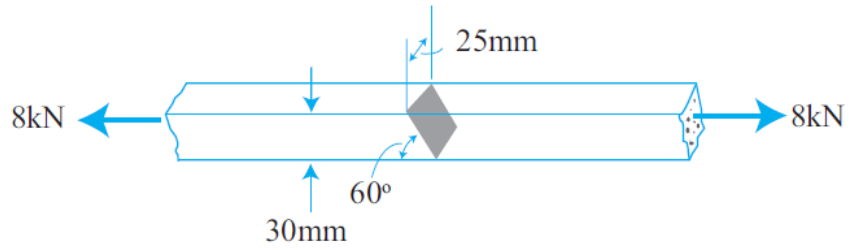
$$\sum F_y = 0 \Rightarrow b_y = 2000(N)$$

$$R_b = \sqrt{(2266.67)^2 + (2000)^2} = 302288$$

$$\text{雙剪 } \tau = \frac{R_b}{2 \times A} = \frac{3022.88}{2 \times \frac{\pi}{4} \times (6)^2} = 53.456(MPa)$$

範例 6

兩鋼構件使用—60°嵌焊結合在一起，求在焊接面的平均正向應力和平均剪應力。



【解析】

$$^{+}\nearrow \Sigma F_x = 0 \ ; \ N - 8\sin 60^\circ = 0 \ ; \ N = 6.928\text{kN}$$

$$\searrow^{+} \Sigma F_y = 0 \ ; \ V - 8\cos 60^\circ = 0 \ ; \ V = 4\text{kN}$$

$$A = (25)\left(\frac{30}{\sin 60^\circ}\right) = 866.03\text{mm}^2$$

$$\sigma = \frac{N}{A} = \frac{6.928(10^3)}{866.03\text{mm}^2} = 8\text{MPa}$$

$$\tau_{avg} = \frac{V}{A} = \frac{4(10^3)}{866.03\text{mm}^2} = 4.62\text{MPa}$$

範例 7

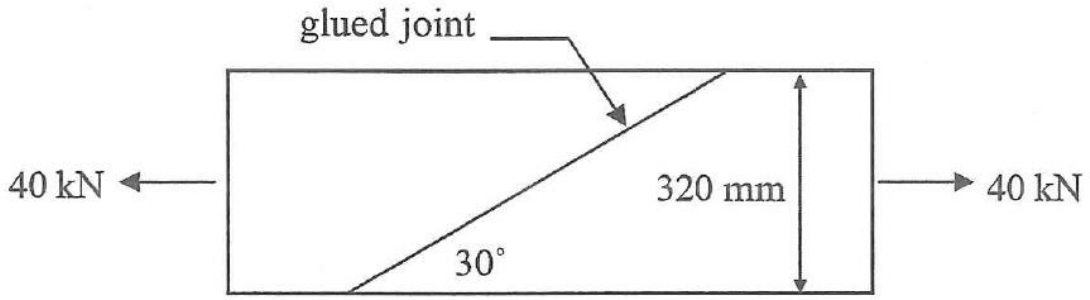
Question 2 (20%)

Two wood plates 320 mm wide and 25 mm thick are glued together at an angle of 30 degrees to the horizontal axis as shown. The glued plate assembly resists an axial load of 40 kN. Use the Mohr's circle solution (**Not** the transformation equations) to determine the following:

a) The normal and shear stress acting on the glued joint. Make sure to show your answer on a properly oriented element.

b) The maximum in-plane shear stress (and associated normal stresses) acting on the wood plates. Once again, show your answer on a sketch of a properly oriented element.

Note: the credit will only be given for a solution using Mohr's circle. This mean that you need to draw a Mohr's circle based on the stress components given in this problem. Remember to show numbers on your circle. Your calculations must be based on the geometry of your circle. In other words, you are expected to use trigonometry to construct your Mohr's circle.



【103 台大土木】

【解析】

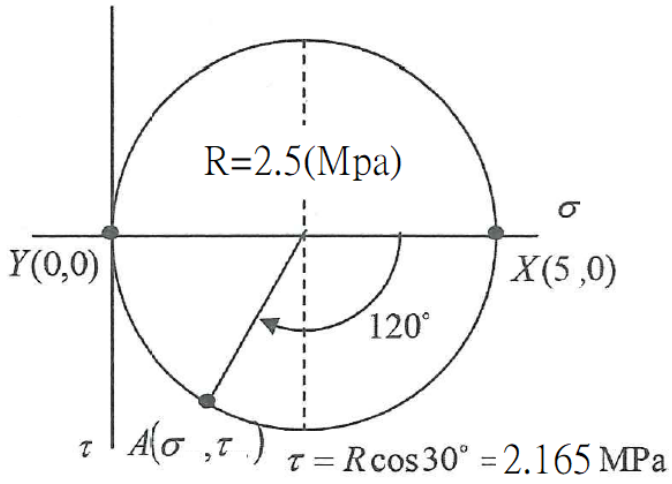
$$\sigma_x = \frac{40000}{320 \times 25} = 5(MPa)$$

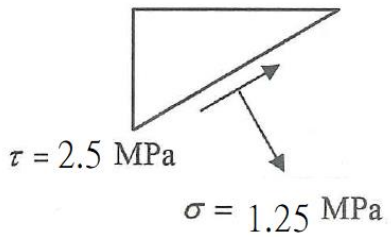
$$\sigma_{av} = \frac{5}{2} = 2.5(MPa) = R = \tau_{\max}$$

畫莫爾圓

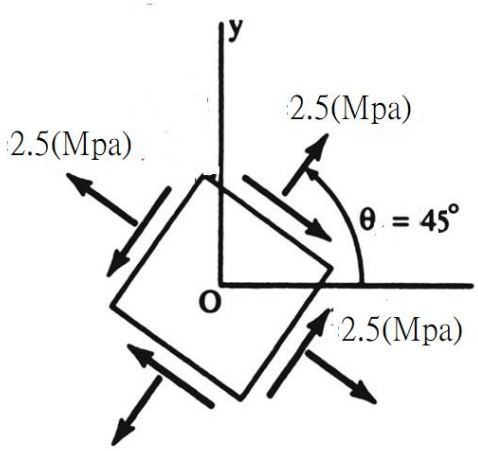
$\sigma = R - \mathrm{Rsin}30^\circ = 1.25(MPa)$

$\tau = \mathrm{Rcos}30^\circ = 2.165(MPa)$





(b)



2-2. 平面應力

一、單軸向負荷之應力

1. 單軸向負荷之應力

(1) 如圖　所示垂直之橫截面上的正應力：$\sigma = \frac{P}{A}$

(2) 斜截面上的應力：$\sigma_p = \frac{P}{A_a} = \frac{P}{A/\cos\alpha} = \sigma\cos\alpha$ (A_a：斜面面積、α：傾斜角)

(3) 斜截面上的正應力 σ_a 和剪應力 τ_a 爲

$$\sigma_a = \sigma_p \cos\alpha = \sigma\cos^2\alpha$$

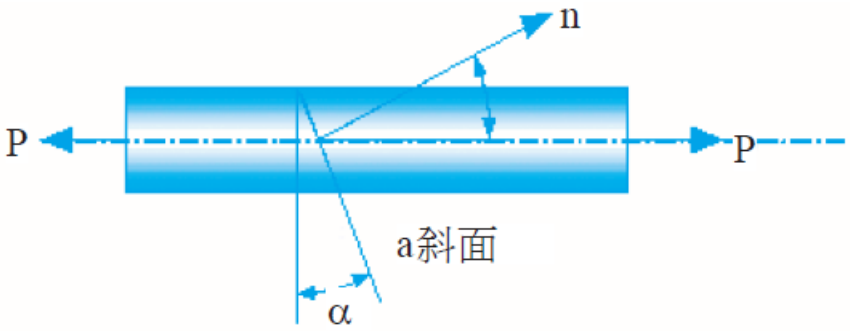
$$\tau_a = \sigma_p \sin\alpha = \frac{\sigma}{2}\sin 2\alpha$$

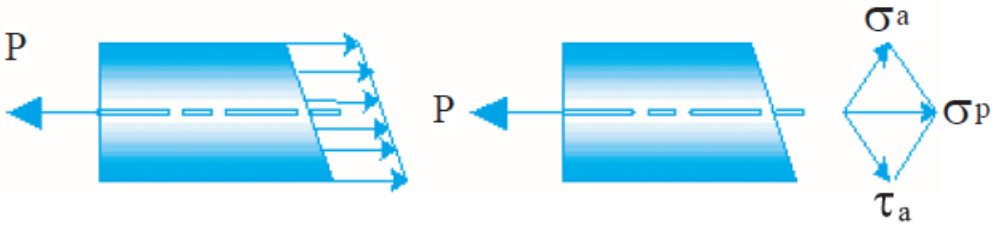
角度	受力狀況
α	$\sigma_a = \sigma_p \cos\alpha = \sigma\cos^2\alpha$ $\tau_a = \sigma_p \sin\alpha = \frac{\sigma}{2}\sin 2\alpha$
$\alpha = 0^\circ$	$\sigma_a = \sigma\cos^2\alpha = \sigma$ $\tau_a = \frac{\sigma}{2}\sin 2\alpha = 0$
$\alpha = 45^\circ$	$\sigma_a = \sigma\cos^2\alpha = \frac{\sigma}{2}$ $\tau_a = \frac{\sigma}{2}\sin 2\alpha = \frac{\sigma}{2}$

備註：

不與外力垂直之任意截面上，必有正交應力與剪應力同時作用於該截面上，該正交應力與剪應力的值與傾斜角α有關，當傾斜角度α＝0°時，cosα＝1，亦即外力P與截面正交，此時會有最大的正交應力，且剪應力$\tau = 0$。

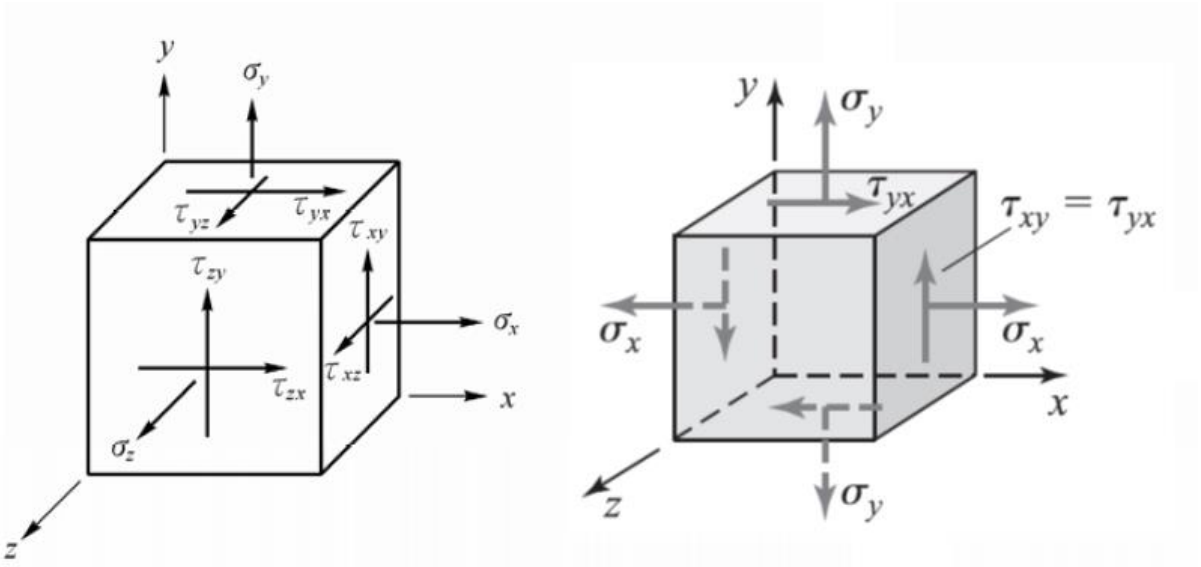
當傾斜角度α＝45°時，亦即外力與截面成45°斜角，將會產生最大剪應力，故$\tau_{max} = \frac{\sigma}{2}$，此時正交應力$\sigma_{max} = \frac{\sigma}{2}$，材料的抗剪強度小於抗拉強度，所以材料在破壞時會沿45°之傾斜面斷裂。





二、雙軸向負荷之應力

在一個結構受到負載時，於結構內部任一點所受應力一般情況下均為三維應力，即在各點三個互相垂直的軸向上均有應力作用，如下圖之左圖所示，圖中 σ_x 為正向應力表示為材料點上單位面積所受的正向力，正向應力 σ_x 的下標表示其作用面的外法線方向；τ_{xy} 及 τ_{xz} 為剪應力表示為材料點上單位面積所受的剪力，τ_{xy} 及 τ_{xz} 剪應力有兩個下標，前一個下標表示其作用面的外法線方向(x 作用面)，後一個下標表示其作用方向沿著哪一個座標軸，若在 Z 作用面上(Z 軸為外法線方向)所有正向應力及剪應力皆為零的情況下，可得到二維應力元素圖(如下圖之右圖:下標 Z 之應力皆為零)。



2. 雙軸向負荷之應力

前節已談過材料承受單軸向應力，本節則討論雙軸向應力（biaxial stress）之情形，即材料內之任一點同時承受兩互相垂直之軸向應力，如圖所示一應力元素，在x及y軸方向僅受軸向應力，而無剪應力，則斜截面pq之法線方向與X軸之夾角爲θ，茲取斜截面pq左側自由體圖，如圖　(b)所示，假設σ_θ作用面上之面積爲A，則σ_x作用面上之面積爲Acosθ，σ_y作用面上之面積爲Asinθ，則斜截面pq上之正交應力σ_θ與剪應力τ_θ，由自由體圖上n與t方向之平衡方程式可得：

$$\sum F_{x'} = 0 \Rightarrow \sigma_\theta A = \sigma_y \sin\theta + \sigma_x \cos\theta \times A\cos\theta$$

$$\Rightarrow \sigma_\theta = \sigma_y \times \sin^2\theta + \sigma_x \times \cos^2\theta$$

$$\sum F_{y'} = 0 \Rightarrow \tau_\theta A = -\sigma_y \cos\theta \times A\sin\theta + \sigma_x \sin\theta \times A\cos\theta$$

$$\Rightarrow \tau_\theta = -\sigma_y \times \cos\theta\sin\theta + \sigma_x \times \cos\theta\sin\theta$$

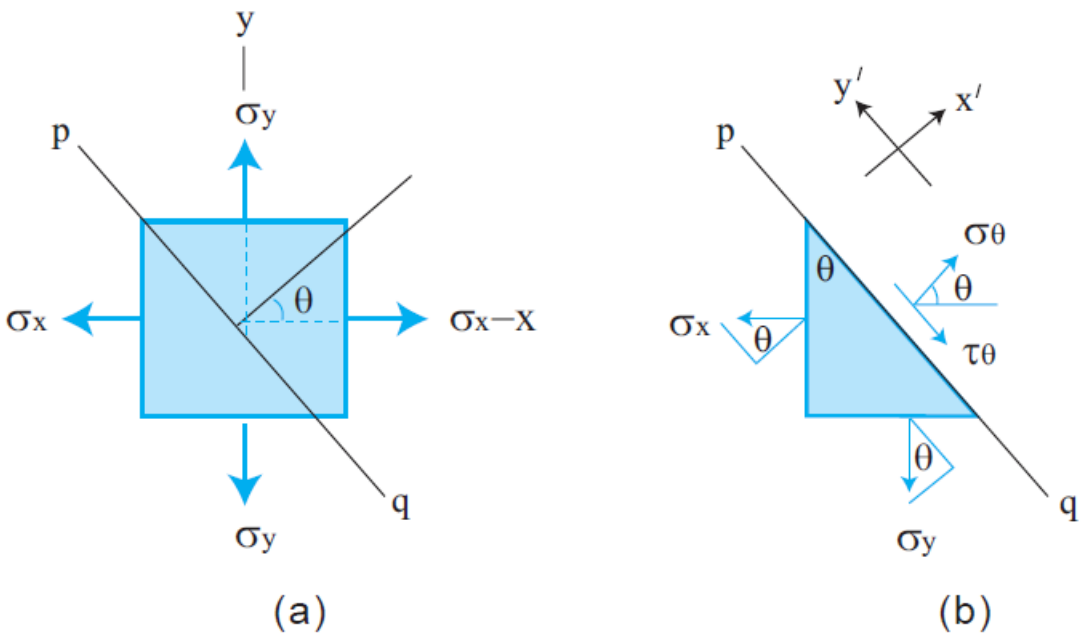
由三角關係

$$\cos^2\theta = \frac{1}{2}(1+\cos 2\theta) \quad , \quad \sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$$

$$\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta$$

最後可得：

$$\sigma_\theta = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta \qquad \tau_\theta = \frac{\sigma_x - \sigma_y}{2}\sin 2\theta$$



(2)雙軸向負荷之說明

A.當θ＝0°時，$\sigma_x = \sigma_\theta$，θ＝90°時，$\sigma_y = \sigma_\theta$；即θ由0°至90°時，σ_θ由σ_x變化至σ_y。

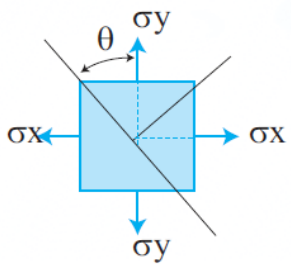
B.若$\sigma_x > \sigma_y$時且當θ＝0°時，正交應力之最大值$(\sigma_\theta)_{max} = \sigma_x$；若$\sigma_x > \sigma_y$時且當θ＝90°，正交應力之最小值$(\sigma_\theta)_{min} = \sigma_y$，一般常將正交應力之最大值及最小值，稱為主應力（principal stress），而其作用面稱為主平面（principal plane）。

C.當θ＝0°及θ＝90°時，亦即主平面上之剪應力$\tau_\theta = 0$，亦即主平面上無剪應力存在。

D.至於任意斜截面上之剪應力τ_θ由θ＝0°時之最小值，隨θ值之增大而逐漸增加，至θ＝45°時剪應力達最大值， 故最大剪力之作用面與主平面相差45°，且其值等於兩主應力差之一半，亦即$\tau_{45°} = \tau_{max} = \frac{\sigma_x - \sigma_y}{2}$。

E.當應力元素承受雙軸向應力時，且若$\sigma_x = -\sigma_y$時，且當θ＝45°時斜面上僅有剪應力，而無正交應力，此方位之應力狀態為純剪。

F.任意兩互相垂直之斜截面上，其正交應力之和恆與雙軸向應力之和相等，但其剪應力之大小相等，方向相反。



雙軸向應力

	應力	剪應力
θ 角	$\sigma_\theta=\frac{1}{2}(\sigma_x+\sigma_y)+\frac{1}{2}(\sigma_x-\sigma_y)\cos 2\theta$	$\tau_\theta=\frac{1}{2}(\sigma_x-\sigma_y)\sin 2\theta$
0°	$\sigma_\theta=\sigma_x$（若$\sigma_x>\sigma_y$則$\sigma_{0°}=\sigma_{max}$）	$\tau_\theta=0$
90°	$\sigma_\theta=\sigma_y$（若$\sigma_x>\sigma_y$則$\sigma_{90°}=\sigma_{min}$）	$\tau_\theta=0$
45°	$\sigma_\theta=\frac{1}{2}(\sigma_x+\sigma_y)$	$\tau_\theta=\tau_{max}=\frac{1}{2}(\sigma_x-\sigma_y)$
θ ，90° + θ 互餘應力	$\sigma_\theta+\sigma'_\theta=\sigma_x+\sigma_y$（互餘應力$\sigma_{\theta'}$）	互餘剪應力$\tau_\theta=-\tau'_\theta$

三、平面應力分析

(一) 任意斜截面上的應力

如圖所示，於應力元素上取任一截面位置，針對斜線部份的應力元素面在外法線 n 和切線 t 上列平衡方程，

$$\sum Fn=0\Rightarrow \sigma_\theta dA-(\tau_{xy}dA\cos\theta)\sin\theta-(\sigma_x dA\cos\theta)\cos\theta$$
$$-(\tau_{yx}dA\sin\theta)\cos\theta-(\sigma_y dA\sin\theta)\sin\theta=0$$

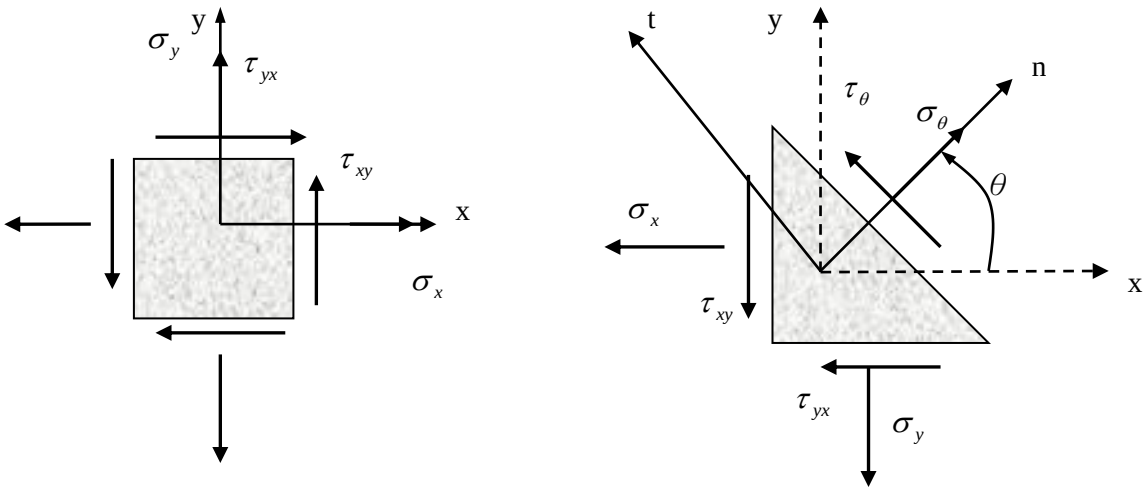
$$\sum F_t=0\Rightarrow \tau_\theta dA-(\tau_{xy}dA\cos\theta)\cos\theta-(\sigma_x dA\cos\theta)\sin\theta$$
$$-(\sigma_y dA\sin\theta)\cos\theta+(\tau_{yx}dA\sin\theta)\sin\theta=0$$

根據剪應力 $\tau_{xy}=\tau_{yx}$ 且

$$\cos^2\theta=\frac{1+\cos 2\theta}{2}\ ,\sin^2\theta=\frac{1-\sin 2\theta}{2}\ ,\ 2\sin\theta\cos\theta=\sin 2\theta$$

得 $\boxed{\sigma_\theta=\frac{\sigma_x+\sigma_y}{2}+\frac{\sigma_x-\sigma_y}{2}\cos 2\theta+\tau_{xy}\sin 2\theta}$

$$\boxed{\tau_\theta=-\frac{\sigma_x-\sigma_y}{2}\sin 2\theta+\tau_{xy}\cos 2\theta}$$



平面應力分析

(二) **主應力及角度**

將正應力公式對 θ 取導數，得

$$\frac{d\sigma_\theta}{d\theta} = 2\left[-(\frac{\sigma_x - \sigma_y}{2})\sin 2\theta + \tau_{xy}\cos 2\theta\right]$$

若 $\theta = \theta_p$ 時，能使導數 $\frac{d\sigma_\theta}{d\theta} = 0$ ，則

$$-\frac{\sigma_x - \sigma_y}{2}\sin 2\theta_p + \tau_{xy}\cos 2\theta_p = 0$$

$$\boxed{\tan(2\theta_p) = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}} \Rightarrow \boxed{\theta_p = \frac{1}{2}\tan^{-1}(\frac{2\tau_{xy}}{\sigma_x - \sigma_y})}$$

上式有兩個解：即 θ_p 和 $\theta_p \pm 90^\circ$ 表示兩個互相垂直平面上，所對應的最大正應力及最小正應力所在的平面，求得最大或最小正應力爲

$$\boxed{\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{(\frac{\sigma_x - \sigma_y}{2})^2 + \tau_{xy}^2}}$$

θ_p 代入剪力公式 $\Rightarrow \tau_{\theta p} = 0$ 稱之為主平面，亦即表示正應力爲最大或最小所在的平面，因此最大或最小的正應力亦可稱為主應力。

(三) **最大剪應力及角度**

將剪應力公式對 θ 求導，令 $\frac{d\tau_\theta}{d\theta} = -(\sigma_x - \sigma_y)\cos 2\theta - 2\tau_{xy}\sin 2\theta = 0$

若$\theta=\theta_s$時，能使導數$\frac{d\tau_\theta}{d\theta}=0$，則在$\theta_s$所確定的截面上，剪應力取得極值。通過求導可得

$$-(\sigma_x-\sigma_y)\cos 2\theta_s-2\tau_{xy}\sin 2\theta_s=0$$

$$\boxed{\tan(2\theta_s)=-\frac{\sigma_x-\sigma_y}{2\tau_{xy}}}\Rightarrow\boxed{\theta_s=\frac{1}{2}\tan^{-1}(\frac{\sigma_y-\sigma_x}{2\tau_{xy}})}$$

求得剪應力的最大值和最小值是：

$$\boxed{\tau_{\max}=\sqrt{(\frac{\sigma_x-\sigma_y}{2})^2+\tau_{xy}^2}=\frac{\sigma_1-\sigma_2}{2}}$$

且$\boxed{\sigma_1+\sigma_2=\sigma_x+\sigma_y}$

因此主平面與最大剪應力平面夾角為$45^\circ\Rightarrow\theta_p=\theta_s\pm 45^\circ$

四、 莫爾圓求斜面之應力

(一) 莫爾圓方程式

將公式$\begin{cases}\sigma_\theta=\frac{\sigma_x+\sigma_y}{2}+\frac{\sigma_x-\sigma_y}{2}\cos 2\theta-\tau_{xy}\sin 2\theta\\ \tau_\theta=\frac{\sigma_x-\sigma_y}{2}\sin 2\theta+\tau_{xy}\cos 2\theta\end{cases}$ 中的θ削掉，得

$$\left(\sigma_\alpha-\frac{\sigma_x+\sigma_y}{2}\right)^2+{\tau_\alpha}^2=\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+{\tau_{xy}}^2$$

由上式確定的以σ_α和τ_α爲變數的圓，這個圓稱作應力圓。 圓心的橫坐標爲$\frac{1}{2}(\sigma_x+\sigma_y)$，縱坐標爲零，圓的半徑爲$\sqrt{\left(\frac{\sigma_x+\sigma_y}{2}\right)^2+\tau_{xy}^2}$ 。

(二) 應力圓的畫法

1. 建立$\sigma-\tau$應力坐標系在坐標系內畫出點(σ_x,τ_{xy})和(σ_y,τ_{yx})，如圖 3.8(b)所示。
2. 此兩點的連線與軸的交點 O 便是圓心，以 O 爲圓心，(σ_x,τ_{xy})到 O 點距離爲半徑畫一應力圓。
3. 如圖所示，若應力元素逆時針旋轉α角，則表現在莫爾圓上為以(σ_x,τ_{xy})座標點逆時針旋轉2α，得到$(\sigma_{x1},\tau_{x1y1})$、$(\sigma_{y1},\tau_{y1x1})$兩座標點，此為$\alpha$平面上之正向應力與剪應力$(\sigma_\theta,\tau_\theta)$。
4. 圓心為$(\frac{\sigma_x+\sigma_y}{2},0)$，半徑為$\frac{\sigma_1-\sigma_2}{2}$＝R
5. 最大剪應力=莫爾圓半徑 (R)，最大剪應力面與主平面夾 45°

(三) 在應力圓上標出極值應力

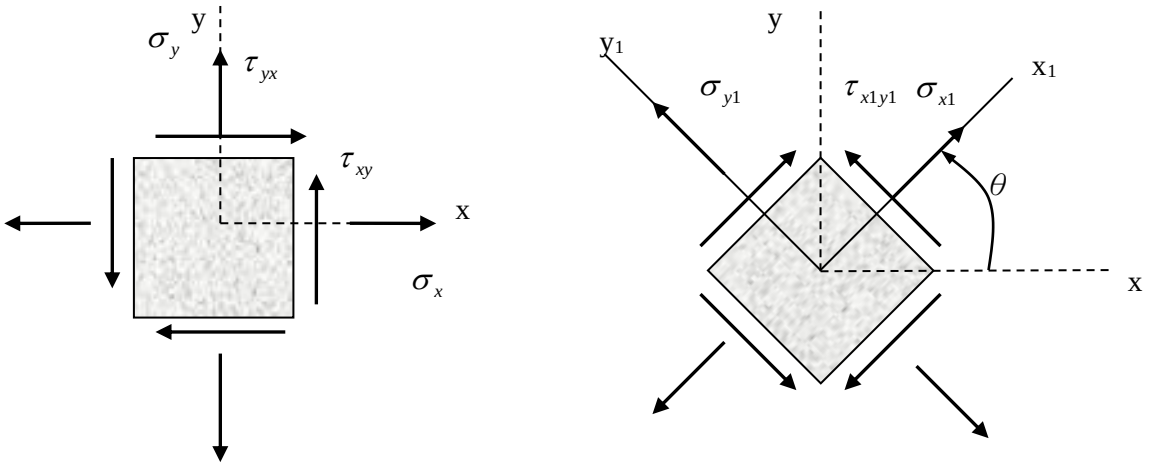
$$\begin{cases}\sigma_1 \\ \sigma_2\end{cases} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}{}^2}$$

$$\begin{cases}\tau_{\max} \\ \tau_{\min}\end{cases} = \pm R = \pm\frac{\sigma_1 - \sigma_2}{2} = \pm\sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}{}^2}$$

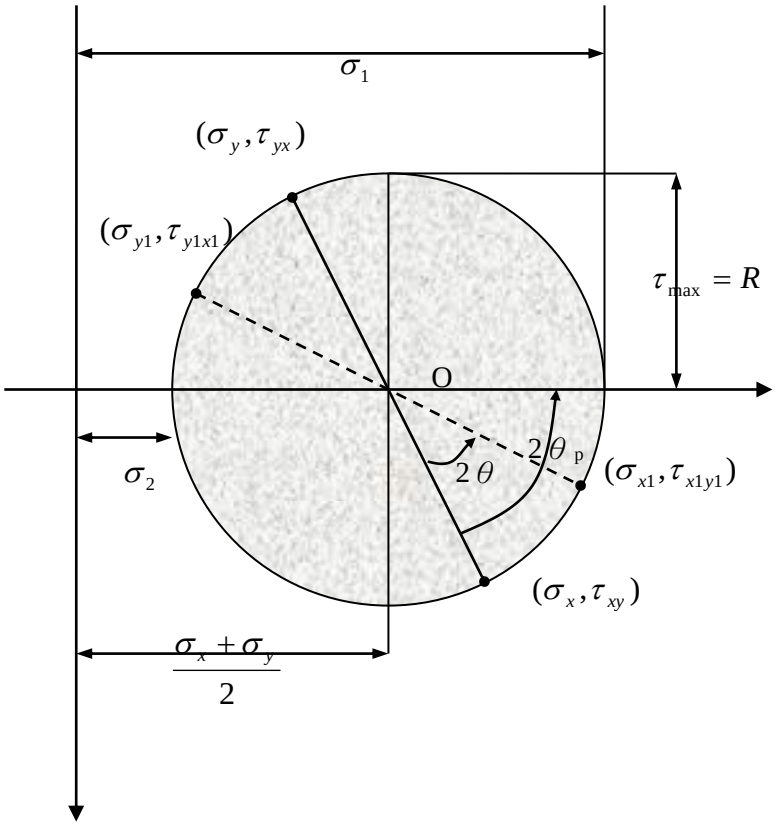
(四) 公式法與莫爾圓法方向

1．本書公式解中剪應力τ_{xy}向上為正、τ_{yx}向右為正、σ_x向右為正、σ_y向上為正。

2．本書莫爾圓中τ_{xy}以向下為正，角度α以逆時針方向為正。

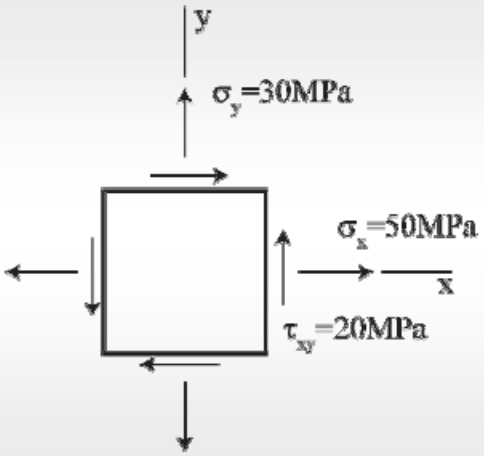


應力元素


σ_1
(σ_y, τ_{yx})
$(\sigma_{y1}, \tau_{y1x1})$
$\tau_{max} = R$
O
σ_2
2θ
$2\theta_p$
$(\sigma_{x1}, \tau_{x1y1})$
(σ_x, τ_{xy})
$\frac{\sigma_x + \sigma_y}{2}$

範例 8

平面上應力元素受應力 $\sigma_x = 50\text{Mpa}$，$\sigma_y = 30\text{MPa}$ 及 $\tau_{xy} = 20\text{MPa}$，如圖所示。求作用從 x 軸指向 $\theta = 50°$ 之元素上的應力；於此，角 θ 逆時針方向為正。示出這些應力於指向 θ 之元素圖上。



【解析】

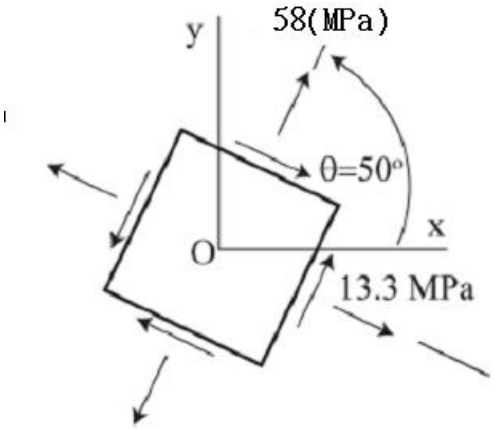
解 $\sigma_x = 50\text{MPa},\ \sigma_y = 30\text{MPa}$

$\tau_{xy} = 20\text{MPa},\ \theta = 50°$

$$\sigma_{x_1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta = 58(\text{MPa})$$

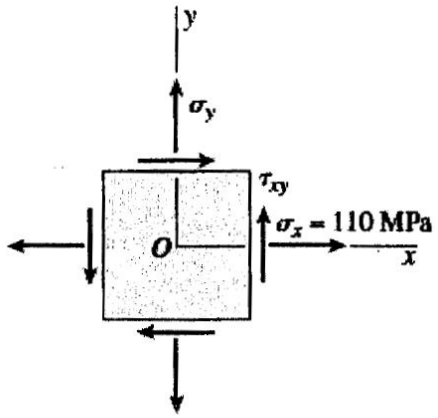
$$\tau_{x_1y_1} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta = -13.3\text{MPa}$$

$$\sigma_{y_1} = \sigma_x + \sigma_y - \sigma_{x_1} = 22(\text{MPa})$$



範例 9

The surface of an airplane wing is subjected to plane stress with normal stresses σ_x and σ_y and shear stress τ_{xy}, as shown in the figure. At a counterclockwise angle θ=32° from the x axis, the normal stress is 37 MPa tension, and at an angle θ=48°, it is 12 MPa compression. If the stress σ_x equals 110 MPa tension, what are the stresses σ_y and τ_{xy}?



【103 成大醫工】【98 台大工科類題】

【中文翻譯】

飛機翼之表面受有平面應力，有正向應力σ_x及σ_y及剪應力τ_{xy}，如圖所示。在從x軸$\theta = 32°$角處，正向應力為37 MPa拉伸，且在θ=48°，係為12MPa壓縮。如果應力σ_x等於110 MPa拉伸，則應力σ_y及τ_{xy}為何？

【解析】

$$\theta = 32°, \sigma_{x1} = 37 \text{ MPa}$$

$$\sigma_{x1} = \frac{110+\sigma_y}{2} + \frac{110-\sigma_y}{2}\cos(2\times 32^\circ) + \tau_{xy}\sin(2\times 32^\circ) = 37(MPa)$$

$$\theta = 48°, \sigma_{x1} = -12 \text{ MPa}$$

$$\sigma_{x1} = \frac{110+\sigma_y}{2} + \frac{110-\sigma_y}{2}\cos(2\times 48^\circ) + \tau_{xy}\sin(2\times 48^\circ) = -12(MPa)$$

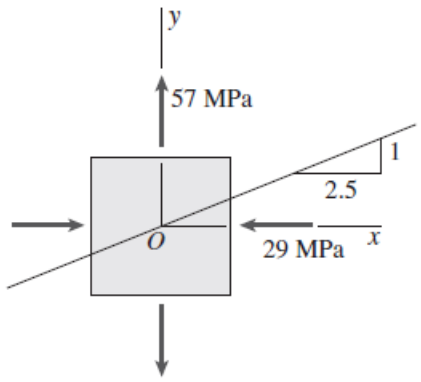
$$\sigma_y = -60.7 \text{ MPa} \qquad \tau_{xy} = -27.9 \text{ MPa} \quad \leftarrow$$

範例 10

An element in biaxial stress is subjected to stresses $\sigma_x = -29(MPa)$ and $\sigma_y = 57(MPa)$, as shown in the figure. Using Mohr's circle, determine:

(a) The stresses acting on an element oriented at a slope of 1 on 2.5 (see figure).

(b) The maximum shear stresses and associated normal stresses.Show all results on sketches of properly oriented elements.



【103 中興機械】

【中文翻譯】

雙軸向應力之元素受有應力 $\sigma_x = -29\,\text{MPa}$ 及 $\sigma_y = 57\ \text{MPa}$，如圖所示，使用摩爾圓求：(a) 作用在斜率為 1 比 2.5 之指向的元素上之應力，及 (b) 最大剪應力及伴隨之正向應力。將所有的結果繪示於正確指向的元素上。

【解析】

(a)

$$\sigma_x = -29\ \text{MPa} \qquad \sigma_y = 57\ \text{MPa} \qquad \tau_{xy} = 0\ \text{MPa}$$

$$\theta = \text{atan}\left(\frac{1}{2.5}\right) \qquad \theta = 21.801° \quad \leftarrow$$

$$2\theta = 43.603° \qquad R = \frac{|\sigma_x| + |\sigma_y|}{2} \qquad R = 43.0\ \text{MPa}$$

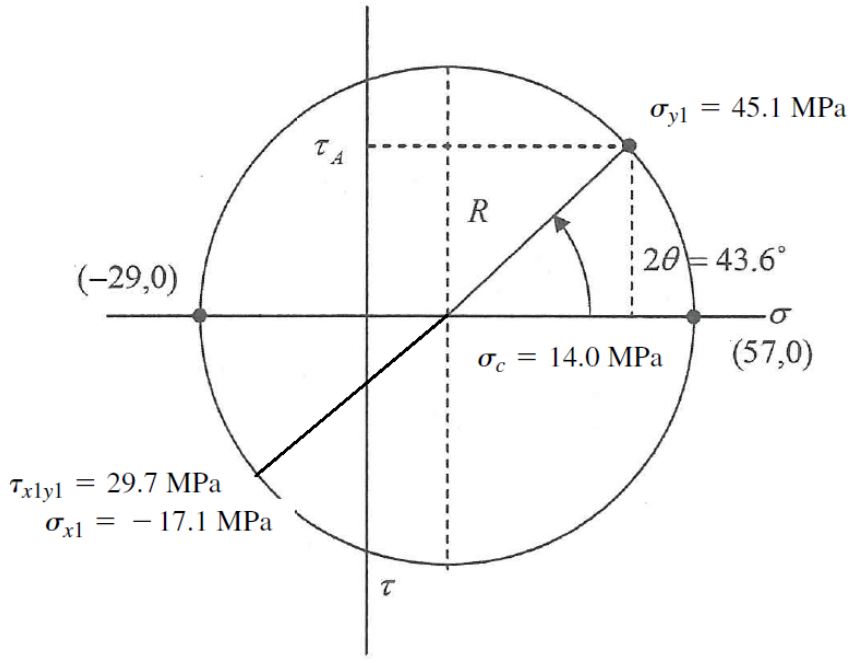
$$\sigma_c = \sigma_x + R \qquad \sigma_c = 14.0\ \text{MPa}$$

$$\sigma_{x1} = \sigma_c - R\cos(2\theta)$$

$$\sigma_{x1} = -17.1\ \text{MPa} \quad \leftarrow$$

$$\tau_{x1y1} = R\sin(2\theta) \qquad \tau_{x1y1} = 29.7\ \text{MPa} \quad \leftarrow$$

$$\sigma_{y1} = \sigma_c + R\cos(2\theta) \qquad \sigma_{y1} = 45.1\ \text{MPa}$$



(b)

$$\theta_{s1} = \frac{90^\circ}{2} \qquad \theta_{s1} = 45.0^\circ$$

$$\tau_{\max} = R \qquad \tau_{\max} = 43.0 \text{ MPa}$$

$$\theta_{s2} = \frac{-90^\circ}{2}$$

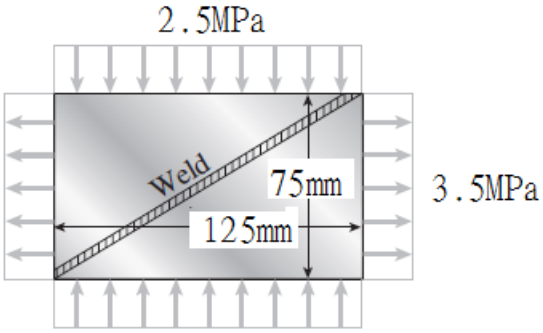
$$\theta_{s2} = -45.0^\circ$$

$$\tau_{\max} = -R \qquad \tau_{\max} = -43.0 \text{ MPa}$$

$$\sigma_{\text{aver}} = \sigma_c \qquad \sigma_{\text{aver}} = 14.0 \text{ MPa}$$

範例 11

A rectangular plate of dimensions $75mm \times 125\text{mm}$ is formed by welding two triangular plates (see figure) .The plate is subjected to a compressive stress of 2.5 MPa in the long direction and a tensile stress of 3.5MPa in the short direction (see figure). Determine the normal stress σ_w acting perpendicular to the line of the weld and the shear τ_w acting parallel to the weld.



【105 台大工科】

【中文翻譯】

一薄矩形板，尺寸為 $75mm \times 125\text{mm}$ ，由兩三角形板焊成（ 見圖 ）。板受壓縮應力2.5MPa於長的方向，而在短的方向受拉伸應力3.5 MPa。求作用在垂直於焊縫之正向應力 σ_w 及平行於焊縫之剪應力 τ_w

【解析】

$\sigma_x = 3.5MPa$ 、 $\sigma_y = -2.5MPa$ 、 $\tau_{xy} = 0$

$$\alpha = \tan^{-1}(\frac{75}{125}) = 59^\circ$$

順時針旋轉 $59^\circ \Rightarrow \theta = -59^\circ$ 代入

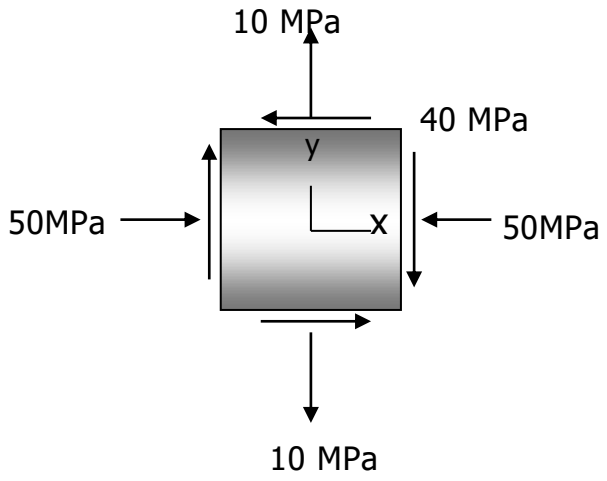
$$\sigma_w = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\Rightarrow \sigma_w = \frac{3.5 + (-2.5)}{2} + \frac{3.5 - (-2.5)}{2}\cos 2(-59^0) + 0 \times \sin 2(-59^\circ) = -0.908(MPa)$$

$$\tau_w = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta = 2.649(MPa)$$

範例 12

A two dimensional stress element is shown below. (1) Construct Mohr's circle. (2) Find stresses acting on an element inclined at an angle $\theta = 45^{o}$. (3) Compute principal stresses. (4) Compute maximum in-plane shear stress. (5) Draw properly oriented principal stress element. (6) Draw properly oriented maximum shear stress element.



【中興機械】

【解析】

(1) **公式解**

$$\begin{cases} \sigma_\theta = \dfrac{\sigma_x + \sigma_y}{2} + \dfrac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta = -60(MPa) \\ \tau_\theta = -\dfrac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta = 30(MPa) \end{cases}$$

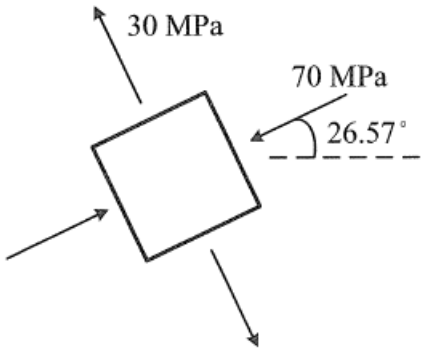
(2)主應力及最大剪應力

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{(\frac{\sigma_x - \sigma_y}{2})^2 + (\tau_{xy})^2} = \frac{-50+10}{2} \pm \sqrt{(\frac{-50-10}{2})^2 + (-40)^2} = -70,30(MPa)$$

$$\tau_{\max} = \sqrt{(\frac{\sigma_x - \sigma_y}{2})^2 + (\tau_{xy})^2} = \sqrt{(\frac{-50-10}{2})^2 + (-40)^2} = 50(\text{MPa})$$

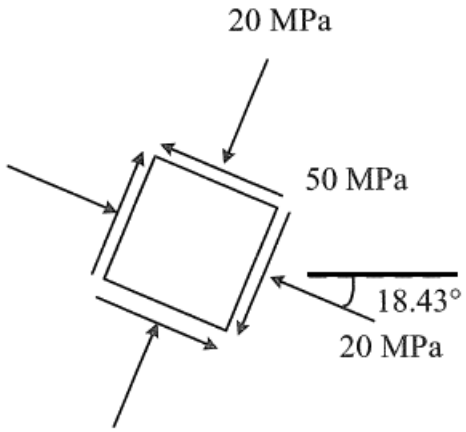
(3)

$$\tan(2\theta_p) = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} \Rightarrow \quad \theta_p = \frac{1}{2}\tan^{-1}(\frac{2\tau_{xy}}{\sigma_x - \sigma_y}) = 26.57^\circ$$



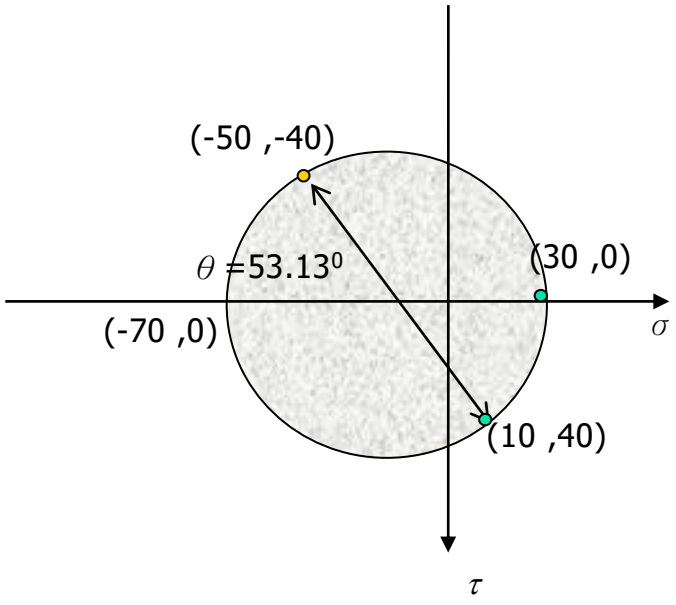
(4)

$\theta_s = \theta_P - 45^\circ = -18.43^\circ$



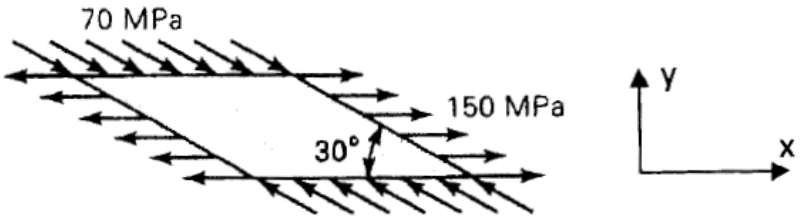
(5)

圓心：(–20,0)；半徑：$R = 50$



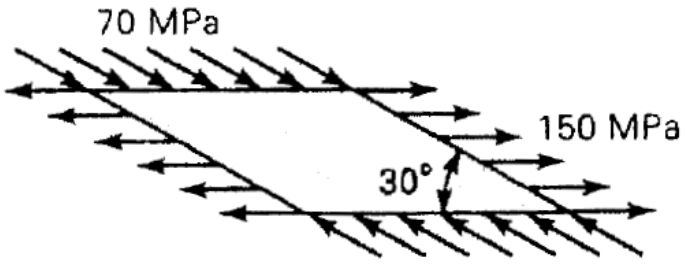
範例 13

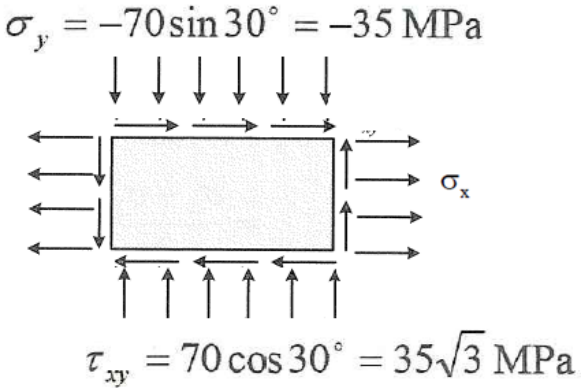
(a) Consider a thin skewed plate subjected to a uniform distribution of stress along its side. Please calculate its σ_x, σ_y, and τ_{xy} and find the principal stresses. (10 Points)



【103 成大機械】

【解析】





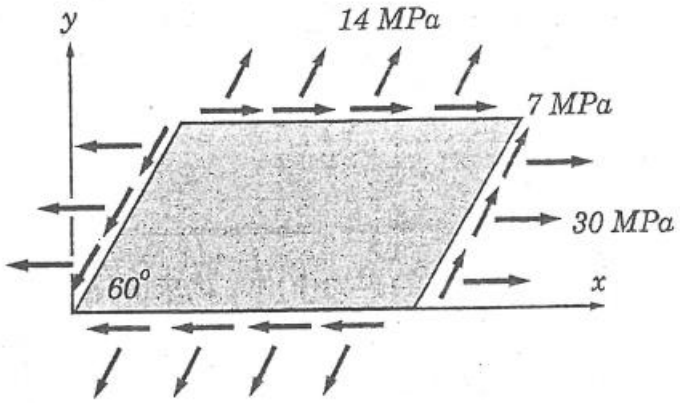
$$\sigma_\theta = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\Rightarrow 75 = \frac{\sigma_x + (-35)}{2} + \frac{\sigma_x - (-35)}{2}\cos(2\times 60^\circ) + \tau_{xy}\sin(2\times 60^\circ) \Rightarrow \sigma_x = 195(MPa)$$

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{(\frac{\sigma_x - \sigma_y}{2})^2 + \tau_{xy}^2} \Rightarrow \sigma_1 = 210(MPa), \sigma_2 = -50(MPa)$$

範例 14

A thin skewed plate is subjected to a uniform distribution of stress along its sides as shown in the figure. Calculate (1) the stresses σ_x, σ_y, τ_{xy}, and (2) the principal stresses.



【交大機械】

【解析】

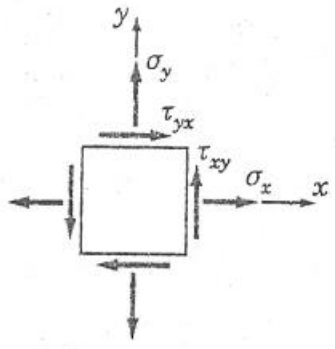
$$\sigma_y = 14\sin 60^o = 7\sqrt{3}\ \ MPa$$

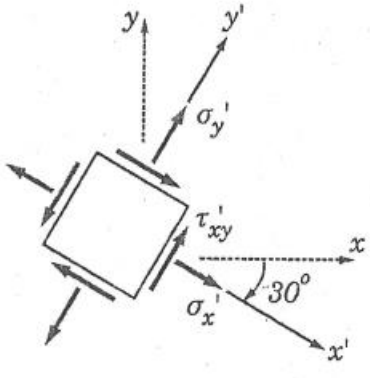
$$\tau_{yx} = \tau_{xy} = 7 + 14\cos 60^o = 14\ \ MPa$$

$$\sigma_{-30^\circ} = 30\cos 30^\circ = 25.98\,(\text{MPa})$$

$$\sigma_{-30^\circ} = \frac{\sigma_x + 12.124}{2} + \frac{\sigma_x - 12.124}{2}\cos(-60^\circ) + 14\sin(-60^\circ)$$

$$\Rightarrow \sigma_x = 46.764\,(\text{MPa})$$

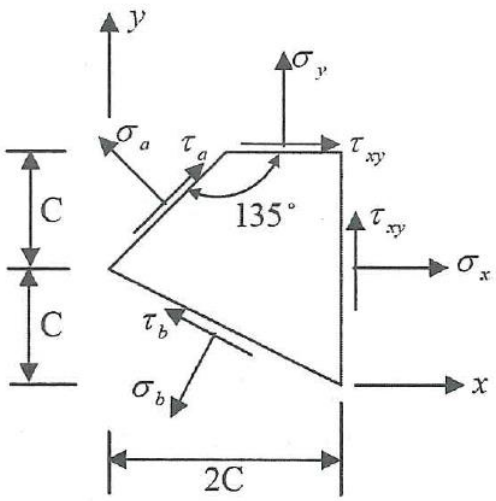




$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{(\frac{\sigma_x - \sigma_y}{2})^2 + \tau_{xy}^2} = 51.715\ ,\ 7.173\ (\text{MPa})$$

範例 15

The state of a uniform plane stress as shown can be described by $\tau_a = 400$ MPa , $\sigma_a = 200$ MPa , and $\tau_b = 100$ MPa. Determine the values of σ_x σ_y σ_b and τ_{xy}



【92 交大機械】【93 土木高考】

【解析】

解 (1) $\sigma_a = \sigma_{-45°} = \dfrac{\sigma_x + \sigma_y}{2} + \dfrac{\sigma_x - \sigma_y}{2}\cos[2\times(-45°)] + \tau_{xy}\sin[2\times(-45°)]$

$\Rightarrow 200 = \dfrac{\sigma_x + \sigma_y}{2} + (\tau_{xy})\times\sin(-90°)$

$\Rightarrow \sigma_x + \sigma_y - 2\tau_{xy} = 400$

同理

$\tau_{-45°} = \tau_a = -\dfrac{\sigma_x - \sigma_y}{2}\sin(-90°) + \tau_{xy}\cos(-90°) = -400$

$\sigma_x - \sigma_y = -800$

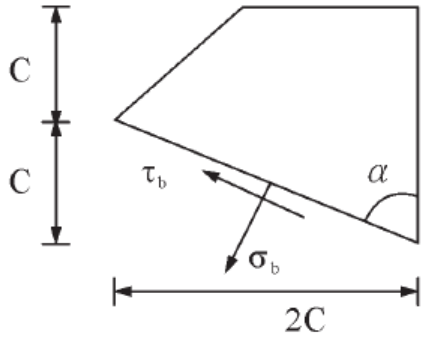
$\tau_b = -\dfrac{\sigma_x - \sigma_y}{2}\sin(126.86°) + \tau_{xy}\cos(126.86°) = -100$

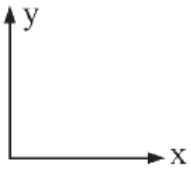
$2\sigma_x - 2\sigma_y + 3\tau_{xy} = 500$

聯立求解可得：

$\sigma_x = 500$ (MPa), $\sigma_y = 1300$ (MPa), $\tau_{xy} = 700$ (MPa)

(2)





$$\alpha = 90° - \tan^{-1}\frac{c}{2c} = 63.435° :$$

$$\sigma_b = \sigma_{63.435°} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos(2\times 63.435°) + \tau_{xy}\sin(2\times 63.435°)$$

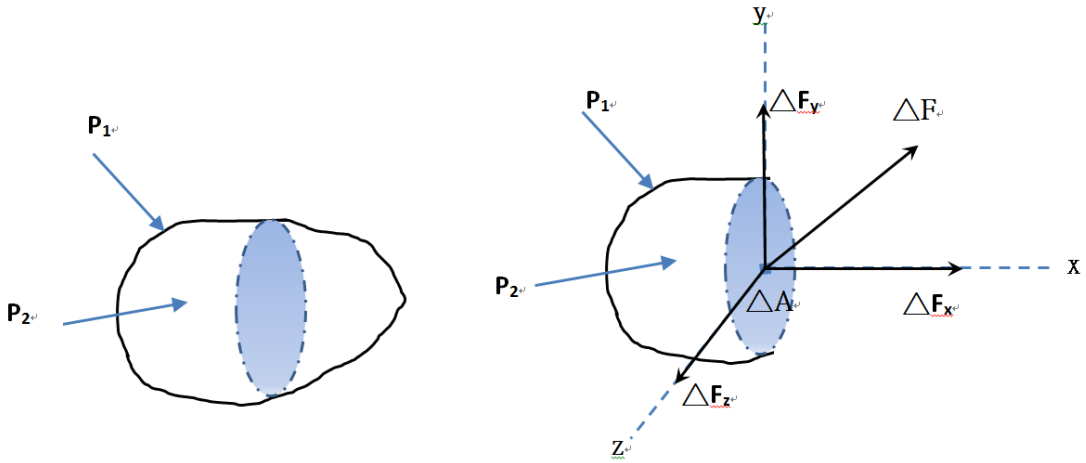
$$= 1700(\text{MPa})$$

2-3. 應力張量(stress tensor)

一、應力向量(stress vector)

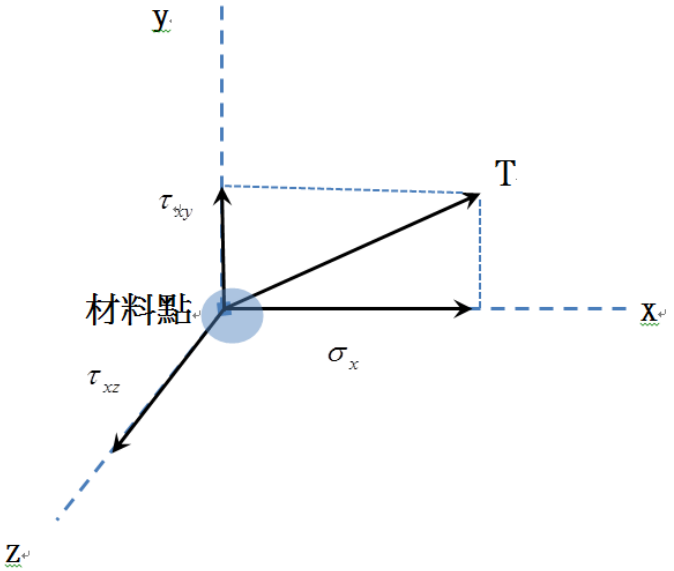
如圖所示當材料體承受外力後，以一假想平面切開，假設以 ΔF 表示此材料點上之內力，考慮於此斷面上點大小之元素面積為 ΔA，則定義應力向量(stress vector)T 為:

$$T = \lim_{\Delta A \to 0} \frac{\Delta F}{\Delta A} = \frac{dF}{dA}$$



應力向量 T 為材料中某切面上單位面積的受力

$$T = \frac{dF_x}{dA} i + \frac{dF_y}{dA} j + \frac{dF_z}{dA} k = \sigma_x i + \tau_{xy} j + \tau_{xz} k$$



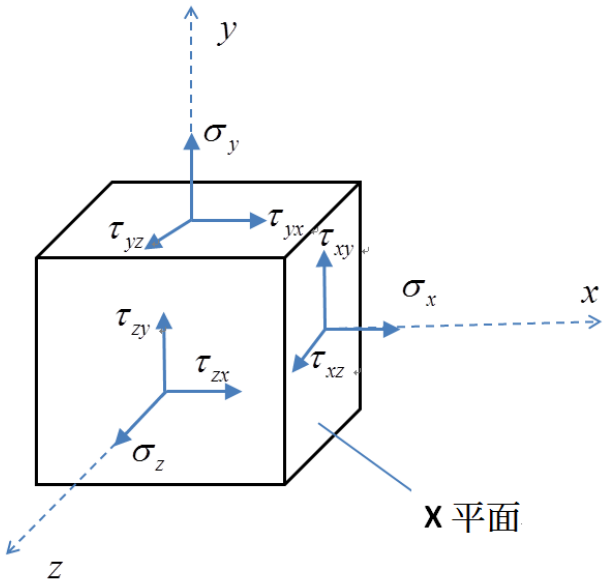
圖中 σ_x 為正向應力表示為材料點上單位面積所受的正向力，正向應力 σ_x 的下標表示其作用面的外法線方向；τ_{xy} 及 τ_{xz} 為剪應力表示為材料點上單位面積所受的剪力，τ_{xy} 及 τ_{xz} 剪應力有兩個下標，前一個下標表示其作用面的外法線方向，後一個下標表示其作用方向沿著哪一個座標軸。如果某一面的外法線沿座標軸的正方向，該面稱為正面，正面上的各應力分量便以指向座標軸正方向為正，反之為負；

如果某一面的外法線沿座標軸的負方向，則稱該面為負面，負面上的各應力便以指向座標軸的負方向為正，反之為負。

二、應力矩陣(stress matrix)

任何需要做應力分析的對象，例如機械零件、塑膠產品和鋼結構等，都是三維的實體，它們內部的應力和應變均是三維的，當它受力後，可由其材料內部之任意位置截取一個點，該點的三維應力狀態如圖所示，其中包括了正向應力σ 與剪應力τ，這些應力可以用下列矩陣來表示:

$$\sigma_{ij}=\begin{bmatrix}\sigma_x & \tau_{xy} & \tau_{xz}\\ \tau_{yx} & \sigma_y & \tau_{yz}\\ \tau_{zx} & \tau_{zy} & \sigma_z\end{bmatrix}$$



此矩陣稱為應力矩陣，主對角線上之值表示垂直於各該平面之應力，稱為法線應力，其餘各值表示作用於各平面上應力，稱為剪應力，適當的旋轉該座標軸，可使應力矩陣上除主對角線之值外其餘均為零；意即剪應力為零，此時之平面稱為主平面，對應之法線應力稱為主應力。

$$\text{主應力}\sigma_{123}=\begin{bmatrix}\sigma_1 & 0 & 0\\ 0 & \sigma_2 & 0\\ 0 & 0 & \sigma_3\end{bmatrix}$$

三、剪應力互等定理(theorem reciprocity)

在沒有徹體力偶矩(body couple monment)的作用時，應力矩陣中的剪應力，會滿足下列關係，此稱為剪應力互等定理(theorem reciprocity):

$\tau_{xy}=\tau_{yx}$ ； $\tau_{yz}=\tau_{zy}$ ； $\tau_{xz}=\tau_{zx}$

證明如下:

利用靜平衡方程式

$\sum F_Z = 0$

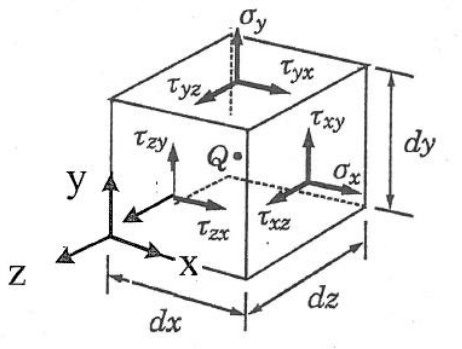
$(\tau_{xy} dydz)dx - (\tau_{yx} dxdz)dy + C_z dxdydz = 0$

其中 C_z 徹體力偶矩的 Z 分量

同除 $dxdydz \Rightarrow \tau_{xy} - \tau_{yx} + C_z = 0$

其中 $C_z = 0$ 可得 $\tau_{xy} = \tau_{yx}$

同理 $\tau_{yz} = \tau_{zy}$ ； $\tau_{xz} = \tau_{zx}$



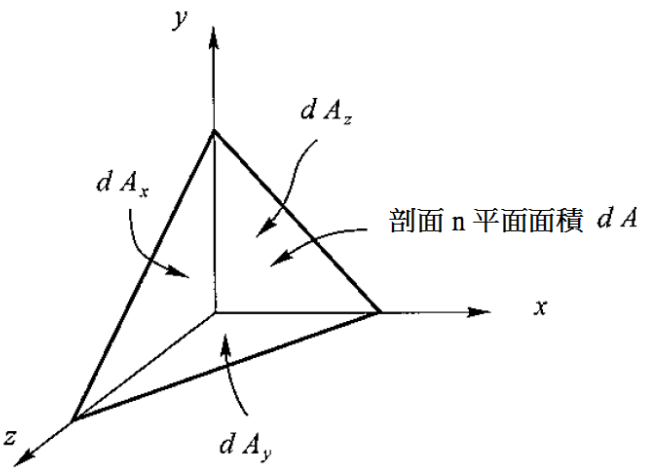
四、柯西公式(Cauchy’s formula)

當已知某材料點的應力矩陣 $[\sigma] = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}$，當通過此材料點任意平面上的應力向量即可推求，如圖所示，n 平面為通過材料點的一個幾何平面，其法向的單位向量為 $\vec{n} = n_x\vec{i} + n_y\vec{j} + n_z\vec{k}$ ，n 平面上的應力向量 $\vec{T} = T_x\vec{i} + T_y\vec{j} + T_z\vec{k}$ ，將剖面平行移至座標軸上與各座標面間所夾面積與剖面面積之關係:

yz平面之面積(x方向為法向量) $dA_x = n_x dA$

xz平面之面積(y方向為法向量) $dA_y = n_y dA$

xy平面之面積(z方向為法向量) $dA_z = n_z dA$



由靜力平衡式

$$\sum F_x = 0$$

$$T_x dA - \sigma_x (n_x dA) - \tau_{yx} (n_y dA) - \tau_{zx} (n_z dA) = 0$$
$$T_x = \sigma_x (n_x) + \tau_{yx} (n_y) + \tau_{zx} (n_z) = 0$$

同理

$$\sum F_y = 0$$

$$T_y dA - \sigma_y (n_y dA) - \tau_{xy} (n_x dA) - \tau_{zy} (n_z dA) = 0$$
$$T_y = \sigma_y (n_y) + \tau_{xy} (n_x) + \tau_{zy} (n_z) = 0$$

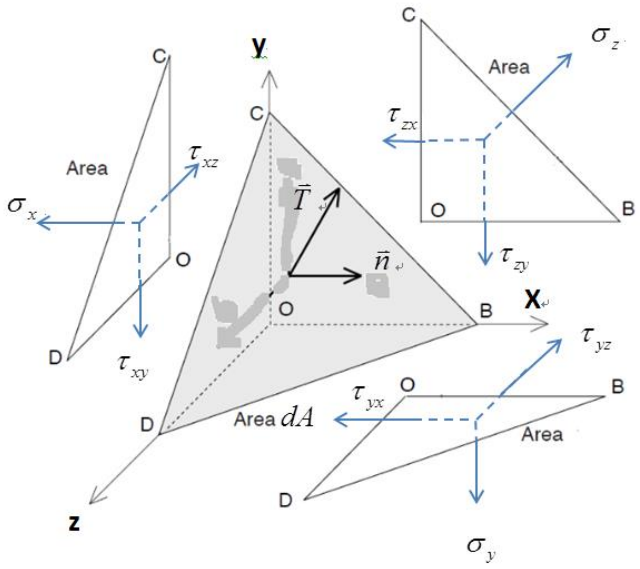
$$\sum F_z = 0$$

$$T_z = \sigma_z (n_z) + \tau_{xz} (n_x) + \tau_{yz} (n_y) = 0$$

寫成矩陣形式即為

$$\begin{bmatrix} T_x & T_y & T_z \end{bmatrix} = \begin{bmatrix} n_x & n_y & n_z \end{bmatrix} \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}$$

$[T] = [n][\sigma]$稱之為柯西公式



五、三維應力轉換

應力矩陣中的元素值(亦即正向應力及剪應力)，與所採用的座標系有關，採用不同之座標系統時，應力值會隨之變化，由柯西公式:

$\vec{T} = [n][\sigma]$

不同之座標系統

$\vec{T}' = [n'][\sigma']$

由座標系方向餘弦所定義轉換矩陣$[A]$，當兩個座標系均為正交座標系時，其間的轉換矩陣$[A]$稱為正交矩陣，正交矩陣有一個很重要的特性為:

$[A]^T = [A]^{-1}$

應力向量可表示為

$$\begin{cases}\vec{T} = \vec{T}'[A] \\ \vec{n} = \vec{n}'[A]\end{cases} \Rightarrow \vec{T}'[A] = \vec{n}'[A][\sigma] \Rightarrow \vec{T}' = \vec{n}'[A][\sigma][A]^T$$

可得$[\sigma]' = [A][\sigma][A]^T$

六、最大主應力及最大主應力平面

依據最大主應力的定義為在任一切面之應力分量，其面上沒有剪應力，僅有正向應力即為最大主應力，該切面為最大主應力面，亦即最大主應力面上的應力向量$\vec{T}$與法向量$\vec{n}$重合，如圖所示，若通過材料點的平面為主平面，其上之應力為$\vec{T}$

$\vec{T} = \sigma\,\vec{n}$

其中σ:主軸方向的主應力、$\vec{n}$:主軸方向的單位向量

由柯西公式可得$\vec{T} = [n][\sigma] = \sigma\vec{n}$

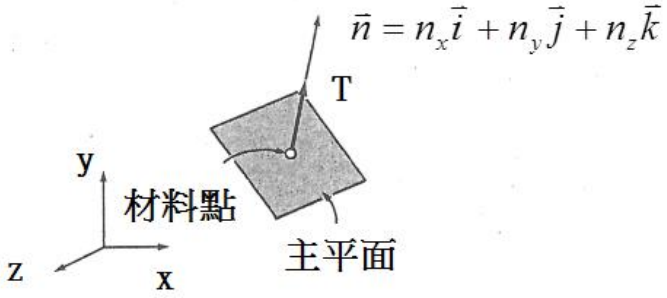
$[n][\sigma] - \sigma\vec{n} = 0 \Rightarrow ([\sigma] - \sigma[I])\vec{n} = 0$

其中上式為一聯立齊性方程式，其中最大主應力σ即為特徵值，相對應之法向量$\vec{n}$即為特徵向量

$([\sigma] - \sigma[I]) = 0$（特徵方程式）

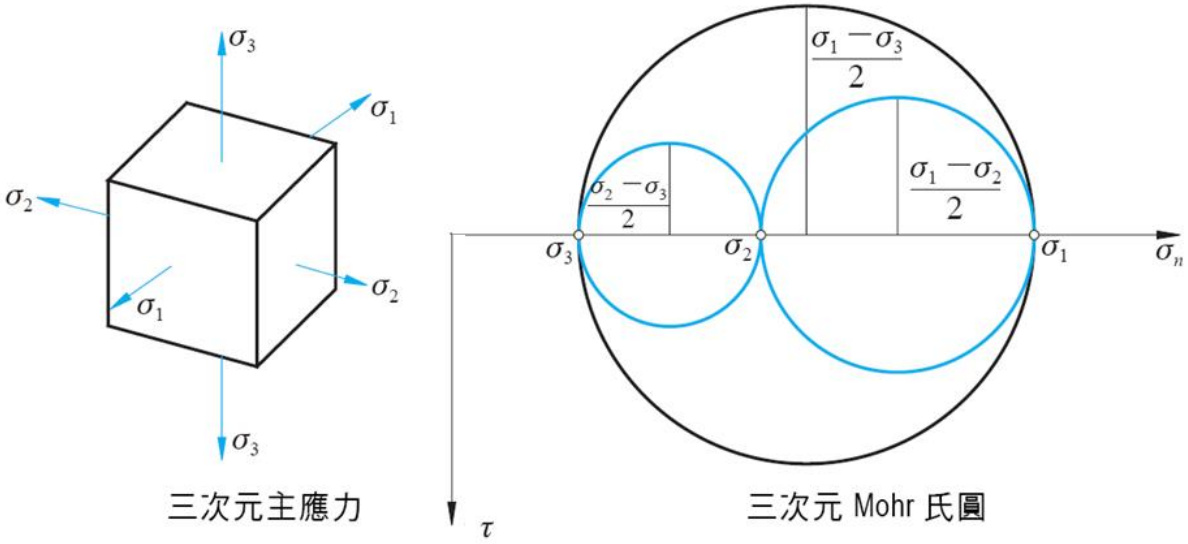
可寫為

$$\begin{vmatrix}\sigma_x - \sigma & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y - \sigma & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z - \sigma\end{vmatrix} = 0$$



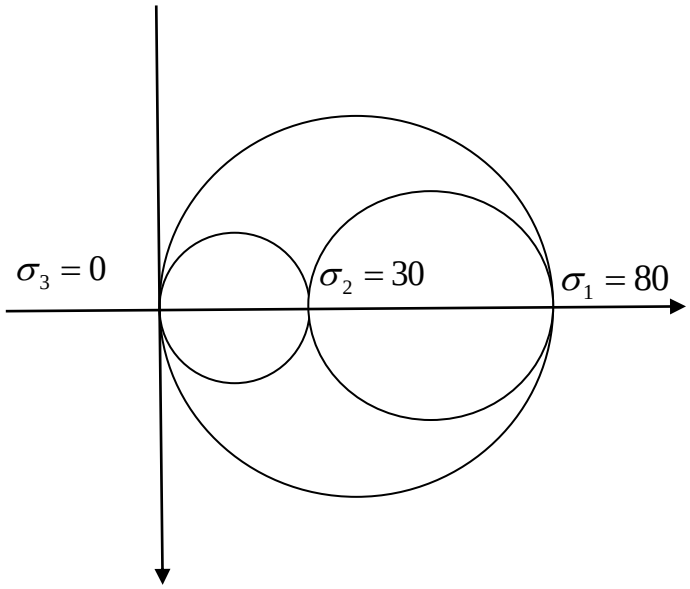
七、 3D 應力之莫爾圓

所謂三軸向應力就是 3D 應力元素在三個軸向只有正交應力存在，而無剪應力，如圖所示，若要求 xy 平面內的面內最大剪應力，就是將應力元素轉角度，方法如上面莫爾圓法所提，然而要是求絕對最大剪應力:



情況一： $\sigma_1>\sigma_2>\sigma_3$

平面應變經計算 $\sigma_1 = 80MPa$ 、 $\sigma_2 = 30\text{MP}a$ ， $\sigma_z = \sigma_3 = 0$ ，如下圖所示

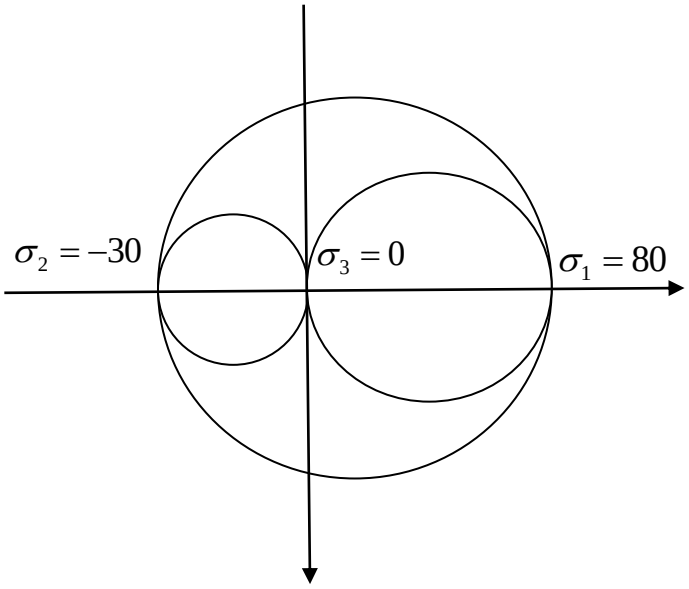


xy 平面最大剪應力 $\tau_{\max}^{in-plane} = \frac{\sigma_1-\sigma_2}{2} = 25(MPa)$

絕對最大剪應變 $\tau_{\max}^{abs} = \frac{\sigma_1-\sigma_3}{2} = 40$

情況二： $\sigma_1>\sigma_3>\sigma_2$

平面應變經計算$\sigma_1 = 80MPa$ 、 $\sigma_2 = -30MPa$ ， $\varepsilon_3 = 0$ ，如下圖所示

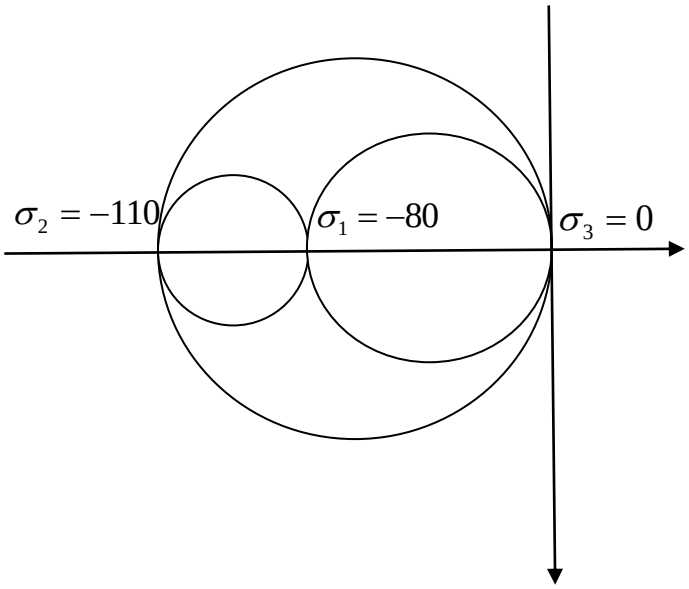


xy 平面最大剪應變 $\tau_{\max}^{in-plane} = \dfrac{\sigma_1 - \sigma_2}{2} = 55MPa$

絕對最大剪應變 $\tau_{\max}^{abs} = \dfrac{\sigma_1 - \sigma_2}{2} = 55MPa$

情況三： $\sigma_3 > \sigma_1 > \varepsilon_2$

平面應變經計算$\sigma_1 = -80MPa$ 、 $\sigma_2 = -110MPa$ ， $\sigma_3 = 0$ ，如下圖所示



xy 平面最大剪應變 $\tau_{\max}^{in-plane} = \dfrac{\sigma_1 - \sigma_2}{2} = 15MPa$

絕對最大剪應變 $\tau_{\max}^{abs} = \dfrac{\sigma_3 - \sigma_2}{2} = 55MPa$

範例 16

At a point in a solid, the stress tensor components referred to (x,y,z) axes are

$$\begin{bmatrix}\sigma_{xx} & \sigma_{xy} & \sigma_{xz}\\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz}\\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz}\end{bmatrix}=\begin{bmatrix}1 & 0 & -4\\ 0 & 3 & 0\\ -4 & 0 & 5\end{bmatrix}$$

Find the stress vector $\vec{T}(\vec{n})$ acting on a plane through P with unit (outer) normal vector $\vec{n}=\frac{1}{2}\vec{i}-\frac{1}{2}\vec{j}+\frac{1}{\sqrt{2}}\vec{k}$.

【中正機械】

【解析】

由柯西公式:

$\vec{T}=[n][\sigma]$

$$\begin{bmatrix}T_x & T_y & T_z\end{bmatrix}=\begin{bmatrix}n_x & n_y & n_z\end{bmatrix}\begin{bmatrix}\sigma_x & \tau_{xy} & \tau_{xz}\\ \tau_{yx} & \sigma_y & \tau_{yz}\\ \tau_{zx} & \tau_{zy} & \sigma_z\end{bmatrix}$$

$$\begin{bmatrix}T_x & T_y & T_z\end{bmatrix}=\begin{bmatrix}\frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}}\end{bmatrix}\begin{bmatrix}1 & 0 & -4\\ 0 & 3 & 0\\ -4 & 0 & 5\end{bmatrix}=\begin{bmatrix}-2.32 & -1.5 & 1.53\end{bmatrix}$$

範例 17

The state of stress at a particular point relative to the *xyz* coordinate system is given by the stress matrix.

$$[\sigma]=\begin{bmatrix}6 & 4 & 3\\ 4 & 4 & -2\\ 3 & -2 & 2\end{bmatrix} psi$$

Determine

(1) the stress vector acting on the outer side (the side away from the origin) of the following plane, which passed through the place in question

$$2x+y-2z=3$$

(2) the normal and tangential components of the stress vector on this plane.

【中正機械】【成大土木】

【解析】

對於平面 $2x+y-2z=3$ 言，其法向單位向量 $\boldsymbol{n}$ 可表成：

$$\boldsymbol{n}=\frac{2\boldsymbol{i}+\boldsymbol{j}-2\boldsymbol{k}}{\sqrt{2^2+1^2+(-2)^2}}=\frac{1}{3}(2\boldsymbol{i}+\boldsymbol{j}-2\boldsymbol{k})$$

依據 *Cauchy* 公式，平面上的應力向量 $\boldsymbol{T}$ 應爲

$$\boldsymbol{T}=[n][\sigma]$$

$$=\frac{1}{3}\begin{bmatrix}2 & 1 & -2\end{bmatrix}\begin{bmatrix}6 & 4 & 3\\ 4 & 4 & -2\\ 3 & -2 & 2\end{bmatrix}=\frac{1}{3}(10\boldsymbol{i}+16\boldsymbol{j})\ psi$$

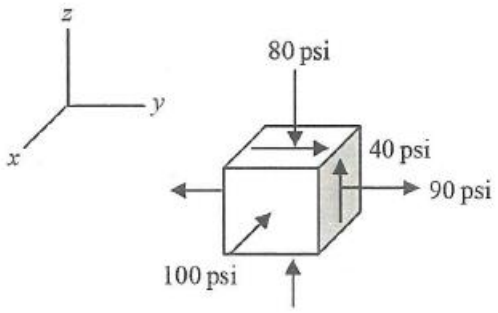
正向應力 σ 是應力向量 $\boldsymbol{T}$ 在 $\boldsymbol{n}$ 方向的投影量

$$\sigma=\boldsymbol{T}\cdot\boldsymbol{n}=\frac{1}{9}(20+16)=4\ psi$$

$$\tau=\sqrt{|\boldsymbol{T}|^2-\sigma^2}=4.85\ psi$$

範例 18

The stress at a point is shown on the element. Draw the three Mohr's circles that describe this state of stress, and find the principal stresses and the absolute maximum shear stress.



【中正機械】

【解析】

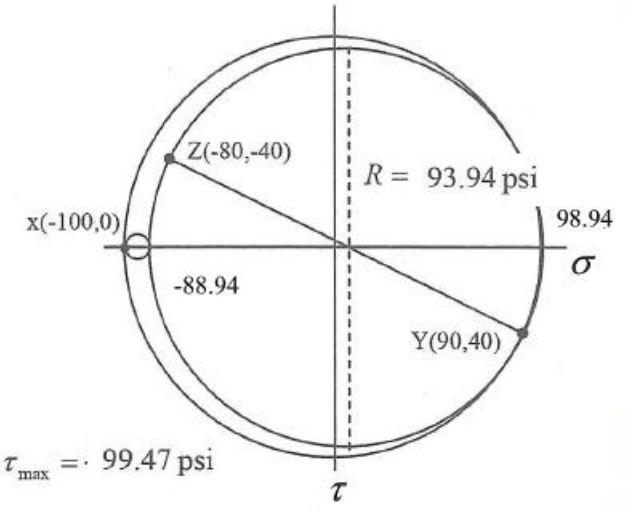
$$\sigma_{1,2}=\frac{\sigma_x+\sigma_y}{2}\pm\sqrt{(\frac{\sigma_x-\sigma_y}{2})^2+(\tau_{xy})^2}=98.94,-88.94(psi)$$

$$\sigma_3=-100(psi)$$

$$\sigma_{av}=\frac{\sigma_x+\sigma_y}{2}=5(psi)$$

$$R=\sqrt{(\frac{\sigma_x-\sigma_y}{2})^2+(\tau_{xy})^2}=93.94(psi)$$

$$\tau_{\max}=\frac{97.94-(-100)}{2}=99.47(psi)$$



範例 19

螺栓牙根處之應力狀態 $\sigma_x = 50\ \text{MPa}$，$\sigma_y = -20\ \text{MPa}$，$\tau_{yz} = 8\ \text{MPa}$，其餘為應力分量 $(\sigma_z, \tau_{xy}, \tau_{xz})$ 為 0，求作用於此處之：（每小題 10 分，共 20 分）

㈠三個主應力。

㈡最大剪應力。

【105 地特三等】

【解析】

(1) $\left|[\sigma] - \sigma[I]\right| = 0$

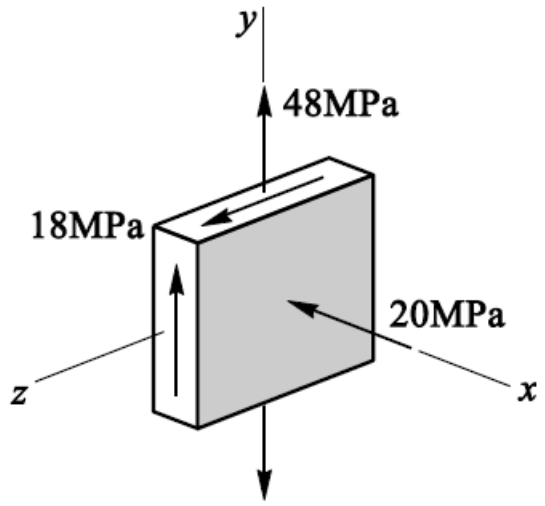
$$\left\| \begin{bmatrix} 50-\sigma & 0 & 0 \\ 0 & -20-\sigma & 8 \\ 0 & 8 & 0-\sigma \end{bmatrix} \right| = 0 \quad \Rightarrow (50-\sigma)(\sigma^2 + 20\sigma - 64) = 0$$

$$\sigma_1 = 50(Mpa) \qquad \sigma_2 = 2.8(Mpa) \qquad \sigma_3 = -22.8(Mpa)$$

(2) $\tau_{\max,abs(out-plane)} = \dfrac{\sigma_1 - \sigma_3}{2} = 36.4(Mpa)$

範例 20

試求圖中應力狀態之最大剪應力。



【台大機械】

【解析】

y - z 平面上之主應力

$$\sigma_{b,c}=\frac{\sigma_y+\sigma_z}{2}\pm\sqrt{\left(\frac{\sigma_y-\sigma_z}{2}\right)^2+\tau_{yz}^2}=\frac{48+0}{2}\pm\sqrt{\left(\frac{48-0}{2}\right)^2+18^2}=24\pm30$$

$\sigma_b=54\ \text{MPa}$，$\sigma_c=-6\ \text{MPa}$

因 x 面上無剪應力，故 $\sigma_x=-20\ \text{MPa}$ 為另一個主應力

將三個主應力按大小次序排列：$\sigma_1=54\ \text{MPa}$，$\sigma_2=-6\ \text{MPa}$，$\sigma_3=-20\ \text{MPa}$

故 $\tau_{\max}=\frac{1}{2}(\sigma_1-\sigma_3)=\frac{1}{2}[54-(-20)]=37\ \text{MPa}$

範例 21

Given a state of stresses at a point with respect to a convenient coordinate system (x, y, z) be $\sigma_x = 100\,\text{MPa}$, $\sigma_y = -60\,\text{MPa}$, $\sigma_z = 40\,\text{MPa}$, $\tau_{xy} = 80\,\text{MPa}$, $\tau_{yz} = \tau_{xz} = 0\,\text{MPa}$.

a.) Determine the principal normal stresses

b.) Determine the principal shear stresses.

【解析】

$$\begin{vmatrix} \sigma_x - \sigma & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y - \sigma & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z - \sigma \end{vmatrix} = 0$$

$$\begin{vmatrix} 100 - \sigma & 80 & 0 \\ 80 & -60 - \sigma & 0 \\ 0 & 0 & 40 - \sigma \end{vmatrix} = 0$$

$$\sigma^3 - I_1\sigma^2 + I_2\sigma - I_3 = 0$$

$$I_1 = 100 - 60 + 40 = 80\,\text{MPa}$$

$$I_2 = 100(-60) + 100(40) + (-60)40 - 80^2 = -10800\,(\text{MPa})^2$$

$$I_3 = 100(-60)40 + 2(80)0(0) - 100(0)^2 - (-60)^2(0)^2 - 40(80)^2 = -496000\,(\text{MPa})^3$$

$$\sigma^3 - 80\sigma^2 - 10800\sigma + 496000 = 0$$

$$\sigma_1 = 133.137\,\text{MPa}$$

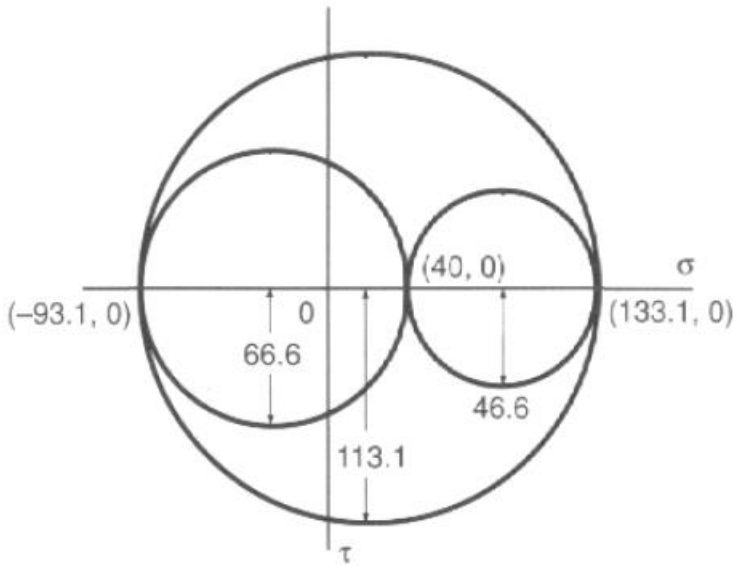
$$\sigma_2 = 40\,\text{MPa}$$

$$\sigma_3 = -93.137\,\text{MPa}$$

$$\tau_{\substack{\max \\ 1-3}} = \left|\frac{\sigma_1 - \sigma_3}{2}\right| = \left|\frac{133.137 - (-93.137)}{2}\right| = 113.14\ \text{MPa}$$

$$\tau_{\substack{\max \\ 2-3}} = \left|\frac{\sigma_2 - \sigma_3}{2}\right| = \left|\frac{40 - (-93.137)}{2}\right| = 66.56\ \text{MPa}$$

$$\tau_{\substack{\max \\ 1-2}} = \left|\frac{\sigma_1 - \sigma_2}{2}\right| = \left|\frac{133.137 - 40}{2}\right| = 46.56\ \text{MPa}$$



範例 22

對參考坐標 xyz 而言，物體內一點的應力張量如下：

$$\sigma_{ij}=\begin{bmatrix}-500 & 0 & -100\\ 0 & 0 & 0\\ -100 & 0 & 800\end{bmatrix}(\text{Pa})$$

若新坐標 $x'y'z'$ 由舊坐標繞 z 軸轉 30°而得。試求此點的新應力張量

【中山機械】

【解析】

$[\sigma]'=[A][\sigma][A]^T$

其中

$$[A]=\begin{bmatrix}\frac{\sqrt{3}}{2} & \frac{1}{2} & 0\\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0\\ 0 & 0 & 1\end{bmatrix}\qquad \sigma=\begin{bmatrix}-500 & 0 & -100\\ 0 & 0 & 0\\ -100 & 0 & 800\end{bmatrix}$$

$$[\sigma]'=[A][\sigma][A]^T=\begin{bmatrix}-375 & 125\sqrt{3} & -50\sqrt{3}\\ 125\sqrt{3} & -125 & 50\\ -50\sqrt{3} & 50 & 800\end{bmatrix}$$

2-4. 應力場分析

一、 應力場的分析

材料體受力後，我們可以利用應力張量描述出材料點的受力狀況，材料體內形成了一應力張量的分布，稱之為應力場(stress field)，隨著位置的不同，應力場中的應力張量也就不同，因此應力場基本上是一個位置函數。

$\sigma_x = \sigma_x(x,y,z)$ $\tau_{xy} = \tau_{xy}(x,y,z)$ 以此類推

考慮如圖所示之體積 $d\forall = dxdydz$ ，假設作用在元素個表面的應力均勻分布，且徹體力(body force)為: $\vec{B} = B_x\vec{i} + B_y\vec{j} + B_z\vec{k}$

由靜力平衡可得

$$\sum F_x = 0; \quad (\sigma_x + \frac{\partial \sigma_x}{\partial x}dx - \sigma_x)dydz + (\tau_{xy} + \frac{\partial \tau_{xy}}{\partial y}dy - \tau_{xy})dxdz$$

$$+ (\tau_{xz} + \frac{\partial \tau_{xz}}{\partial z}dz - \tau_{xz})dxdy + B_x(dxdydz) = 0$$

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} + B_x = 0$$

同理可求得

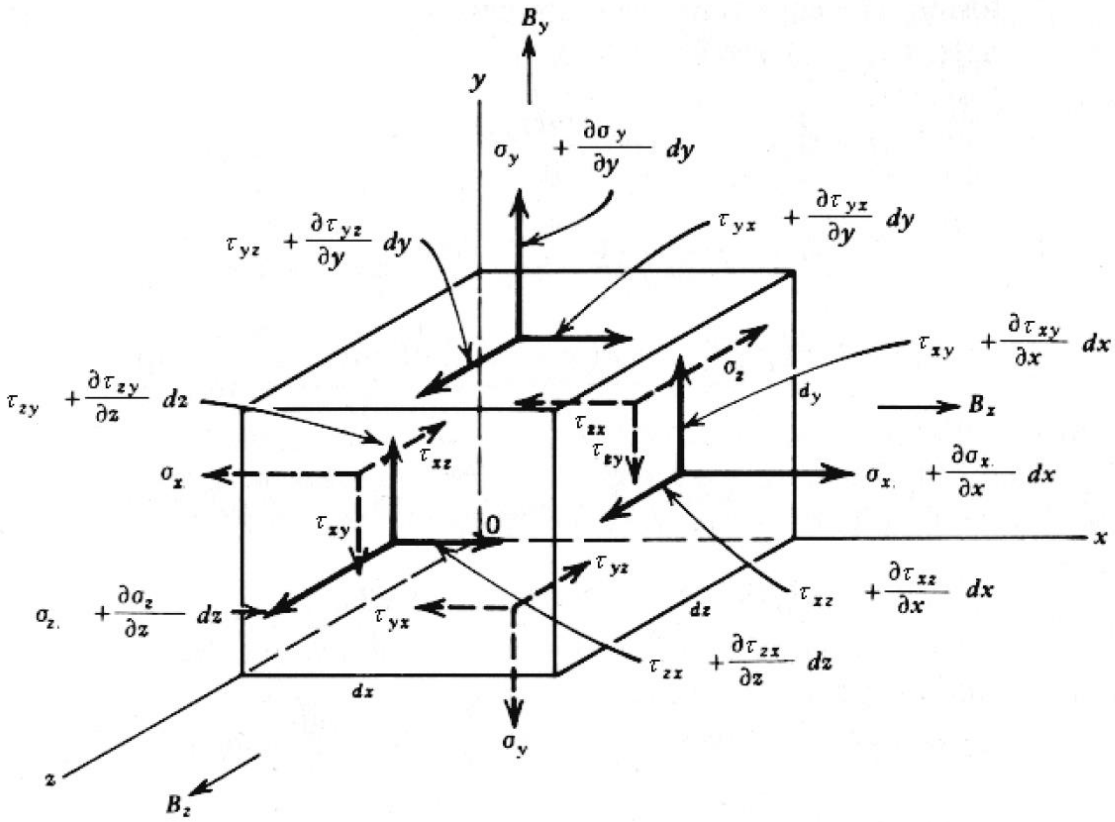
$$\sum F_y = 0; \quad \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} + B_y = 0$$

$$\sum F_z = 0; \quad \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} + B_z = 0$$

應力場的平面分析

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + B_x = 0$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + B_y = 0$$


B_y
y
$\sigma_y + \frac{\partial \sigma_y}{\partial y} dy$
$\tau_{yz} + \frac{\partial \tau_{yz}}{\partial y} dy$
$\tau_{yx} + \frac{\partial \tau_{yx}}{\partial y} dy$
$\tau_{xy} + \frac{\partial \tau_{xy}}{\partial x} dx$
$\tau_{zy} + \frac{\partial \tau_{zy}}{\partial z} dz$
σ_z
d_y
B_x
τ_{zx}
τ_{zy}
τ_{xz}
σ_x
$\sigma_x + \frac{\partial \sigma_x}{\partial x} dx$
τ_{xy}
0
x
τ_{yz}
$\sigma_z + \frac{\partial \sigma_z}{\partial z} dz$
τ_{yx}
$\tau_{xz} + \frac{\partial \tau_{xz}}{\partial x} dx$
dz
$\tau_{zx} + \frac{\partial \tau_{zx}}{\partial z} dz$
dx
σ_y
z
B_z

範例 23

The stress field within an elastic structural member is expressed as follows:

$$\sigma_x = -x^3 + y^2, \qquad \tau_{xy} = 5z + 2y^2, \qquad \tau_{xz} = xz^3 + x^2y$$
$$\sigma_y = 2x^3 + \tfrac{1}{2}y^2, \qquad \tau_{yz} = 0, \qquad \sigma_z = 4y^2 - z^3 \qquad \textbf{(d)}$$

Determine the body force distribution required for equilibrium.

【解析】

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} + B_x = 0$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} + B_y = 0$$

$$\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} + B_z = 0$$

$$(-3x^2) + (4y) + (3xz^2) + B_x = 0$$
$$(y) + (0) + (0) + B_y = 0$$
$$(-3z^2) + (z^3 + 2xy) + (0) + B_z = 0$$

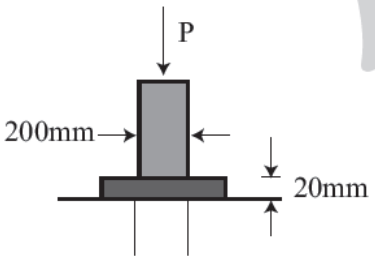
$$B_x = 3x^2 - 4y - 3xz^2,$$

$$B_y = -y, \qquad B_z = -2xy + 3z^2 - z^3$$

實戰演練

例 1

某板厚度20mm，以直徑200mm的衝頭衝孔，若板料剪應力強度 τ = 800MPa，請算出：負荷P值為何？若安全係數為1.5，衝頭之抗壓強度最少需多少。(96年機械關務四等)



【解析】

(1) 板材受到剪力 $V = P$

$$\text{剪應力 } \tau = \frac{P}{\pi \times (200 \times 10^{-3}) \times (20 \times 10^{-3})} = 800 \times 10^{6}$$

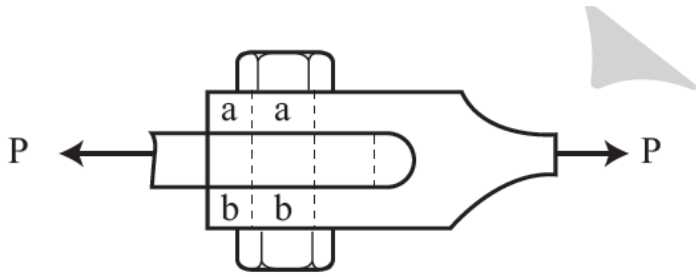
$$\Rightarrow P = 10053096.49(N)$$

(2) $$n = \frac{\sigma_L}{\sigma_W} \Rightarrow \sigma_L = 1.5 \times \frac{P}{\frac{\pi}{4} \times D^2}$$

$$= 1.5 \times \frac{10053096.49}{\frac{\pi}{4} \times (200 \times 10^{-3})^2} = 480000000\,\text{Pa} = 480(\text{MPa})$$

例 2

如圖所示，有一螺栓接頭受外力P作用，請問該接頭有那些可能之破壞方式。另若P＝5000公斤，螺栓之直徑為2.5 公分，螺栓所受之剪應力為若干？(92年機械地特四等)

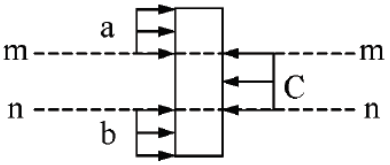


【解析】

答：(1) 取螺栓受力圖如圖所示

A. 接頭於a、b、c受到壓力而破壞

B. 接頭於m－m、n－n截面受到剪力而破壞



(2) 板材受到壓力而破壞

(3) P = 5000kg , d = 2.3cm

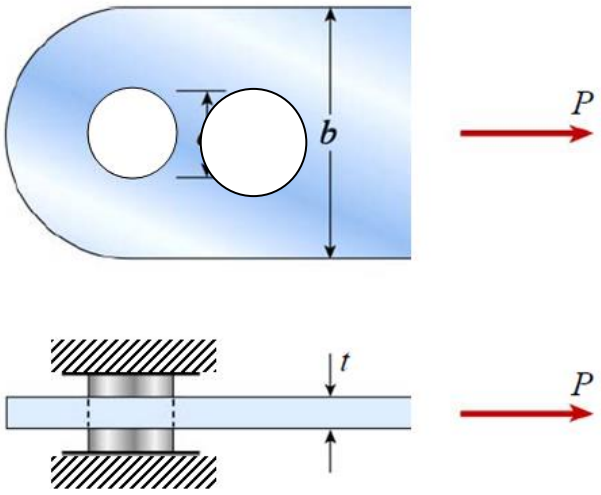
$$\tau = \frac{P}{2A} = \frac{P}{[\frac{\pi}{4}d^2]\times 2} = \frac{5000}{[\frac{\pi}{4}\times(2.5\times10^{-2})^2]\times 2} = 5092958.2(kg/m^2)$$

例 3

A bar of rectangular cross section is subjected to an axial load P (see figures below). The bar has width $b = 60$ mm and thickness $t = 10$ mm. A hole of diameter d is drilled through the bar to provide for a pin support. The allowable tensile stress in the bar is 140 MPa, and the allowable shear stress in the pin is 80 MPa. The allowable bearing stress between the pin and the bar is 200 MPa .

(a) Given the pin diameter $d = 10$ mm, determine the allowable load P_{allow}.

(b) Determine the pin diameter d_m for which the load P will be a maximum.



【101 機械技師】【99 成大航太】

【中文翻譯】

如圖一所示平板桿件寬度$b = 60$ mm，厚度$t = 10$ mm，承受P之拉伸負載。桿件一端是由直徑同為d之銷與孔配合固定，且沒有裝配所產生之應力。若桿件容許之拉伸應力$\sigma_T = 140$ MPa，銷之容許剪應力$\tau_S = 80$ MPa，銷與孔間容許之承壓應力（bearing stress）$\sigma_B = 200$ MPa， (a) 若d=10mm試求允許的作用力P(b) 試求容許之最大負載P與對應之銷的直徑d。

【解析】

(a)

考慮鋼板的拉應力

$$\sigma = \frac{P}{A} \Rightarrow P = \sigma(b - d)t = 140(60 - 10) \times 10 = 70000(N)$$

考慮插銷之剪應力

$$\tau = \frac{P}{2A} \Rightarrow P = \tau \times 2 \times \frac{\pi}{4} d^2 = 80 \times \frac{\pi}{2} \times 10^2 = 12566.37(N)$$

考慮插銷之壓應力

$$\sigma = \frac{P}{A} \Rightarrow P = \sigma(d)t = 200(10)\times 10 = 20000(N)$$

比較作用力取小值

$$P = 12566.37(N)$$

(b)

考慮鋼板的拉應力

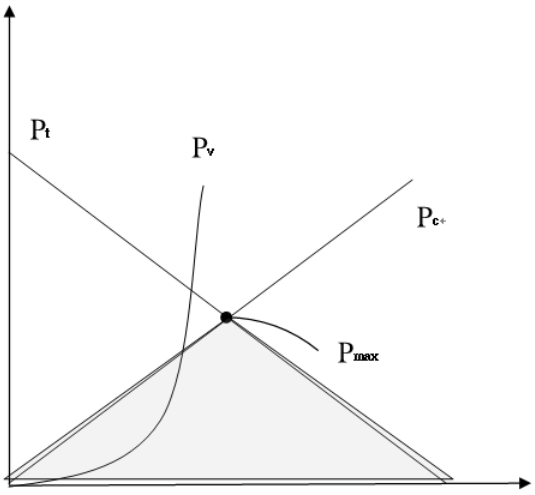
$$\sigma = \frac{P_t}{A} \Rightarrow P_t = \sigma(b-d)t = 140(60-d)\times 10$$

考慮插銷之剪應力

$$\tau = \frac{P_v}{2A} \Rightarrow P_v = \tau \times 2 \times \frac{\pi}{4} d^2 = 80 \times \frac{\pi}{2} \times d^2$$

考慮插銷之壓應力

$$\sigma = \frac{P_c}{A} \Rightarrow P_c = \sigma(d)t = 200(d)\times 10$$



比較P_C 、P_V、 P_t皆為d的函數可繪圖得知陰影部分為安全區域得知

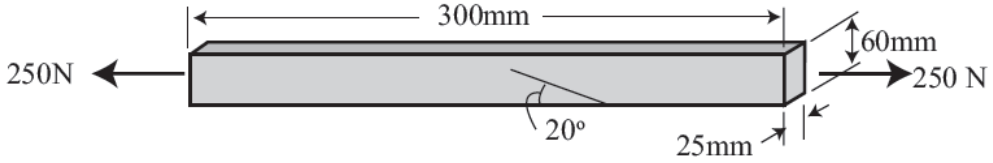
當P_C=P_t有最大值

$$P_t = 140(60-d)\times 10 = P_{.c} = 200(d)\times 10 \Rightarrow d = 24.7mm$$

$$P_{.max} = 200(24.7)\times 10 \Rightarrow d = 49400(N)$$

例 4

,木塊的木紋與水平 20° 角如圖所示。若木塊承受一軸向負載250N，求作用垂直於木紋的正向和剪應力。



【解析】

$$\sigma_x = \frac{P}{A} = \frac{250}{(0.06)(0.025)} = 166.67\,\text{kPa} \qquad \sigma_y = 0 \qquad \tau_{xy} = 0 \qquad \theta = 70^\circ$$

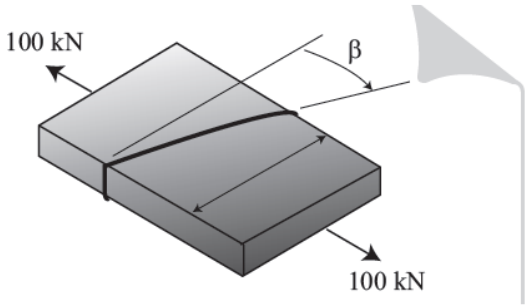
$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$= \frac{166.67 + 0}{2} + \frac{166.67 - 0}{2}\cos 140^\circ + 0 = 19.5\,\text{kPa}$$

$$\tau_{x'y'} = -\left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta + \tau_{xy}\cos 2\theta = -\left(\frac{166.67 - 0}{2}\right)\sin 140^\circ + 0 = 53.6\,\text{kPa}$$

例 5

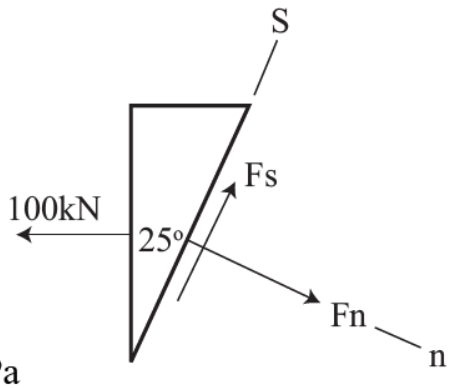
某板之寬度及厚度之截面為80mm×10mm，受到負荷P=100kN，且該角度β=25°，試求沿著該角度β斜面上所受之正向應力及剪應力多少。



【解析】

Area of wed

$$A_w = \frac{\left(10\times10^{-3}\right)\left(80\times10^{-3}\right)}{\cos 25^\circ} = 882.7\times10^{-6}\,m^2$$



(1) $\sum F_s = 0$

$F_s = 100\sin 25^\circ = 0 \qquad F_s = 42.26kN$

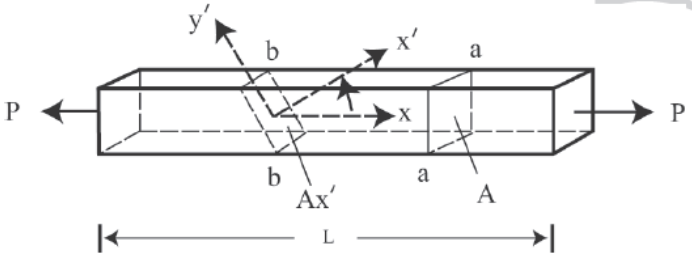
$$\tau_w = \frac{F_s}{A_w} = \frac{42.26\times10^3}{882.7\times10^{-6}} = 47.9\times10^6\,Pa = 47.9\,MPa$$

(2) $\sum F_w = 0 \qquad F_n - 100\cos 25^\circ = 0 \qquad F_n = 90.63kN$

$$\sigma_w = \frac{F_a}{A_w} = \frac{90.63\times10^3}{882.7\times10^{-6}} = 102.7\times10^6\,Pa = 102.7\,MPa$$

例 6

如右圖所示，為一均勻方桿受到軸向拉力P的作用。已知方桿的截面積為A、長度為L、楊氏係數為E，試回答下列問題：



(1) 若以δ代表方桿受P力作用產生的長度伸長量，試推導δ和P的關係式；亦即，$\delta = \frac{PL}{AE}$ 。

(2) 試推導下列兩式 $\sigma' = \frac{P}{A}\cos^2\theta$ 及 $\tau' = -\frac{P}{2A}\sin^2\theta$

其中σ'及τ'分別代表作用在b－b截面上的正向應力及剪應力，θ代表b－b截面的法線方向x'和軸向x的夾角。(99普考)

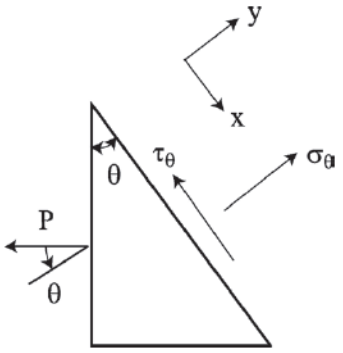
【解析】

(1) $\sum Fg = 0$

$\sigma_\theta A_W = P \times \cos\theta$ ，

其中 $A_W \cos = A$

$\sigma_\theta = \frac{P \times \cos^2\theta}{A} = \sigma'$

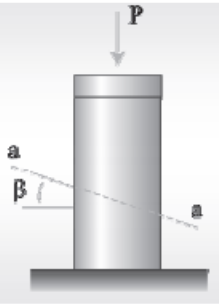


(2) $\sum F_X = 0$

$\tau_\theta A_W = -P \times \sin\theta \Rightarrow \tau_\theta = \frac{-P \times \cos\theta \times \sin\theta}{A} = \frac{-P\sin 2\theta}{2A} = \tau'$

例 7

右圖顯示一力量 P 加諸於一柱中心，已知圖中 a-a 斷面所受到之壓應力為 100MPa、剪應力為 35MPa，請問 a-a 與水平面的夾角β為何？而該柱所受到之最大壓應力又為何？ (97 年鐵路特考員級)



【解析】

解 $\tau_a = \frac{\sigma_x}{2}\sin 2\beta = 35 \cdots\cdots$

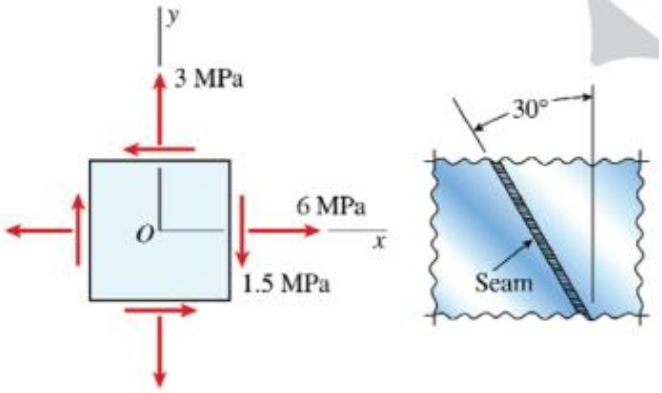
$\sigma_a = \sigma_x(\cos\beta)^2 = 100 \cdots\cdots$

由　　可得 $\sigma_x = 112.2\,(\text{MPa})$

$\beta = 19.3°$

例 8

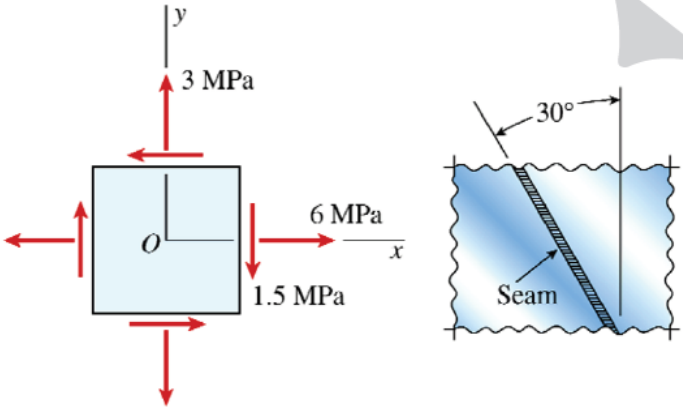
The polyethylene liner of a settling pond is subjected to stresses as shown by the plane-stress element in the first part of the figure. Determine the normal and shear stresses acting on a seam oriented at an angle of 30° to the element, as shown in the second part of the figure. Show these stresses on a sketch of an element having its sides parallel and perpendicular to the seam.



【104 地特四等】

【中文翻譯】

有一沈澱池所用之PE膜承受如圖左側所示之應力狀態，若假設其為在平面應力（plane stress）下之狀態，試求在與y軸夾30° 之焊縫邊上之應力狀態，並繪出對應之應力元素圖。

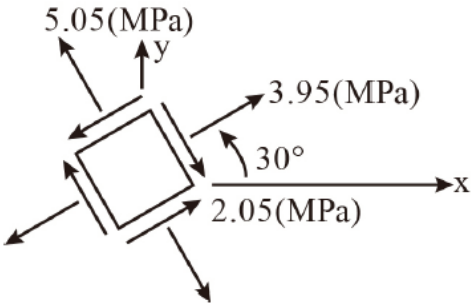


【解析】

$$\sigma_x = \frac{6+3}{2} + \frac{6-3}{2}\cos 2\times 30^\circ + (-1.5)\times \sin(2\times 30^\circ) = 3.95(\text{MPa})$$

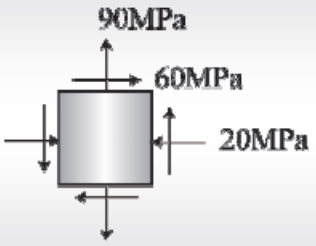
$$\tau_{30^\circ} = \frac{-(6-3)}{2}\sin 60^\circ + (-1.5)\times \cos 60^\circ = -2.05(\text{MPa})$$

$$\sigma_y = 6+3-3.95 = 5.05(\text{MPa})$$



例 9

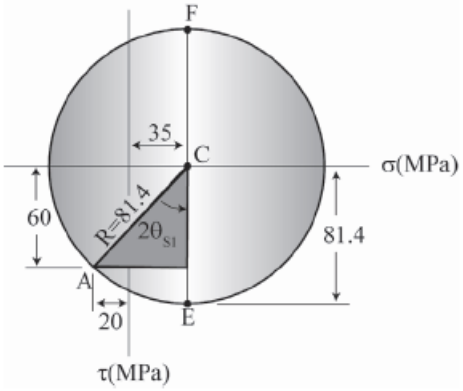
材料上某點之平面應力狀態為 $\sigma_x=-20\text{MPa}$、$\sigma_y=90\text{MPa}$ 及 $\tau_{xy}=60\text{MPa}$，如圖所示。試利用莫耳圓求該點之主應力大小及方向，並以應力元素圖表示其最大剪應力狀態。



【解析】

解 1. $\sigma_x=-20\,\text{MPa}$　　$\sigma_y=90\,\text{MPa}$　　$\tau_{xy}=60\,\text{MPa}$

σ與τ軸建立在圖中。

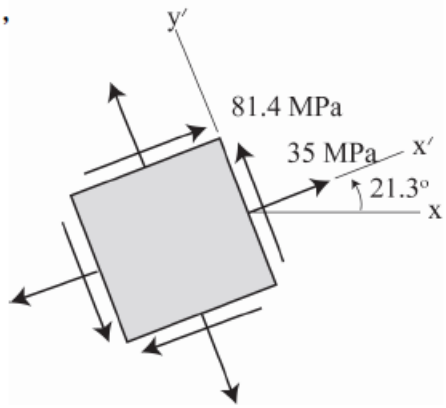


圓心 C 位於σ軸上，在點 $\sigma_{avg}=\dfrac{-20+90}{2}=35\,\text{MPa}$

繪出 C 點及參考點 A(−20,60)。

利用畢氏定理於陰影三角形求出圓之半徑 CA，

得 $R=\sqrt{(60)^2+(55)^2}=81.4\,\text{MPa}$



2. $\tau_{max}=81.4\,\text{MPa}$　　$\sigma_{avg}=35\,\text{MPa}$

由圓上 $2\theta_{s1}$ 可得逆時鐘角度 θ_{s1}，

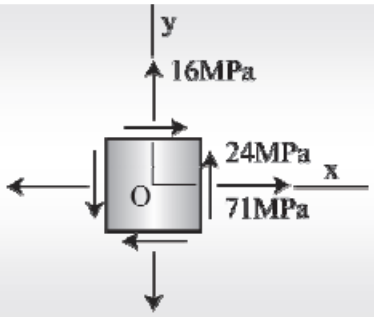
得 $2\theta_{s1}=\tan^{-1}\left(\dfrac{20+35}{60}\right)=42.5°$　　$\theta_{s1}=21.3°$

則平均正向應力及最大

平面剪應力均皆作用在正 x'與 y'方向上，如圖示。

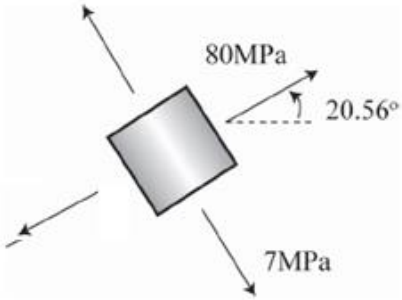
例 10

材料上某點之平面應力狀態為 σ_x＝71MPa、σ_y＝16 MPa 及 σ_{xy}＝24 MPa，如圖所示。試求該點之主應力大小及方向，並以應力元素圖表示其應力狀態。最大剪應力之大小及方向，並以應力元素圖表示其應力狀態。(93 年機械關務三等)



【解析】

解 1. 求主應力大小及方向：



$$\sigma_{1,2} = \frac{\sigma_x+\sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+(\sigma_{xy})^2}$$

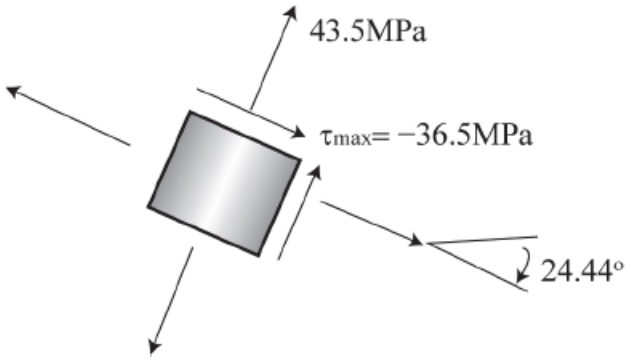
$$= \frac{71+16}{2} \pm \sqrt{\left(\frac{71-16}{2}\right)^2+(24)^2}$$

$$\sigma_1 = 80(\text{MPa}),\ \sigma_2 = 7(\text{MPa})$$

$$\tan 2\theta_P = \frac{\sigma_{xy}}{\left(\frac{\sigma_x-\sigma_y}{2}\right)} = \frac{24}{\left(\frac{71-16}{2}\right)}$$

$$\theta_P = 20.56°$$

2. 求最大剪應力之大小及方向：



$$\tau_{max} = \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+(\sigma_{xy})^2}$$

$$= \sqrt{\left(\frac{71-16}{2}\right)^2+(24)^2}$$

$$= 36.5(\text{MPa})$$

$$\tan 2\theta = -\frac{(\sigma_x-\sigma_y)}{2\sigma_{xy}}$$

$$= -\frac{(71-16)}{2\times 24}$$

$$\theta = -24.44°$$

$$\sigma_{aver} = \frac{\sigma_x+\sigma_y}{2} = 43.5(\text{MPa})$$

例 11

有個應力狀態為 $\sigma_x = -15$ MPa，$\sigma_y = 9$ MPa 及 $\tau_{xy} = 5$ MPa。

㈠求解主應力（principal stresses）及其方向。（10 分）

㈡繪製主應力元素（principal stress element）。（10 分）

【106 機械普考】

【解析】

答：圓心C：$\sigma_{x1} = \sigma_{aver} = \dfrac{\sigma_x + \sigma_y}{2} = \dfrac{-15+9}{2} = -3(\text{MPa})$

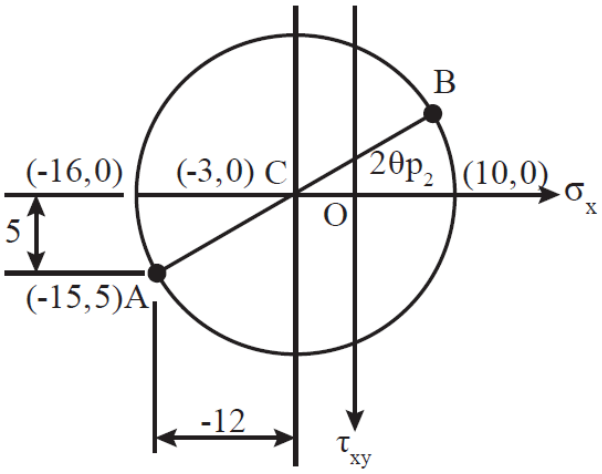
半徑R：$R = \sqrt{(\dfrac{\sigma_x + \sigma_y}{2})^2 + (\tau_{xy})^2} = \sqrt{(\dfrac{-15-9}{2})^2 + (5)^2} = 13(\text{MPa})$

在莫耳圓上其主應力是由 σ_1 與 σ_2 表示

1.較大的主應力 σ_1：$\sigma_1 = C + R = -3 + 13 = 10(\text{MPa})$

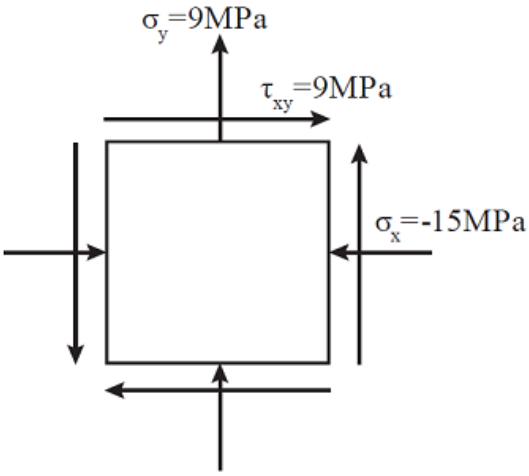
2.較小的主應力 σ_2：$\sigma_2 = C - R = -3 - 13 = -16(\text{MPa})$

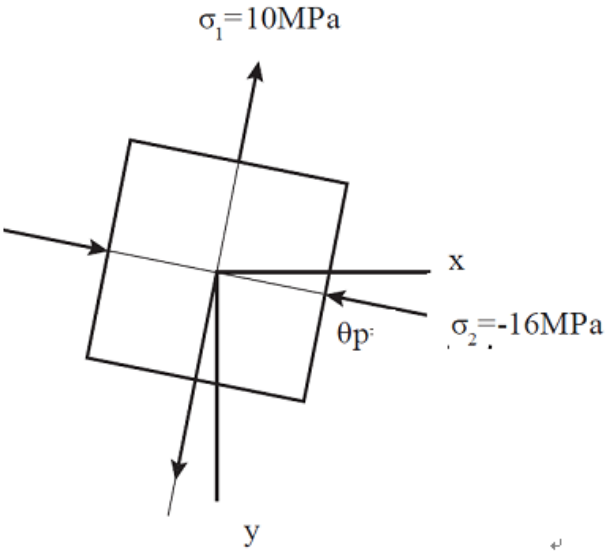
3. $\tan 2\theta_P = \dfrac{-5}{12}$ $\therefore 2\theta_P = -22.62° \Rightarrow \theta_P = -11.31°$



A點座標：σ ＝－15MPa及τ ＝5MPa；

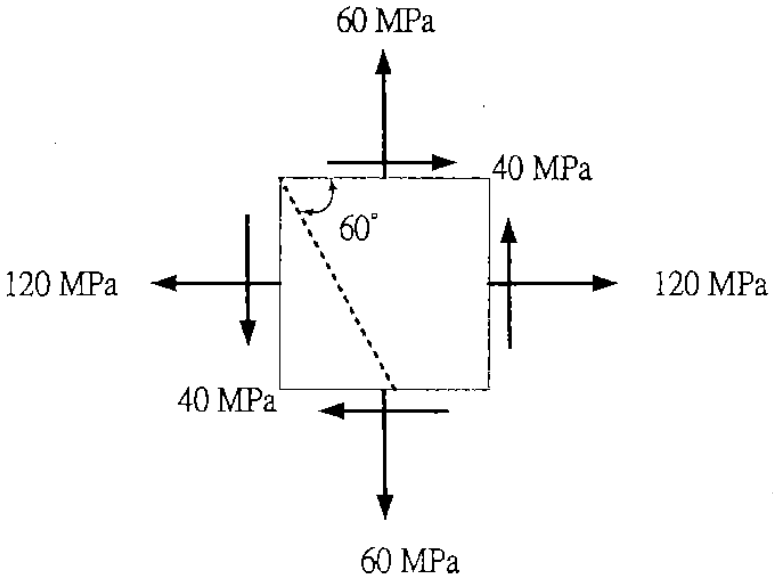
B點座標：σ ＝9MPa及τ ＝－5MPa





例 12

、如圖四所示的應力狀態，請計算(a)最大最小主應力(principal stresses) (5 分)；(b)最大剪應力(maximum shear stress)及其正向應力 (5 分)；(c)圖中旋轉 70 度後之平面（虛線所示）上之正向應力與剪應力 (10 分)。　　(共 20 分)



【106 中央土木】

【解析】

(1) 主應力

$$\sigma_{1,2} = \frac{\sigma_x+\sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= 140,\ 40\ (MPa)$$

(2) 最大剪應力

$$\tau_{max} = \frac{\sigma_1-\sigma_2}{2} = 50\ (MPa)$$

$$\sigma_{av} = \frac{\sigma_x+\sigma_y}{2} = 90\ (MPa)$$

(3) 斜面上之 σ' 及 τ'

↳ $\theta = 30°$

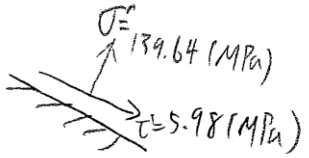
由平面應力轉換公式

$$\sigma' = \frac{\sigma_x+\sigma_y}{2} + \frac{\sigma_x-\sigma_y}{2}\cos 60° + \tau_{xy}\sin 60°$$

$$= 139.64\ (MPa)$$

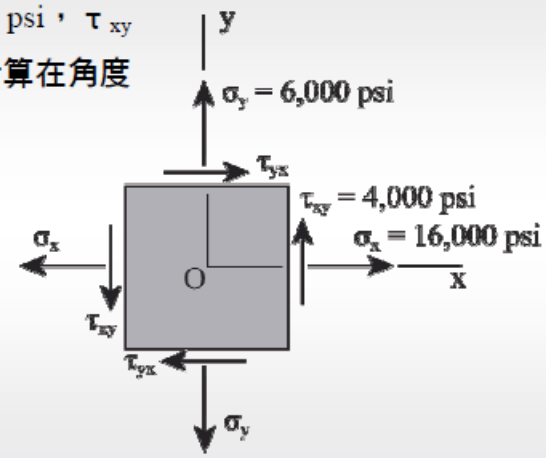
$$\tau' = -\frac{\sigma_x-\sigma_y}{2}\sin 60° + \tau_{xy}\cos 60°$$

$$= -5.98\ (MPa)$$



例 13

一個元素之面內應力分別為 $\sigma_x = 16{,}000$ psi，$\tau_{xy} = \tau_{yx} = 4{,}000$ psi，$\sigma_y = 6{,}000$ psi，請計算在角度 $\theta = 45^\circ$ 面上的正向應力和剪切應力。



【解析】

解 $\sigma_{45^\circ} = \dfrac{\sigma_x + \sigma_y}{2} + \dfrac{\sigma_x - \sigma_y}{2}\cos(2\times 45^\circ) + \tau_{xy}\sin(2\times 45^\circ)$

$= \dfrac{16000+6000}{2} + \left(\dfrac{16000-6000}{2}\right)\cos 90^\circ + 4000\sin 90^\circ$

$= 11000 + 4000$

$= 15000(\text{PSi})$

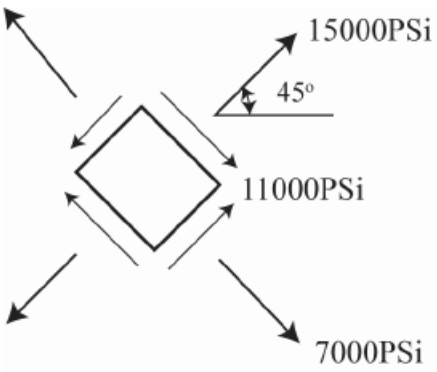
$\tau_{45^\circ} = -\dfrac{\sigma_x - \sigma_y}{2}\sin(2\times 45^\circ) + \tau_{xy}\cos(2\times 45^\circ)$

$= -11000(\text{psi})$

$\sigma_{135^\circ} = \dfrac{\sigma_x + \sigma_y}{2} + \dfrac{\sigma_x - \sigma_y}{2} + \cos(2\times 135^\circ) + \tau_{xy}\times\sin(2\times 135^\circ)$

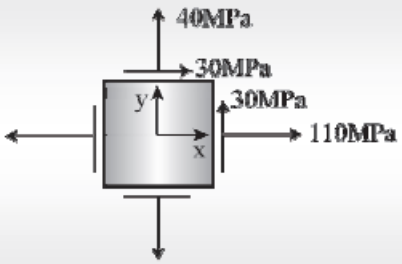
$= 11000 - 4000$

$= 7000$



例 14

有一材料試體之拉力強度為 160 MPa，剪力強度為 60 MPa，此試體之受力狀態如右圖所示，若 σ_y 與 τ_{xy} 保持不變，而 σ_x 逐漸增加，試問 σ_x 達何值時產生破壞？(96 年土木普考)



【解析】

解 (1) 當材料之拉力強度為 160MPa 時：

$$\sigma_1 = 160 = \frac{\sigma_x + 40}{2} + \sqrt{(\frac{\sigma_x - 40}{2})^2 + 30^2} \quad \Rightarrow \sigma_x = 152.5(\text{MPa})$$

(2) 當材料之剪力強度為 60MPa 時：

$$\tau_{max} = \sqrt{(\frac{\sigma_x - 40}{2})^2 + 30^2} \quad \Rightarrow 60 = \sqrt{(\frac{\sigma_x - 40}{2})^2 + 30^2} \quad \sigma_x = 143.923(\text{MPa})$$

所以 σ_x 達到 143.923(MPa)時材料產生破壞

例 15

(10%) Consider a state of plane stress described by σ_{11}, σ_{22} and σ_{12}. It is in a state of pure shear if $\sigma_{11}+\sigma_{22}=0$. Suppose $\sigma_{11}=100\text{MPa}$, $\sigma_{22}=-100\text{MPa}$, $\sigma_{12}=-100\text{MPa}$. Find a plane oriented at an angle θ such that the normal stress on that plane is zero. In addition, what is the shear stress on that plane?

【105 台大應力】

【解析】

$\sigma_{11}=100\,MPa=\sigma_x$

$\sigma_{22}=-100\,MPa=\sigma_y$

$\sigma_{12}=-100\,MPa=\tau_{xy}$

$$\sigma_{x1}=\frac{100+(-100)}{2}+\frac{100-(-100)}{2}\cos(2\times\theta)+\tau_{xy}\sin(2\theta)=0(MPa)\Rightarrow\theta_{s1}=22.5^\circ$$

$$\theta_{s1}=22.5^\circ\Rightarrow\tau=-\frac{100-(-100)}{2}\sin(2\times22.5^\circ)+(-100)\cos(2\times22.5^\circ)=-141.4(MPa)$$

$$\theta_{s2}=-67.5^\circ\Rightarrow\tau=-\frac{100-(-100)}{2}\sin(2\times-67.5^\circ)+(-100)\cos(2\times-67.5^\circ)=141.4(MPa)$$

例 16

The stress state at a point is shown in Fig. 5.

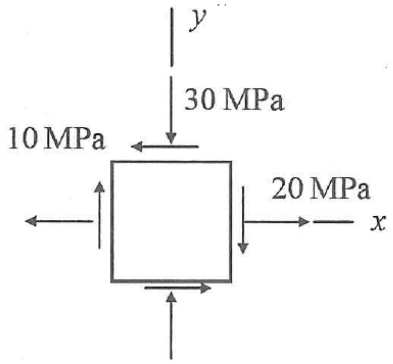


Fig. 5

(a) Please determine the principal stresses and show them on a sketch of a properly oriented stress element (10%).

(b) Please determine the value of the maximum in-plane shear stress and the corresponding normal stress. Please show the results on a sketch of a properly oriented stress element (10%).

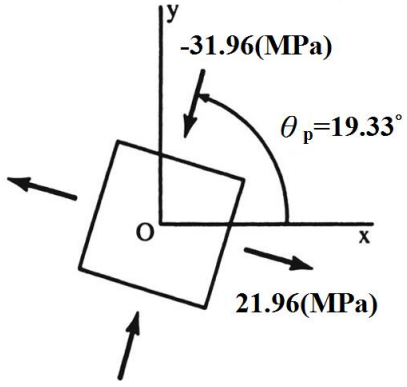
【102 交大機械】

【解析】

(a)主應力及最大剪應力

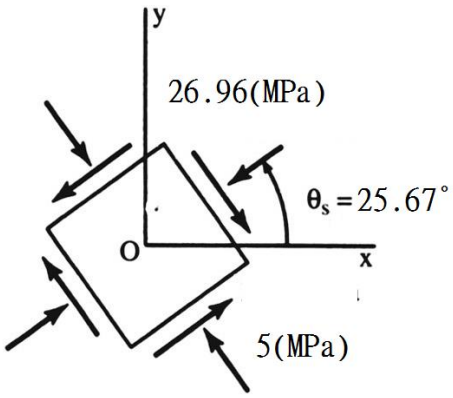
$$\sigma_{1,2}=\frac{\sigma_x+\sigma_y}{2}\pm\sqrt{(\frac{\sigma_x-\sigma_y}{2})^2+(\tau_{xy})^2}=\frac{20+(-30)}{2}\pm\sqrt{(\frac{20-(-30)}{2})^2+(-10)^2}=21.96,-31.96(MPa)$$

$$\tan(2\theta_p)=\frac{2\tau_{xy}}{\sigma_x-\sigma_y}\Rightarrow\ \theta_p=\frac{1}{2}\tan^{-1}(\frac{2\tau_{xy}}{\sigma_x-\sigma_y})=-19.33^\circ$$



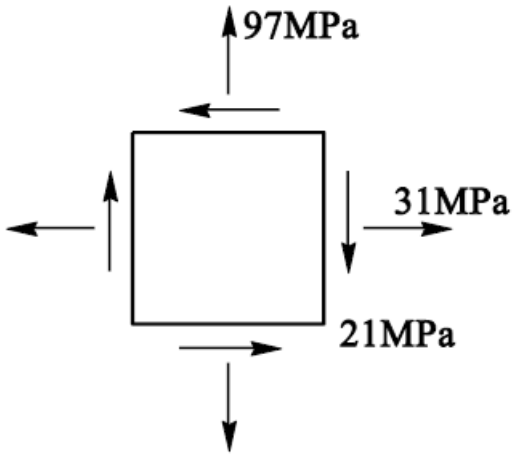
$$\tau_{\max}=\sqrt{(\frac{\sigma_x-\sigma_y}{2})^2+(\tau_{xy})^2}=26.96(\text{MPa})$$

$$\theta_s=\theta_p+45^\circ=25.67^\circ$$


y
26.96(MPa)
θs=25.67°
O
x
5(MPa)

例 17

中平面應力狀態σ_x = 31 MPa，σ_y = 97 MPa，τ_{xy} = –21 MPa，試求(a)θ = –55°(順時針)方位之應力狀態，(b)主應力狀態，(c)同平面最大剪應力狀態，(d)絕對最大剪應力。



【解析】

$\sigma_x = 31$ MPa，$\sigma_y = 97$ MPa，$\tau_{xy} = -21$ MPa

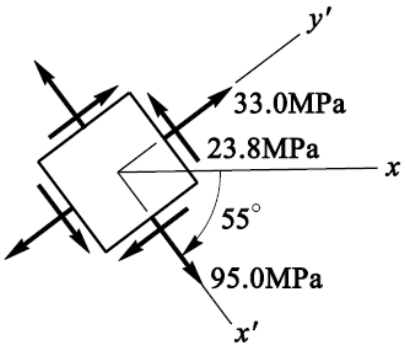
(a) $\theta = -55°$

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$= \frac{31+97}{2} + \frac{31-97}{2}\cos(-110°) + (-21)\sin(-110°) = 95.0 \text{ MPa}$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta = -\frac{31-97}{2}\sin(-110°) + (-21)\cos(-110°) = -23.8 \text{ MPa}$$

$$\sigma_{y'} = (\sigma_x + \sigma_y) - \sigma_{x'} = (31+97) - 95.0 = 33.0 \text{ MPa}$$

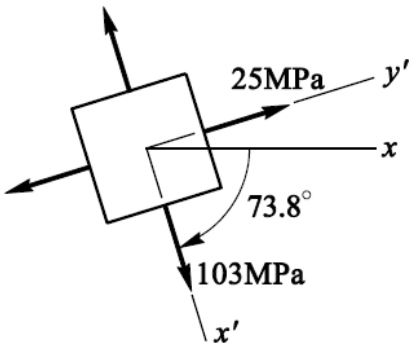


(b) $\tan 2\theta_p = \dfrac{\tau_{xy}}{\dfrac{\sigma_x - \sigma_y}{2}} = \dfrac{-21}{\dfrac{31-97}{2}} = \dfrac{-21}{-33}$

$\sin 2\theta_{p1} < 0$ ， $\cos 2\theta_{p1} < 0$ ，故 $2\theta_{p1} = -147.5°$ ， $\theta_{p1} = -73.8°$

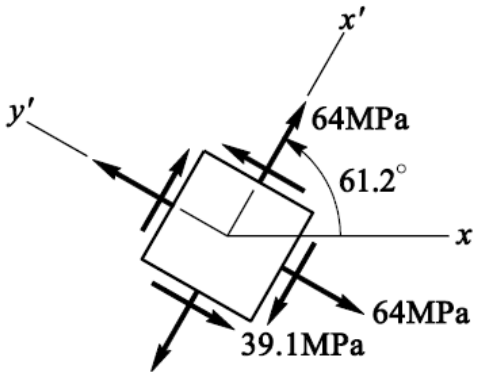
$$\sigma_{1,2}=\frac{\sigma_x+\sigma_y}{2}\pm\sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}^2}=\frac{31+97}{2}\pm\sqrt{\left(\frac{31-97}{2}\right)^2+(-21)^2}=64\pm39.1$$

$\sigma_1=103$ MPa ，$\sigma_2=25$ MPa



(c) $\theta_{s1}=\theta_{p1}-45°=-118.8°$ ，或 $\theta_{s1}=61.2°$

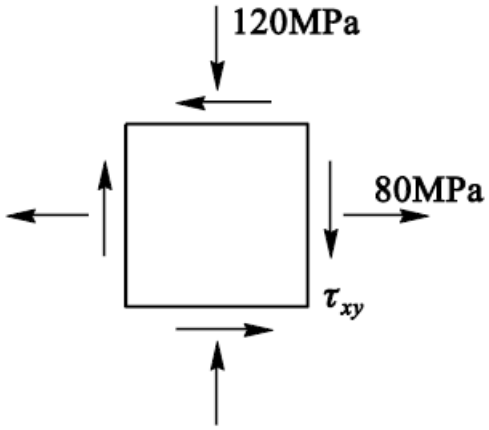
$$\tau_{\max}=\sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}^2}=39.1\text{ MPa}\quad，\quad\sigma_{x'}=\sigma_{y'}=\frac{\sigma_x+\sigma_y}{2}=64\text{ MPa}$$



(d) $\sigma_1>0$ ，$\sigma_2>0$ ，故 $(\tau_{\max})_{abs}=\frac{\sigma_1}{2}=51.5$ MPa

例 18

圖中所示為物體上某一點之平面應力狀態，σ_x = 80 MPa，σ_y = −120 MPa，τ_{xy}大小未知(方向如圖所示)，已知該點之同平面最大剪應力為 125 MPa，試求(a)τ_{xy}之大小，(b)主應力狀態(以應力元素圖表示)。



【機械高考】

【解析】

(a) $\tau_{\max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

$125 = \sqrt{\left(\frac{80+120}{2}\right)^2 + \tau_{xy}^2}$ ，$\tau_{xy} = \pm 75$ MPa

已知τ_{xy}為順時針方向，

故$\tau_{xy} = -75$ MPa

(b) $\tan 2\theta_p = \frac{\tau_{xy}}{\frac{\sigma_x - \sigma_y}{2}} = \frac{-75}{100}$

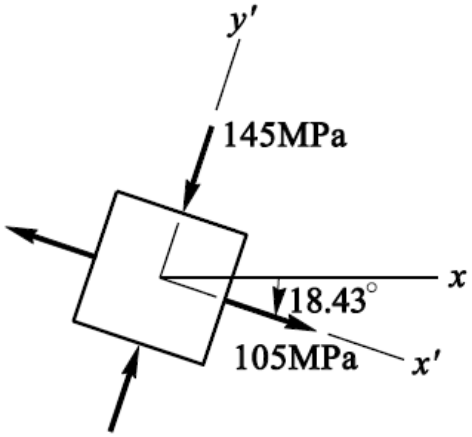
$\sin 2\theta_{p1} < 0$ ，$\cos 2\theta_{p1} > 0$ ，故$2\theta_{p1} = -36.87°$

$\theta_{p1} = -18.43°$ (順時針方向)

$\sigma_1 + \sigma_2 = \sigma_x + \sigma_y = -40$ ……①

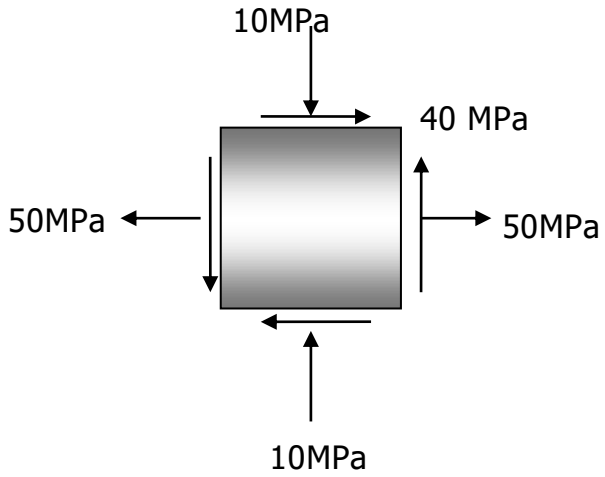
$\sigma_1 - \sigma_2 = 2\tau_{\max} = 250$ ……②

由①②σ_1= 105 MPa，$\sigma_2 = -145$ MPa


y′
145MPa
x
18.43°
105MPa
x′

例 19

For the state of plane stress shown in Fig., determine (a) the principal planes, (b) the principal stresses, (c) the maximum shearing stress and the corresponding normal stress.(d)The corresponding stress tensor



【93、96 中山機械】【交大土木】【中興機械】

【解析】

(a)

$$\sigma_x = +50 \text{ MPa} \qquad \sigma_y = -10 \text{ MPa} \qquad \tau_{xy} = +40 \text{ MPa}$$

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} = \frac{2(+40)}{50 - (-10)} = \frac{80}{60}$$

$$2\theta_p = 53.1°$$

$$\theta_p = 26.6°$$

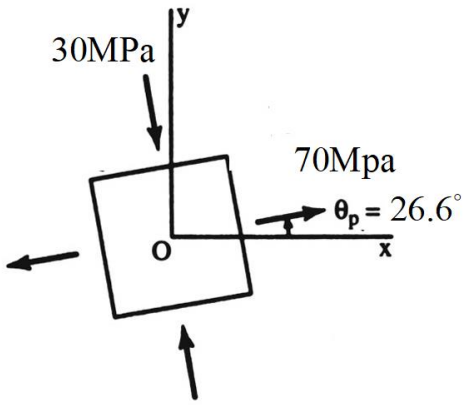
(b)

$$\sigma_{\max,\,\min} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= 20 \pm \sqrt{(30)^2 + (40)^2}$$

$$\sigma_{\max} = 20 + 50 = 70 \text{ MPa}$$

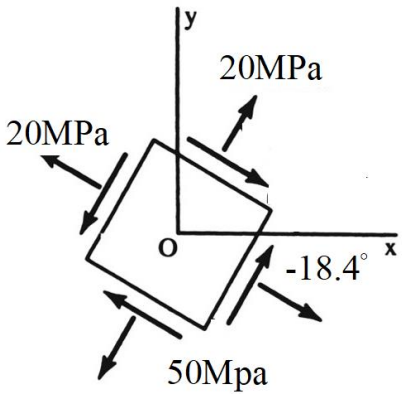
$$\sigma_{\min} = 20 - 50 = -30 \text{ MPa}$$



(c)

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \sqrt{(30)^2 + (40)^2} = 50 \text{ MPa}$$

$$\sigma' = \sigma_{\text{ave}} = \frac{\sigma_x + \sigma_y}{2} = \frac{50 - 10}{2} = 20 \text{ MPa}$$



(d)

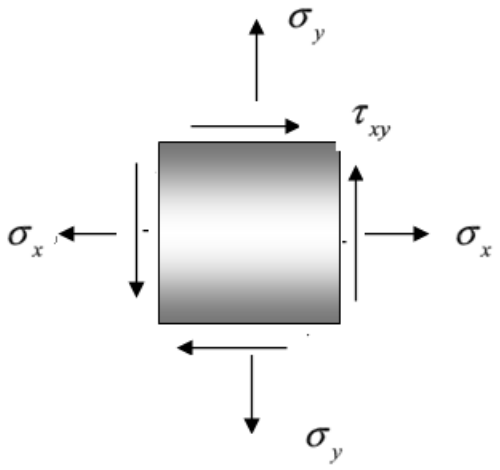
$$\begin{bmatrix} 50 & 40 & 0 \\ 40 & -10 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

例 20

(15%) 4. An element in *plane stress* (see figure) is subjected to stresses $\sigma_x = 2150$ kPa, $\sigma_y = 375$ kPa, $\tau_{xy} = -460$ kPa

(a) Determine the principal stresses and show them on a sketch of a properly oriented element. (8%)

(b) Determine the maximum shear stresses and associated normal stresses and show them on a sketch of a properly oriented element. (7%)



【106 台大工科】【108 台大工科】

【解析】

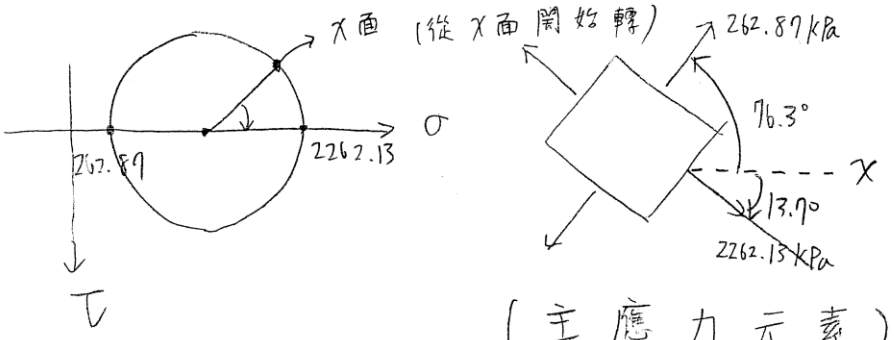
(1) 主 應 力

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \underbrace{2262.13}_{-13.7^\circ}, 267.87 \text{ (kPa)}$$

$$\theta_p = \frac{1}{2}\tan^{-1}\left(\frac{2\tau_{xy}}{\sigma_x - \sigma_y}\right)$$

$$= -13.7^\circ, 76.3^\circ$$



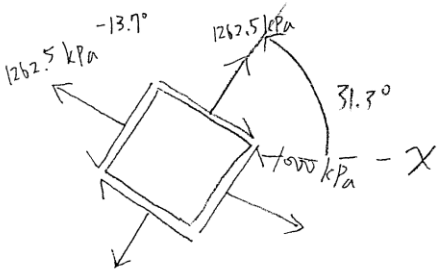
(主應力元素)

(2) 最大剪應力

$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2} = 1000\ (kPa)$$

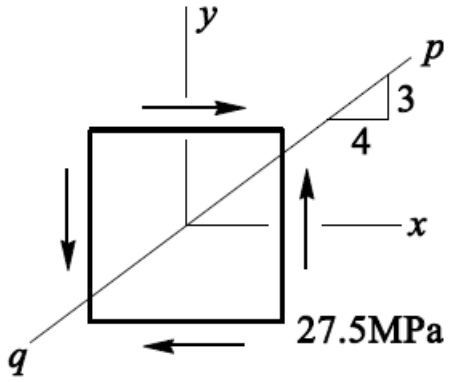
$$\sigma_{ave} = \frac{\sigma_x + \sigma_y}{2} = 1262.5\ (kPa)$$

$$\theta_s = \theta_p + 45° = 31.3°$$



例 21

圖中純剪之應力狀態，τ_{xy} = 27.5 MPa，試求(a)*pq* 傾斜面上之應力，(b)主平面及主應力。



【解析】

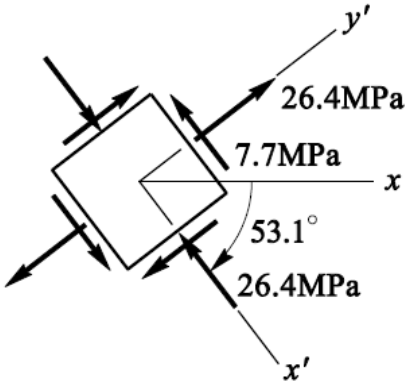
$\sigma_x = 0$，$\sigma_y = 0$，$\tau_{xy} = 27.5$ MPa，$\sigma_{av} = 0$，$R = 27.5$ MPa

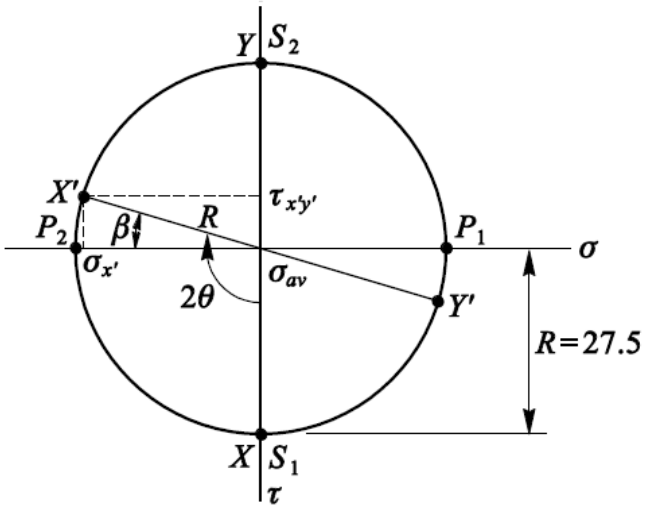
(a) $\theta = \tan^{-1}\left(\dfrac{4}{3}\right) = 53.1°$ (順時針方向)，$\beta = 2\theta - 90° = 16.2^\circ$

$\sigma_{x'} = -R\cos\beta = -(27.5)\cos 16.2° = -26.4$ MPa

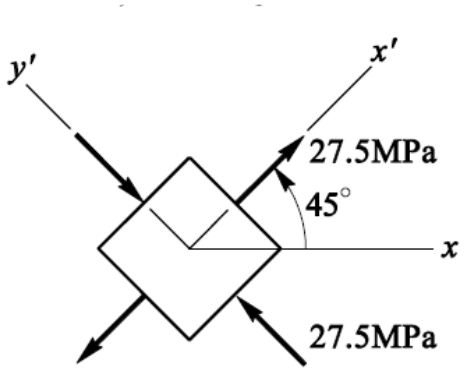
$\tau_{x'y'} = -R\sin\beta = -(27.5)\sin 16.2° = -7.7$ MPa

$\sigma_{y'} = (\sigma_x + \sigma_y) - \sigma_{x'} = 0 - (-26.4) = 26.4$ MPa



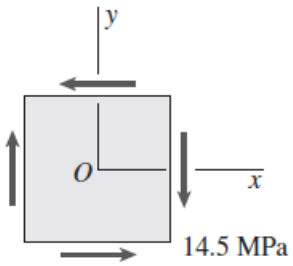


(b) $\theta_{s1} = 0°$，$\theta_{p1} = \theta_{s1} + 45° = 45°$ (逆時針)，$\sigma_1 = R = 27.5$ MPa，$\sigma_2 = -R = -27.5$ MPa


y′
x′
27.5MPa
45°
x
27.5MPa

例 22

An element in pure shear is subjected to stresses $\tau_{xy} = 14.5(MPa)$, as shown in the figure. Using Mohr's circle, determine: (a) The stresses acting on an element oriented at a counterclockwise angle $\theta = 22.5^\circ$ from the x axis (b) The principal stresses. Show all results on sketches of properly oriented elements



【105 成大醫工】

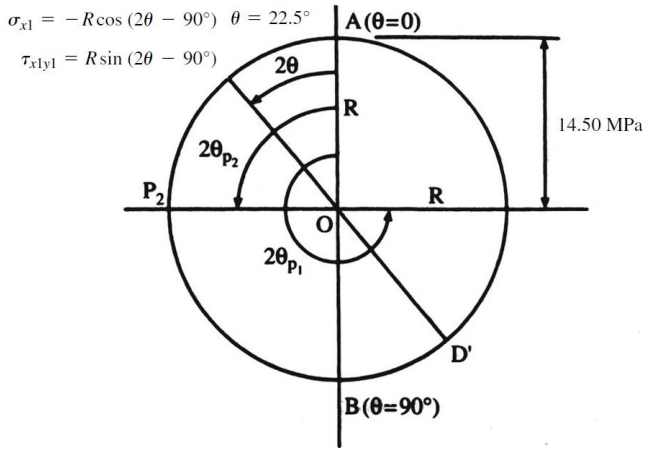
【解析】

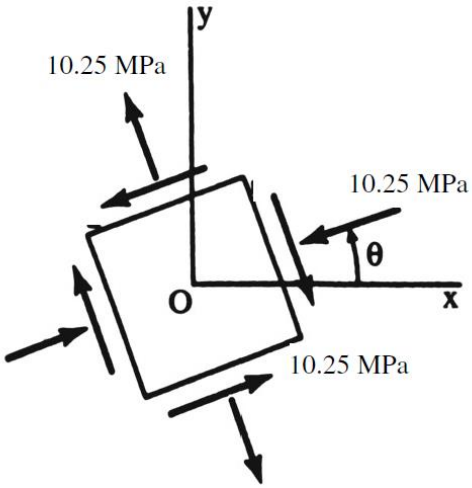
(a)

$\sigma_x = 0$ MPa $\quad \sigma_y = 0$ MPa $\quad \tau_{xy} = -14.5$ MPa

$R = |\tau_{xy}| \quad R = 14.50$ MPa

$$\sigma_{x1} = -R\cos(2\theta - 90^\circ)$$
$$\sigma_{x1} = -10.25 \text{ MPa} \quad \leftarrow$$
$$\tau_{x1y1} = R\sin(2\theta - 90^\circ)$$
$$\tau_{x1y1} = -10.25 \text{ MPa} \quad \leftarrow$$
$$\sigma_{y1} = R\cos(2\theta - 90^\circ)$$
$$\sigma_{y1} = 10.25 \text{ MPa} \quad \leftarrow$$





(b)

$$\theta_{p1} = \frac{270°}{2} \quad \theta_{p1} = 135.0°$$

$$\sigma_1 = R \qquad \sigma_1 = 14.50 \text{ MPa}$$

$$\theta_{p2} = \frac{-270°}{2}$$

$$\theta_{p2} = -135.0°$$

$$\sigma_2 = -R$$

例 23

4. An element in *biaxial stress* is subjected to stresses $\sigma_x = 28$ MPa and $\sigma_y = -7$ MPa, as shown in the figure (Fig. 3). Using Mohr's circle, determine:

(a) The stresses acting on an element oriented at a counterclockwise angle $\theta = 60^0$ from the x axis.

(b) The maximum shear stresses and associated normal stresses.

Show all results on sketches of properly oriented elements. (15%)

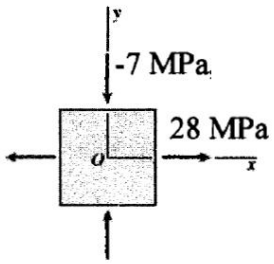


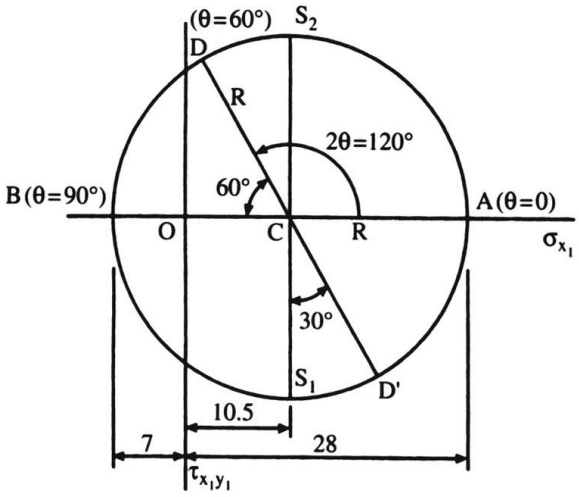
Fig. 3 An element in biaxial stress is subjected to stresses

【104 成大醫工】

【中文翻譯】

雙軸向應力之元素有應力 $\sigma_x = 28$ MPa 及 $\sigma_y = -7$MPa ，如圖所示。使用摩爾圓求：(a) 作用在從 x 軸 $\theta = 60°$ 之指向的元素上之應力，及 (b) 最大剪應力及伴隨的正向應力。將所有的結果繪示於正確指向的元素上。

【解析】



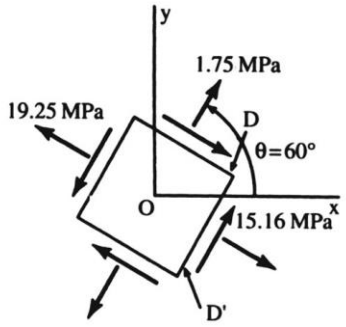
(a)

$$\sigma_x = 28 \text{ MPa},\ \sigma_y = -7 \text{ MPa},\ \tau_{xy} = 0$$

$$R = 17.5 \text{ MPa},\ \sigma_{x_1} = 10.5 \text{ MPa}$$

$$\sigma_{x_1} = 10.5 - R - R\cos 60 = 1.75 \text{ MPa}$$

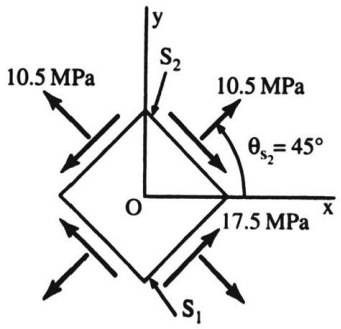
$$\tau_{x_1y_1} = -R\sin 60 = -15.16 \text{ MPa}$$



$$\sigma_{x_1} = 10.5 - R + R\cos 60 = 19.25 \text{ MPa}$$

$$\tau_{x_1 y_1} = -R\sin 60 = 15.16 \text{ MPa}$$

(b)



$$S_1 : 2\theta_{s_1} = -90°, \quad \theta_{s_1} = -45°$$

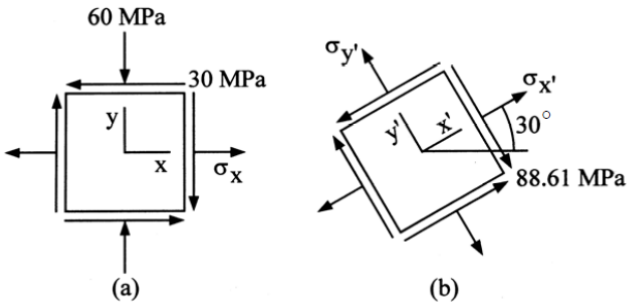
$$\tau_{max} = R = 17.5 \text{ MPa}$$

$$S_2 : 2\theta_{s_2} = 90°, \quad \theta_{s_2} = 45°$$

$$\tau_{min} = -17.5 \text{ MPa}$$

例 24

有一平面應力元素受應力如圖四(a)所示，當此元素逆時鐘方向旋轉 30°後，其應力狀況如圖四(b)所示。請計算σ_x、$\sigma_{x'}$、$\sigma_{y'}$及此元素之主軸應力（principal stresses）與主軸應力方向，並將主軸應力標示於旋轉至主軸應力方向之應力元素上。（25 分）



【106 土木高考】

【解析】

$$\sigma_x = \frac{\sigma_x - 60}{2} + \frac{\sigma_x + 60}{2}\cos 60^0 - 30\sin 60^0$$

$$-88.61 = -\frac{\sigma_x + 60}{2}\sin 60^0 - 30\cos 60^0$$

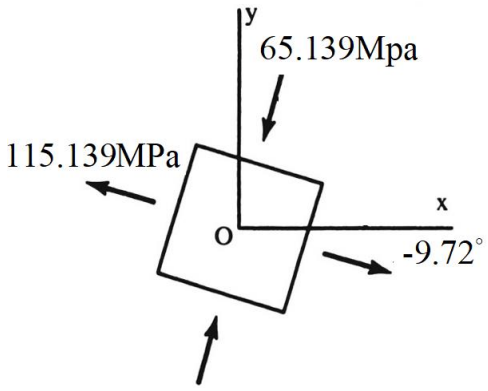
$$\sigma_{x'} = 41.519(MPa)$$

$$\sigma_x = 110(MPa)$$

$$\begin{Bmatrix}\sigma_x \\ \sigma_{x'} \\ \sigma_{y'}\end{Bmatrix} = \begin{Bmatrix}110 \\ 41.519 \\ 8.481\end{Bmatrix}(MPa)$$

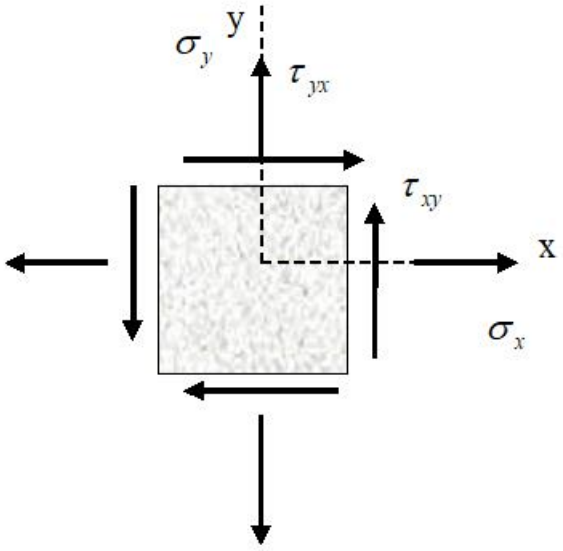
$$\begin{Bmatrix}\sigma_1 \\ \sigma_2\end{Bmatrix} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}{}^2} \Rightarrow \sigma_1 = 115.139 \text{、} \sigma_2 = -65.139$$

$$\tan 2\theta_P = -\frac{30}{85} \Rightarrow 2\theta_P = -19.44^0 \Rightarrow \theta_P = -9.72^0$$


y
65.139Mpa
115.139MPa
x
O
-9.72°

例 25

An element in plane stress (see figure) is subjected to stresses $\sigma_x = 2100(KPa), \sigma_y = 300(KPa)$ and $\tau_{xy} = -560(KPa)$.(a) Determine the principal stresses and show them on a sketch of a properly oriented element. (b) Determine the maximum shear stresses and associated normalstresses and show them on a sketch of a properly oriented element.



【解析】

平面上應力之一元素（ 見圖 ）受有應力 $\sigma_x = 2100(KPa), \sigma_y = 300(KPa)$ 及 $\tau_{xy} = -560(KPa)$ 。(a) 求主應力，並且在正確方向的元素上圖示它們。(b) 最大剪應力及伴隨的正向應力，並且在正確方向的元素上圖示它們。

【解析】

(a)

$$\sigma_x = 2100 \text{ kPa} \quad \sigma_y = 300 \text{ kPa} \quad \tau_{xy} = -560 \text{ kPa}$$

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} = -0.6222$$

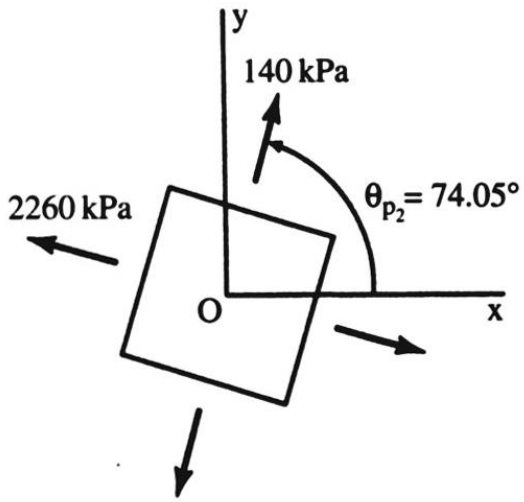
$$2\theta_p = -31.89° \quad \text{and} \quad \theta_p = -15.95°$$
$$2\theta_p = 148.11° \quad \text{and} \quad \theta_p = 74.05°$$

$$\sigma_{x_1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$2\theta_p = -31.89°: \quad \sigma_{x_1} = 2260 \text{ kPa}$$
$$2\theta_p = 148.11°: \quad \sigma_{x_1} = 140 \text{ kPa}$$

$\sigma_1 = 2260$ kPa and $\theta_{p_1} = -15.95°$

$\sigma_2 = 140$ kPa and $\theta_{p_2} = 74.05°$



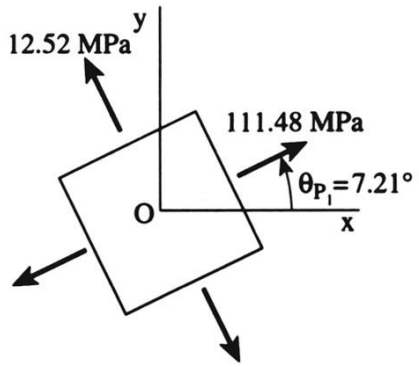
(b)

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = 1060 \text{ kPa}$$

$\theta_{s_1} = \theta_{p_1} - 45° = -60.95°$ and $\tau = 1060$ kPa

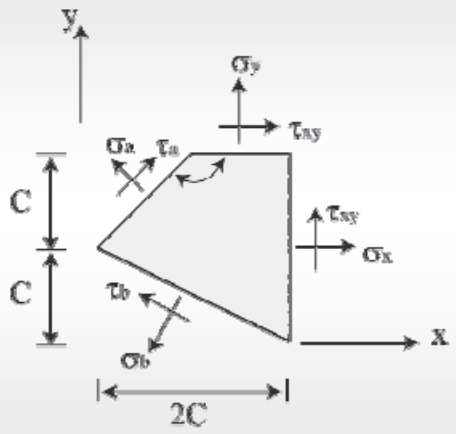
$\theta_{s_2} = \theta_{p_1} + 45° = 29.05°$ and $\tau = -1060$ kPa

$$\sigma_{\text{aver}} = \frac{\sigma_x + \sigma_y}{2} = 1200 \text{ kPa}$$



例 26

如圖所示，一厚度為 t 的薄板受一均勻應力作用，若已知 τ_{xy} = 500MPa、σ_a = 200MPa、τ_a = 300MPa，**試求** σ_b、τ_b、σ_x、σ_y **的大小。**(93 年土木高考)



【解析】

解 (1) $\sigma_a = \sigma_{-45°} = \dfrac{\sigma_x+\sigma_y}{2} + \dfrac{\sigma_x-\sigma_y}{2}\cos[2\times(-45°)] + \tau_{xy}\sin[2\times(-45°)]$

$\Rightarrow 200 = \dfrac{\sigma_x+\sigma_y}{2} + (500)\times\sin(-90°)$

$\Rightarrow \sigma_x+\sigma_y = 1400\cdots\cdots$

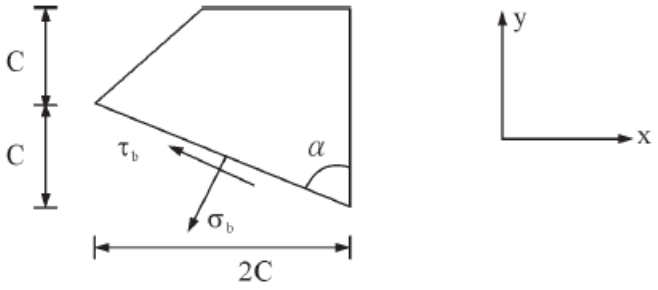
同理

$\tau_{-45°} = \tau_a = \dfrac{\sigma_x-\sigma_y}{2}\sin(-90°) - \tau_{xy}\cos(-90°)$

$\Rightarrow 300 = \dfrac{\sigma_x-\sigma_y}{2}\sin(-90°) - 500\cos(-90°) \quad \Rightarrow 600 = -\sigma_x+\sigma_y\cdots\cdots$

由　　得 $\sigma_x = 400\text{MPa}$, $\sigma_y = 1000\text{MPa}$

(2)



$$\alpha = 90° - \tan^{-1}\frac{c}{2c} = 63.435°\ ;$$

$$\sigma_b = \sigma_{63.435°} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos(2\times 63.435°) + \tau_{xy}\sin(2\times 63.435°)$$

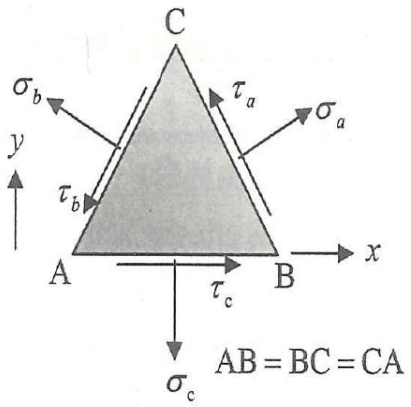
$$= \frac{400+1000}{2} + \frac{400-1000}{2}\cos(126.87°) + 500\sin(126.87°) = 1280(\text{MPa})$$

同理

$$\tau_b = \frac{400-1000}{2}\sin(126.87°) - 500\cos(126.87°) = 60(\text{MPa})$$

例 27

The state of a uniform plane stress as shown can be described by $\sigma_a = 200$ MPa ， $\sigma_b = 500$ MPa and $\tau_a = 300$ MPa . Determine the value of σ_c 、 τ_b 、 τ_c 、 σ_x 、 σ_y and τ_{xy}



【交大機械】

【解析】

$$\sigma_a = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 60^\circ + \tau_{xy}\sin 60^\circ$$

$$200 = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{4} + \frac{\sqrt{3}}{2}\tau_{xy}$$

$$\tau_a = -\frac{\sigma_x - \sigma_y}{2}\sin(60^\circ) + \tau_{xy}\cos(60^\circ)$$

$$300 = -\frac{\sqrt{3}(\sigma_x - \sigma_y)}{4} + \frac{\tau_{xy}}{2}$$

$$\sigma_b = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos(-60^\circ) + \tau_{xy}\sin(-60^\circ)$$

$$500 = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{4} - \frac{\sqrt{3}}{2}\tau_{xy}$$

$\sigma_x = 126.8$ MPa　　$\sigma_y = 1019.6$ MPa　　$\tau_{xy} = -173.2$ MPa

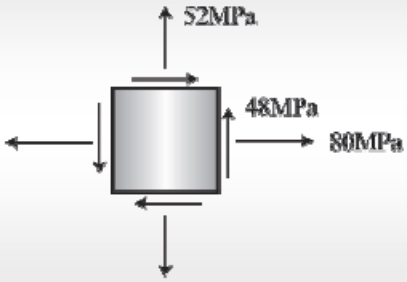
$$\tau_b = -\frac{\sigma_x - \sigma_y}{2}\sin(-60^\circ) + \tau_{xy}\cos(-60^\circ) = -473.2\,\text{MPa}$$

$$\sigma_c = \sigma_y = 1019.6\,\text{MPa}$$

$$\tau_c = -\tau_{xy} = 173.2\,\text{MPa}$$

例 28

如圖所示，有一受平面應力(Plane Stress)作用之平板元素，其在 x 軸與 y 軸方向之應力分別為＋80MPa 及＋52MPa，剪應力為＋48MPa，試求出其主應力(Principal Stresses)之大小與方向為何？(95 年機械高考)



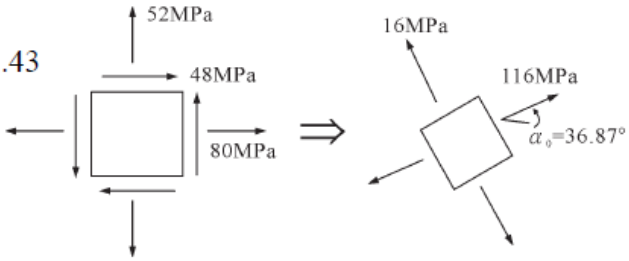
【解析】

解 $\sigma_{1,2} = \dfrac{\sigma_x + \sigma_y}{2} \pm \sqrt{(\dfrac{\sigma_x - \sigma_y}{2})^2 + \tau_{xy}^2} = \dfrac{80+52}{2} \pm \sqrt{(\dfrac{80-52}{2})^2 + 48^2}$

$\sigma_1 = 116(\text{MPa})$, $\sigma_2 = 16(\text{MPa})$

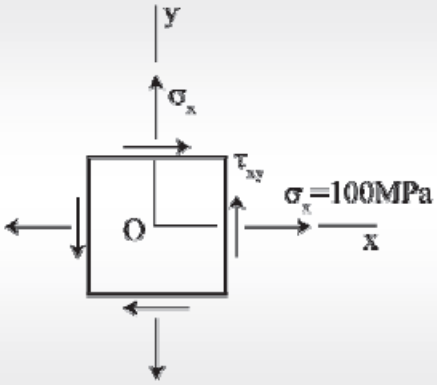
$\tan 2\alpha_0 = \dfrac{\tau_{xy}}{\frac{1}{2}(\sigma_x - \sigma_y)} = \dfrac{48}{\frac{1}{2}(80-52)} = 3.43$

$\alpha_0 = 36.87°$



例 29

飛機翼之表面受有平面應力，有正向應力 σ_x 及 σ_y 及剪應力 τ_{xy} ，如圖所示。在從 x 軸 $\theta = 30°$ 角處，正向應力為 3.5 MPa 拉伸，且在 $\theta = 50°$ ，係為 10 MPa 壓縮。如果應力 σ_x 等於 100 MPa 拉伸，則應力 σ_y 及 τ_{xy} 為何？



【解析】

解 Find σ_y and τ_{xy}

$$\sigma_{x_1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

(1) For $\theta = 30°$

$$\sigma_{x_1} = 35 = \frac{100 + \sigma_y}{2} + \frac{100 - \sigma_y}{2}\cos 60° = \tau_{xy}\sin 60°$$

or $0.25\sigma_y + 0.86603\tau_{xy} = -40$ MPa

(2) For $\theta = 50°$

$$\sigma_{x_1} = -10 = \frac{100 + \sigma_y}{2} + \frac{100 - \sigma_y}{2}\cos 100° = \tau_{xy}\sin 100°$$

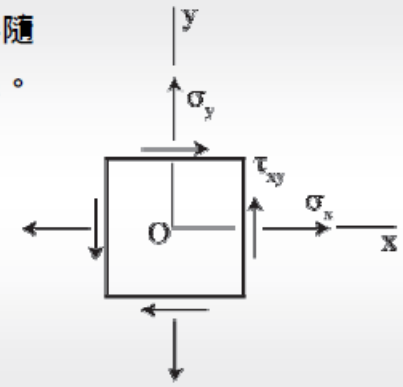
$0.58682\sigma_y + 0.98481\tau_{xy} = -51.318$ MPa

$\sigma_y = -19.3$MPa, $\tau_{xy} = -40.6$ MPa

例 30

利用摩爾圓求(a)主軸應力與(b)最大剪應力，及其伴隨之正向應力，並在正確方向的元素繪出所有結果。

σ_x =12.75MPa, σ_y =43.75MPa, τ_{xy} =21MPa

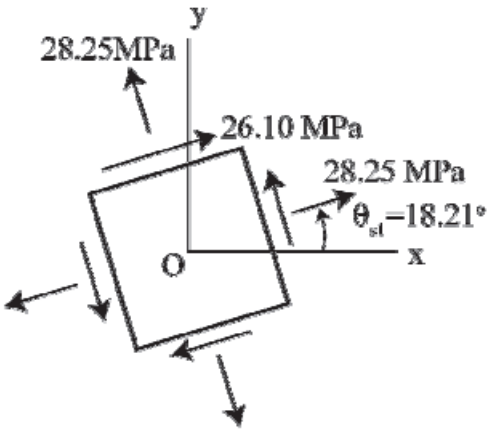


【解析】

解 σ_x =12.75MPa, σ_y =43.75MPa, τ_{xy} =21MPa

$$R=\sqrt{(15.5)^2+(21)^2}=26.10\text{MPa}$$

$$\alpha=\tan^{-1}(\frac{21}{15.5})=53.13°$$



(1) **主平面應力**

$2\theta_{p_1}=180°-\alpha=126.43°$, $\theta_{p_1}=63.21°$

$2\theta_{p_2}=-\alpha=-53.57°$, $\theta_{p_2}=-26.78°$

$\sigma_1=28.25+R=54.35\text{MPa}$

$\sigma_2=28.25-R=-2.15\text{MPa}$

(2) **最大剪**應力

$2\theta_{S_1}=90°-\alpha=36.43°$, $\theta_{S_1}=18.21°$

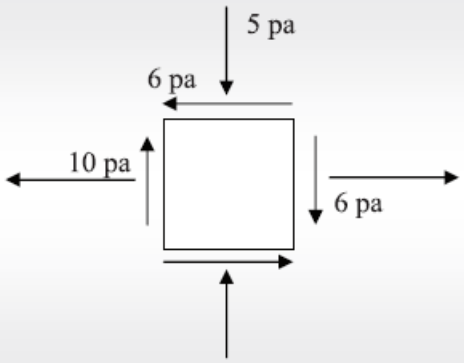
$2\theta_{S_2}=270°-\alpha=216.43°$, $\theta_{S_2}=108.21°$

$\tau_{max}=R=26.10\text{MPa}$

$\tau_{min}=-26.10\text{MPa}$

例 31

在結構中有一點之應力如圖，試求此應力之主軸的方向及應力。（103 高員）



【解析】

解

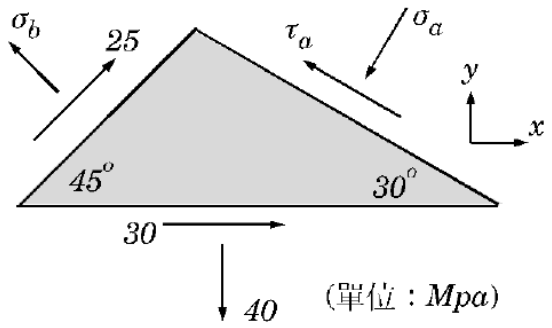
$$\sigma_{1,2}=\frac{10-5}{-2}\mathrm{L}\pm\sqrt{\frac{10+5}{2}+(-6)^2}\ ，故\sigma_1=12.1(\mathrm{pa})\ ，\ \sigma_2=-7.1(\mathrm{Pa})$$

$$\tan 2\theta=\frac{2\tau_{xy}}{\sigma_x-\sigma_y}=\frac{2(-6)}{10-(-5)}=-0.8$$

$\theta=19.33°$ 及 $\theta=199.33°$

例 32

某材料所受之平面應力如圖所示，試求圖面上(一) σ_a、τ_a、σ_b(二)主應力及主軸方向(三)平面上最大剪應力及其相對應方向



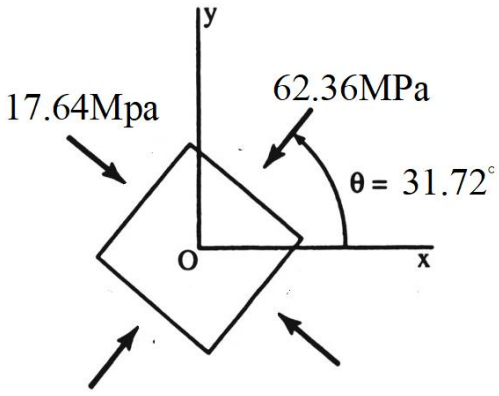
【解析】

$$\tau_\theta = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$$

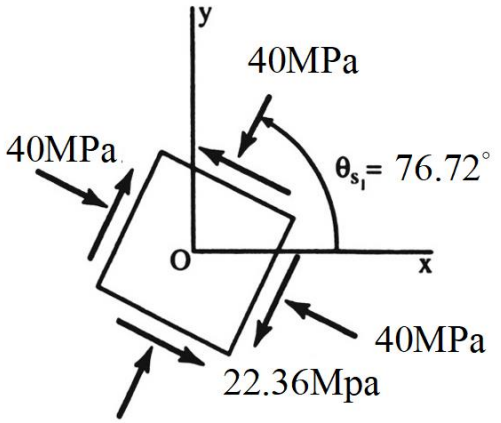
$$\Rightarrow -25 = -\frac{\sigma_x - 40}{2}\sin 270^\circ + (-30)\cos 270^\circ \Rightarrow \sigma_x = -10MPa$$

$$\sigma_b = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta = 45(MPa)$$

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{(\frac{\sigma_x - \sigma_y}{2})^2 + (\tau_{xy})^2} = 17.64, -62.36(MPa)$$

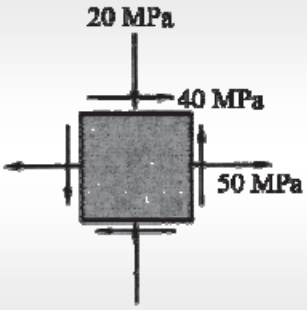


$$\tau_{\max} = \sqrt{(\frac{\sigma_x - \sigma_y}{2})^2 + (\tau_{xy})^2} = 22.36(\text{MPa})$$


y
40MPa
40MPa
θ_{s_1}= 76.72°
O
x
40MPa
22.36Mpa

例 33

如圖所示之平面應力狀態，試求其主應力及最大剪應力之值。（102 關三）



【解析】

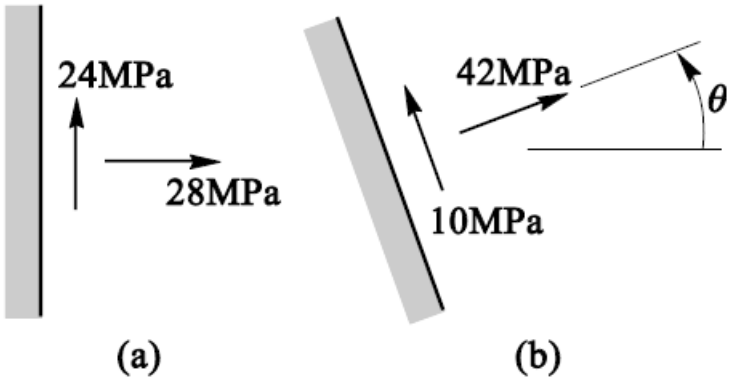
解 $\sigma_x=50$，$\sigma_y=-20$，$\tau_{xy}=40$

$$\sigma_{1,2}=\frac{\sigma_x+\sigma_y}{2}\pm\sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\left(\tau_{xy}\right)^2}=\frac{50-20}{2}\pm\sqrt{\left(\frac{50-(-20)}{2}\right)^2+(40)^2}=15\pm53.15$$

故 $\sigma_1=68.15(\text{MPa})$，$\sigma_2=-38.15(\text{MPa})$，$\tau_{max}=53.15(\text{MPa})$

例 35

件上之某一點為平面應力狀態，已知此應力元素在垂直面上之正交應力為 28 MPa(拉應力)剪應力為 24 MPa(逆時針)，如(a)圖所示，而在某一θ角傾斜面上之正交應力為 42 MPa(拉應力)剪應力為 10 MPa(逆時針)，如(b)圖所示，試此傾斜面之角度θ。



【解析】

$$\sigma_{x'}=\frac{1}{2}(\sigma_x+\sigma_y)+\frac{1}{2}(\sigma_x-\sigma_y)\cos 2\theta+\tau_{xy}\sin 2\theta$$

$$42=\frac{1}{2}(28+\sigma_y)+\frac{1}{2}(28-\sigma_y)\cos 2\theta+24\sin 2\theta$$

$$\text{得}\frac{1}{2}\sigma_y=\frac{28-14\cos 2\theta-24\sin 2\theta}{1-\cos 2\theta}\quad\cdots\cdots①$$

$$\tau_{x'y'}=-\frac{1}{2}(\sigma_x-\sigma_y)\sin 2\theta+\tau_{xy}\cos 2\theta$$

$$10=-\frac{1}{2}(28-\sigma_y)\sin 2\theta+24\cos 2\theta$$

$$\text{得}\frac{1}{2}\sigma_y=\frac{10+14\sin 2\theta-24\cos 2\theta}{\sin 2\theta}\quad\cdots\cdots②$$

$$\text{由①②，}\frac{28-14\cos 2\theta-24\sin 2\theta}{1-\cos 2\theta}=\frac{10+14\sin 2\theta-24\cos 2\theta}{\sin 2\theta}$$

$$17-7\sin 2\theta-17\cos 2\theta=0$$

$$17-7\sin 2\theta-17\sqrt{1-\sin^2 2\theta}=0$$

$$238\sin 2\theta=338\sin^2 2\theta$$

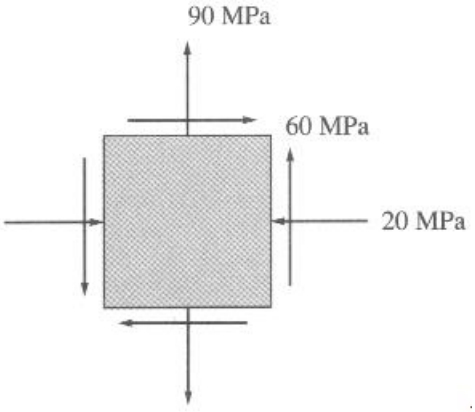
$$\sin 2\theta=\frac{238}{338}\ ,\ 2\theta=44.76^\circ\ ,\ \theta=22.38^\circ$$

例 36

Given the state of stresses at the dark spot on the surface of a pressure vessel as shown in Fig.

a.) Determine the state of principal stresses.

b.) Determine the state of maximum in-plane shear stress.



【解析】

$\sigma_x = -20\,\text{MPa}$ $\sigma_y = +90\,\text{MPa}$ $\tau_{xy} = 60\,\text{MPa}$

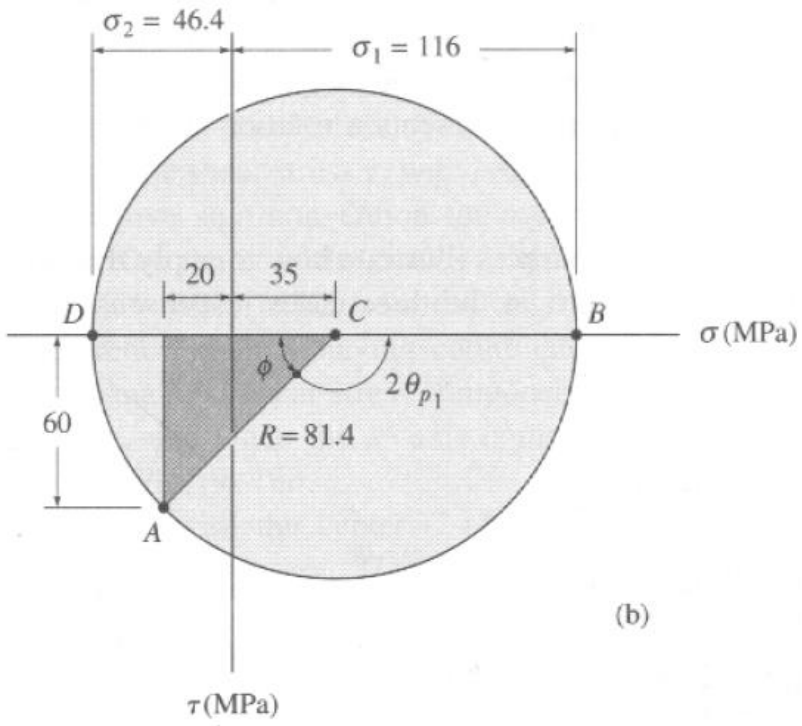
$$\sigma_{avg} = \frac{\sigma_x + \sigma_y}{2} = \frac{-20+90}{2} = 35\,\text{MPa}$$

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \sqrt{\left(\frac{-20-90}{2}\right)^2 + 60^2} = 81.4\,\text{MPa}$$

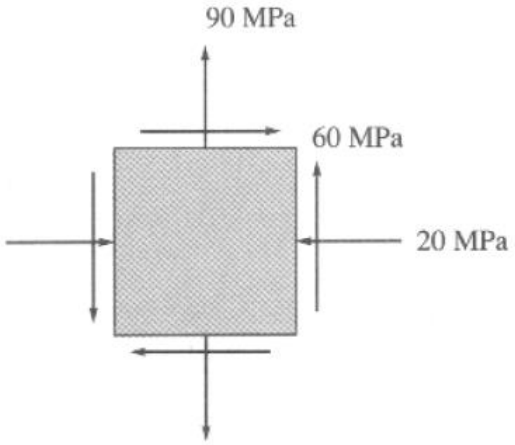
$$\sigma_1 = 35 + 81.4 = 116.4\,\text{MPa}$$

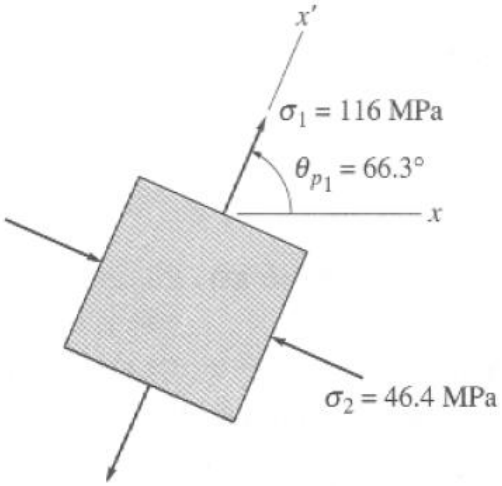
$$\sigma_2 = 35 - 81.4 = -46.4\,\text{MPa}$$

$$2\theta_{p_1} = 180^o - \phi = 180^o - \tan^{-1}\frac{60}{55} = 132.5^o$$



$$\theta_{p1} = \frac{132.5^o}{2} = 66.3^o$$

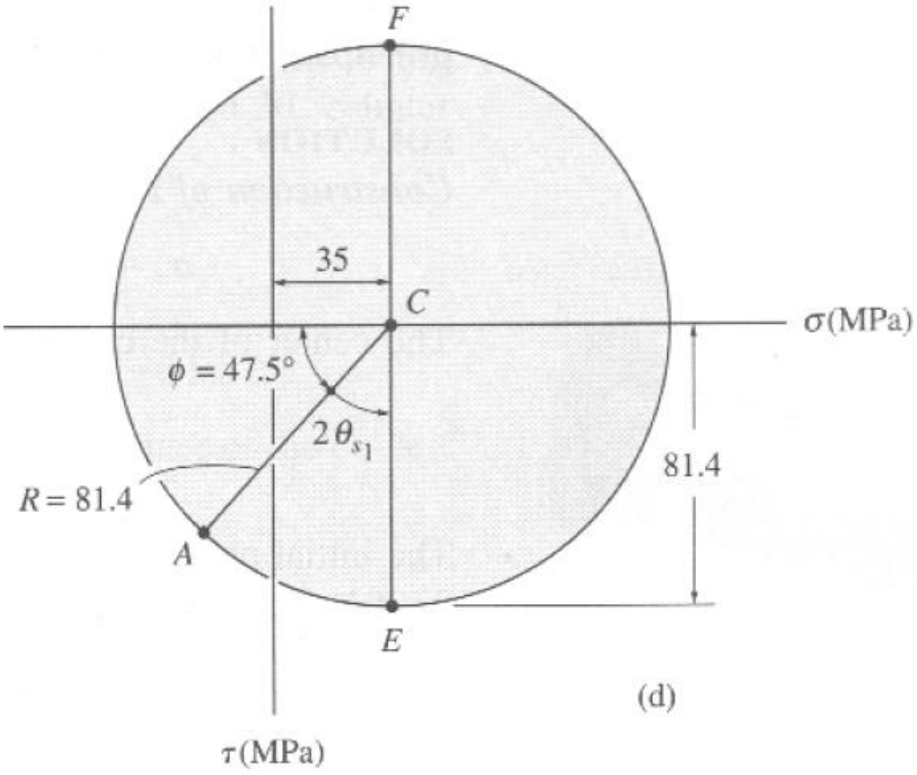




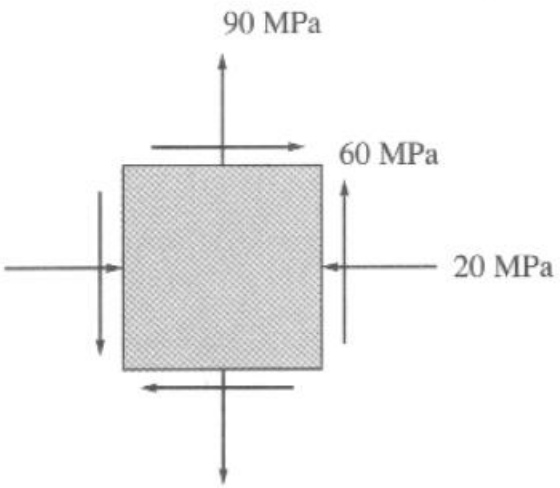
$$\tau_{\substack{\max \\ in-plane}} = 81.4\,\text{MPa}$$

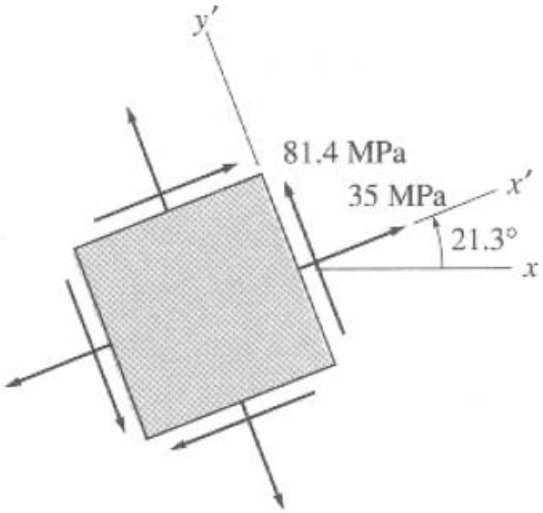
$$2\theta_{s_1} = 90^\circ - \phi = 90^\circ - 47.5^\circ = 42.5^\circ$$

$$\theta_{s1} = \frac{42.5^\circ}{2} = 21.3^\circ$$



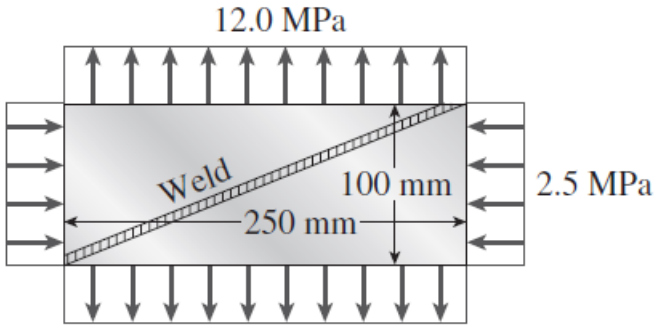
(d)





例 37

A rectangular plate of dimensions $100mm \times 250\text{mm}$ is formed by welding two triangular plates (see figure) .The plate is subjected to a compressive stress of 2.5MPa in the long direction and a tensile stress of 12.0 MPa in the short direction (see figure). Determine the normal stress σ_{x1} acting perpendicular to the line of the weld and the shear τ_{x1} acting parallel to the weld.



【中文翻譯】

一薄矩形板，尺寸為$100mm \times 250\text{mm}$，由兩三角形板焊成（ 見圖 ）。板受壓縮應力2.5MPa於長的方向，而在短的方向受拉伸應力12 MPa。求作用在垂直於焊縫之正向應力σ_{x1}及平行於焊縫之剪應力τ_{x1}

【解析】

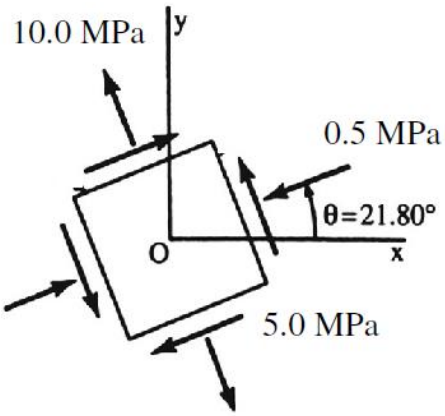
$$\sigma_x = -2.5 \text{ MPa} \qquad \sigma_y = 12.0 \text{ MPa} \qquad \tau_{xy} = 0$$

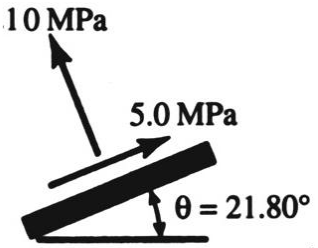
$$\sigma_{x_1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$= -0.5 \text{ MPa}$$

$$\tau_{x_1 y_1} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta = 5.0 \text{ MPa}$$

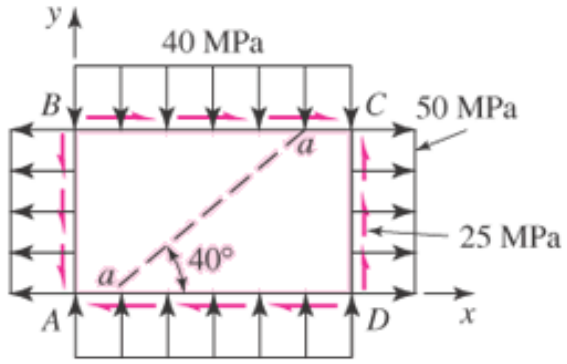
$$\sigma_{y_1} = \sigma_x + \sigma_y - \sigma_{x_1} = 10.0 \text{ MPa}$$


10.0 MPa
y
0.5 MPa
θ=21.80°
O
x
5.0 MPa


10 MPa
5.0 MPa
θ = 21.80°

例 38

矩形平板上的應力分布如圖所示，試求（a）平行與垂直斜面 $a-a$ 上的正交和剪應力（b）最大剪應力



【解析】

$\theta = 40 + 90 = 130^o$

$\sigma_y = -40\ MPa, \quad \sigma_x = 50\ MPa, \quad \tau_{xy} = 25\ MPa$

$\sigma_{x'} = \frac{50-40}{2} + \frac{50+40}{2}\cos 2(130^o) + 25\sin 2(130^o)$

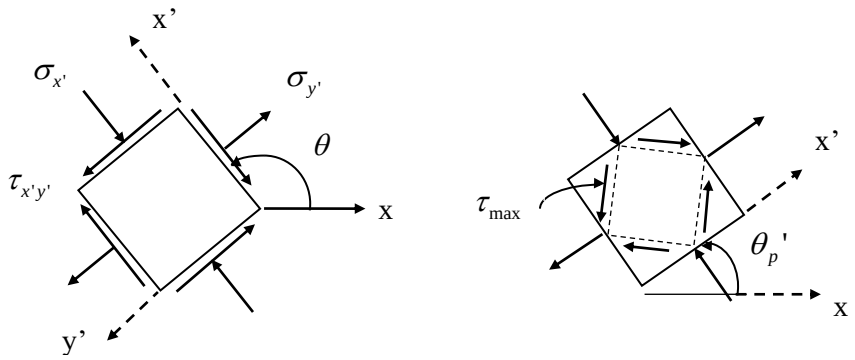
$= 5 - 7.814 - 24.62 = -27.43\ MPa$

$\tau_{x'y'} = -45\sin 260^o + 25\cos 260^o = 44.316 - 4.341 = 39.975\ MPa$

$\sigma_{y'} = 5 + 7.814 + 24.62 = 37.434\ MPa$

(b) $\tau_{\max} = [(\frac{50+40}{2})^2 + 25^2]^{\frac{1}{2}} = 51.48\ MPa$

$\theta_p' = \frac{1}{2}\tan^{-1}\frac{25}{45} = 14.53^o$



例 39

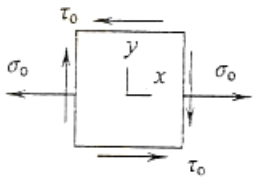
有關右圖之平面應力狀態，請問以下敘述何者錯誤：

(A) 最大剪應力方向與最大主應力方向相差 45°。

(B) 兩主應力合爲 $\sigma_o/2$。

(C) z 方向之應力爲零。

(D) 最大剪應力爲 $\sqrt{(\sigma_o/2)^2+\tau_o^2}$。 (E) 兩主應力方向垂直。



【100 台大機械】

【解析】

(B)

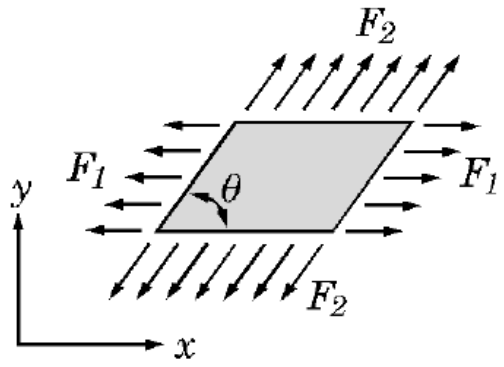
兩主應力合爲σ_0

例 40

如圖所示之材料體，厚度為一個單位，若 F1 及 F2 為單位長度的作用力，令σ_x、σ_y、τ_{xy}、τ_{yx}為元素內某點應力，下列何者正確

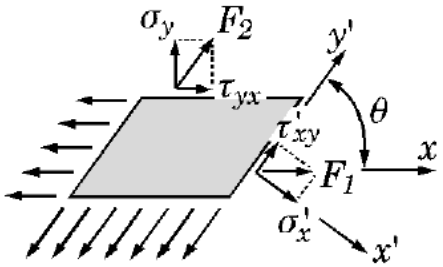
$(a)\ \sigma_x = \left(F_1 + F_2 \cos^2\theta\right) / \sin\theta$　　$(b)\ \sigma_y = \left(F_1 \sin^2\theta + F_2\right) / \cos\theta$

$(c)\ \tau_{xy} = F_2 \cos\theta$　　$(d)\ \tau_{yx} = F_2 \sin\theta$　　(e)以上皆非



【台大機械】

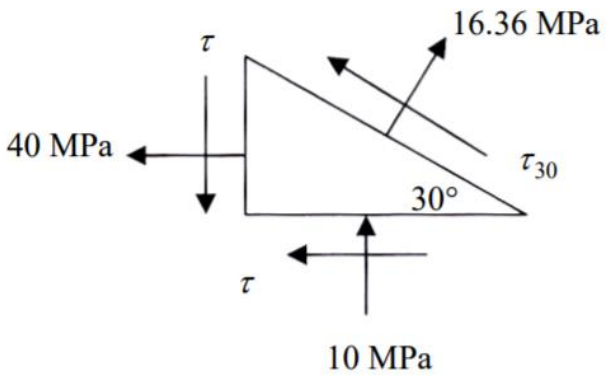
【解析】



正確答案為(a)及(c)

例 41

由圖示某點之應力狀況，計算未知應力τ與τ_{30}。（20 分）



【100 土木高考】

【解析】

應力轉換公式

$$\sigma_\theta = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

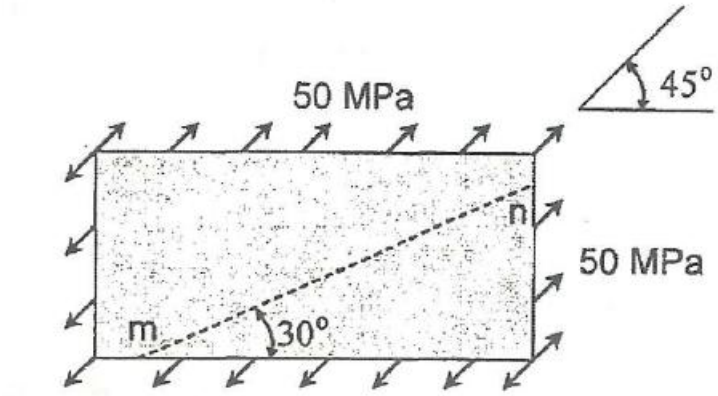
$$\sigma_{\theta=60^0} = \frac{40+(-10)}{2} + \frac{40-(-10)}{2}\cos 120 + \tau_{xy}\sin 120 = 16.36$$

$$\tau_{xy} = 16.00 Mpa = \tau$$

$$\tau_{30} = \tau_{\theta=60} = -\frac{40-(-10)}{2}\sin(120) + 16\cos(120) = -29.65(Mpa)$$

例 42

The stress acting uniformly over the sides of a rectangular block is shown in the figure. Please calculate the stress components on the planes parallel and perpendicular to mn which is shown in the figure.



【交大機械】

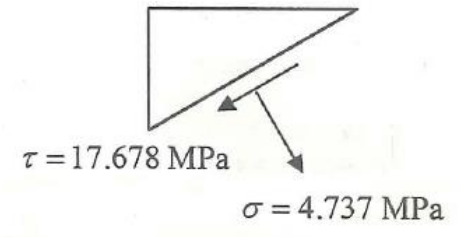
【解析】

$\sigma_x = 25\sqrt{2}\ \text{MPa} \quad \sigma_y = 25\sqrt{2}\ \text{MPa} \quad \tau_{xy} = 25\sqrt{2}\ \text{MPa}$

$$\sigma_\theta = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

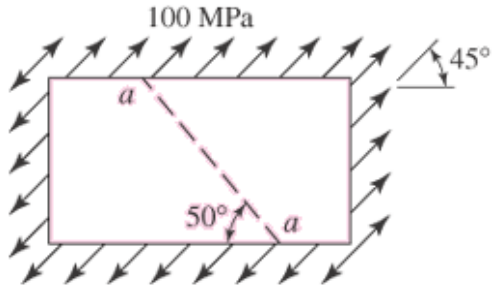
$$\Rightarrow \sigma_\theta = \frac{25\sqrt{2} + (25\sqrt{2})}{2} + \frac{25\sqrt{2} - (25\sqrt{2})}{2}\cos(2\times -60^\circ) + 25\sqrt{2}\sin(2\times -60^\circ) \Rightarrow \sigma_\theta = 4.727(MPa)$$

同理: $\tau_\theta = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta = -17.678(MPa)$



例 43

作用在機翼結構上的均勻負載如圖所示，試求平行與垂直斜面 $a-a$ 的應力組成，並繪出其方向性。



【交大機械類題】

【解析】

$\theta = 40^o$

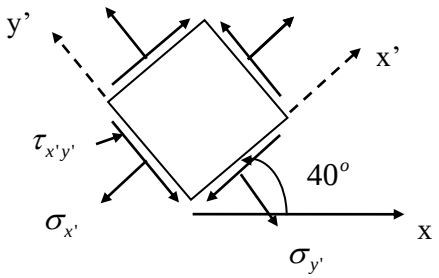
$\sigma_x = \sigma_y = 100\cos 45^o = 70.71 \ MPa$

$\tau_{xy} = 100\sin 45^o = 70.71 \ MPa$

$\sigma_{x'} = \frac{70.71+70.71}{2} + 0 + 70.71\sin 80^o = 140.3 \ MPa$

$\sigma_{y'} = 70.71 - 0 - 69.64 = 1.07 \ MPa$

$\tau_{x'y'} = -0 + 70.71\cos 80^o = 12.28 \ MPa$



例 44

機械元件上某點所承受之應力狀態，其應力值為：

$$\begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_{yy} & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_{zz} \end{bmatrix} = \begin{bmatrix} 11 & -1 & 0 \\ -1 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \ (MPa)$$

㈠試問此點對應之主應力 σ_1，σ_2 與 σ_3 各若干？（9 分）

㈡試問此點對應之最大剪應力 τ_{max} 為若干？（6 分）

㈢試問此點是否屬平面應力（Plane Stress）狀況？（5 分）

【101 身障四等】

【解析】

(1) $\left|[\sigma]-\sigma[I]\right|=0$

$$\left\|\begin{bmatrix} 11-\sigma & -1 & 0 \\ -1 & 4-\sigma & 0 \\ 0 & 0 & 0-\sigma \end{bmatrix}\right\|=0$$

$$\Rightarrow \sigma(\sigma^2-15\sigma+43)=0$$

$$\sigma_{1,2}=\frac{1}{2}(15\pm\sqrt{53}) \qquad \sigma_3=0$$

(2) $\tau_{\max,abs(out-plane)}=\dfrac{\sigma_1-\sigma_3}{2}=11.14(Mpa)$

$$\tau_{\max,(in-plane)}=\frac{\sigma_1-\sigma_2}{2}=7.28(Mpa)$$

(3)因為 $\sigma_{zz}=\tau_{xz}=\tau_{yz}=0$ 因此屬於平面應力**(plane stress)**

例 45

設某材料點的應力矩陣如下：

$$\sigma=\begin{bmatrix}14 & 7 & -7\\ 7 & 10 & 0\\ -7 & 0 & 35\end{bmatrix}(\text{MPa})$$

試求(a)過此點而在平面 $2x-y+3z=9$ 上的應力向量，以及(b)該平面上的正向應力 σ 與剪應力 τ 。

【解析】

平面 $2x-y+3z=9$ 之單位法線向量

$$\hat{n}=\frac{1}{\sqrt{2^2+(-1)^2+3^2}}(2\hat{i}-\hat{j}+3\hat{k})=\frac{2}{\sqrt{14}}\hat{i}-\frac{1}{\sqrt{14}}\hat{j}+\frac{3}{\sqrt{14}}\hat{k}$$

$$\vec{T}=[\sigma]\{n\}=\begin{bmatrix}14 & 7 & -7\\ 7 & 10 & 0\\ -7 & 0 & 35\end{bmatrix}\begin{Bmatrix}\frac{2}{\sqrt{14}}\\ \frac{-1}{\sqrt{14}}\\ \frac{3}{\sqrt{14}}\end{Bmatrix}=\begin{Bmatrix}0\\ 1.07\\ 24.32\end{Bmatrix}\text{ MPa}$$

$$=1.07\hat{j}+24.32\hat{k}\text{ MPa}$$

正應力及剪應力分量

$$\sigma=\hat{n}\cdot\vec{T}=\left(\frac{2}{\sqrt{14}}\hat{i}-\frac{1}{\sqrt{14}}\hat{j}+\frac{3}{\sqrt{14}}\hat{k}\right)\cdot(1.07\hat{j}+24.32\hat{k})$$

$$=19.21\text{ MPa}$$

$$\tau=\sqrt{|\vec{T}|^2-\sigma^2}$$

$$=\sqrt{(1.07)^2+(24.32)^2-(19.21)^2}=14.95\text{ MPa}$$

例 46

設某材料點的應力矩陣如下：

$$\sigma=\begin{bmatrix}14 & 7 & -7\\ 7 & 10 & 0\\ -7 & 0 & 35\end{bmatrix}(\text{MPa})$$

試求(a)過此點而在平面 $2x-y+3z=9$ 上的應力向量，以及(b)該平面上的正向應力 σ 與剪應力 τ 。

【解析】

平面 $2x-y+3z=9$ 之單位法線向量

$$\hat{n}=\frac{1}{\sqrt{2^2+(-1)^2+3^2}}(2\hat{i}-\hat{j}+3\hat{k})=\frac{2}{\sqrt{14}}\hat{i}-\frac{1}{\sqrt{14}}\hat{j}+\frac{3}{\sqrt{14}}\hat{k}$$

$$\vec{T}=[\sigma]\{n\}=\begin{bmatrix}14 & 7 & -7\\ 7 & 10 & 0\\ -7 & 0 & 35\end{bmatrix}\begin{Bmatrix}\frac{2}{\sqrt{14}}\\ \frac{-1}{\sqrt{14}}\\ \frac{3}{\sqrt{14}}\end{Bmatrix}=\begin{Bmatrix}0\\ 1.07\\ 24.32\end{Bmatrix}\text{ MPa}$$

$$=1.07\hat{j}+24.32\hat{k}\text{ MPa}$$

正應力及剪應力分量

$$\sigma=\hat{n}\cdot\vec{T}=\left(\frac{2}{\sqrt{14}}\hat{i}-\frac{1}{\sqrt{14}}\hat{j}+\frac{3}{\sqrt{14}}\hat{k}\right)\cdot(1.07\hat{j}+24.32\hat{k})$$

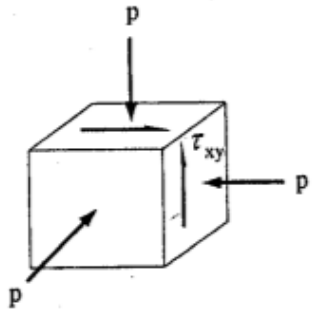
$$=19.21\text{ MPa}$$

$$\tau=\sqrt{|\vec{T}|^2-\sigma^2}$$

$$=\sqrt{(1.07)^2+(24.32)^2-(19.21)^2}=14.95\text{ MPa}$$

例 47

As shown in the following figure, a block is subjected to a confining pressure of $p = 100$ MPa on all sides, along with a shear stress τ_{xy}. (*a*) If the maximum allowable shear stress against yielding for this block is 150 MPa, what is the largest value of τ_{xy} that can be applied with a safety factor of 2? (b) Is there a large effect of the pressure p on the τ_{xy} required to cause yielding? Briefly discuss the effect of p as to whether the effect is large, small, or absent, and explain why. (15%)



【96 中央機械】

【解析】

(a)

$$\tau_{\max} = \frac{\tau_y}{F.S.} = \frac{150}{2} = 75(MPa)$$

$$\sigma_x = \sigma_y = \sigma_z = -P$$

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{(\frac{\sigma_x - \sigma_y}{2})^2 + (\tau_{xy})^2} = -P \pm \tau_{xy}$$

$$\sigma_3 = -P$$

$$\tau_{\max} = \frac{\sigma_1 - \sigma_2}{2} = \tau_{xy}$$

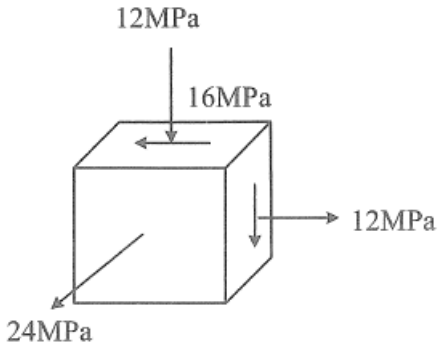
(b)

$$\tau_{\max} = \frac{\tau_y}{F.S.} = \frac{150}{2} = 75(MPa) = \tau_{xy}$$

所以與P無關

例 48

For the given stress state shown in the figure, please determine the absolute maximum shear stress.



【交大機械】

【解析】

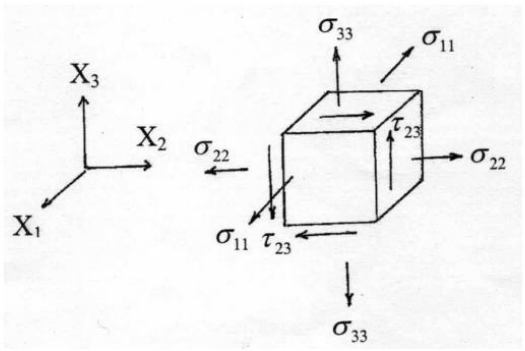
$$\sigma_{1,2}=\frac{\sigma_x+\sigma_y}{2}\pm\sqrt{(\frac{\sigma_x-\sigma_y}{2})^2+(\tau_{xy})^2}=20,-20(MPa)$$

$$\sigma_3=24(MPa)$$

$$\tau_{\max}=\frac{24-(-20)}{2}=22(MPa)$$

例 52

一固體材料承受多軸應力作用，如圖所示，其中 $\sigma_{11}=11\,\text{MPa}$，$\sigma_{22}=\sigma_{33}=4\,\text{MPa}$，$\tau_{23}=5\,\text{MPa}$，於此多軸應力作用下，求此固體材料所承受之最大剪應力。(25 分)



【107 土木特考三等】

【解析】

依據主應力公式

考慮 x_1 平面為主平面

$\therefore\ \sigma_{P_1}=11\,\text{MPa}$

再考慮 x_2-x_3

$$A=\frac{\sigma_{22}+\sigma_{33}}{2}=\frac{(4)+(4)}{2}=4\,\text{MPa}$$

$$B=\frac{\sigma_{22}-\sigma_{33}}{2}=\frac{(4)-(4)}{2}=0\,\text{MPa}$$

$$C=\tau_{23}=5\,\text{MPa}$$

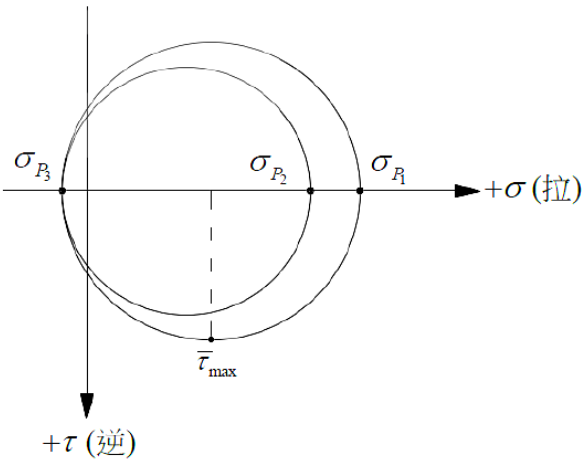
$$R=\sqrt{B^2+C^2}=\sqrt{(0)^2+(5)^2}=5\,\text{MPa}$$

$$\sigma_{P_2}=A+R=4+5=9\,\text{MPa}$$

$$\sigma_{P_3}=A-R=4-5=-1\,\text{MPa}$$

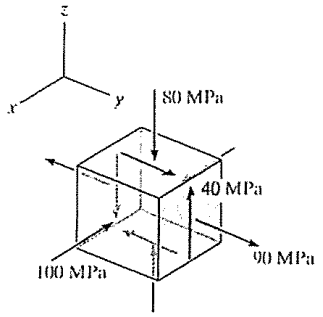
依據最大莫爾圓的半徑

$$\overline{\tau}_{\max}=\frac{\sigma_{P_1}-\sigma_{P_3}}{2}=\frac{(11)-(-1)}{2}=6\ \ \text{MPa}$$


σ_{P_3}
σ_{P_2}
σ_{P_1}
$+\sigma$ (拉)
$\overline{\tau}_{\max}$
$+\tau$ (逆)

例 49

附圖為某構件受力時某點之應力狀態，以下敘述何者正確？



(A) y 軸為應力主軸（a principal axis of stress）

(B) z 軸為應力主軸

(C)最大剪應力位於 yz 平面

(D) x 軸為應變主軸（a principal axis of strain）

(E)其中一主應力為+90 MPa

【105 台大機械】

【解析】

(A)y軸不是應力主軸

(B)Z軸不是應力主軸

(C)最大剪應力不在yz平面

(E)主應力

$$\sigma_{1,2}=\frac{\sigma_x+\sigma_y}{2}\pm\sqrt{(\frac{\sigma_x-\sigma_y}{2})^2+(\tau_{xy})^2}=98.94,-88.94(MPa)$$

$$\sigma_3=100(MPa)$$

故選(D)

例 50

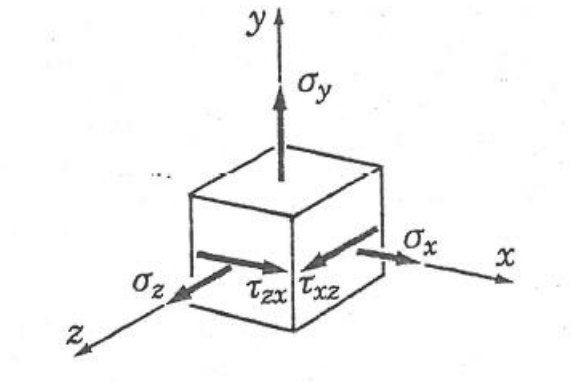
To find the principal stresses and the maximum shear stress for the following stress state：

$\sigma_x = 3$ MPa ； $\tau_{xy} = 0$ MPa ； $\sigma_z = 12$ MPa ； $\sigma_y = 10$ MPa

$\tau_{yz} = 0$ MPa ； $\tau_{zx} = 6$ MPa

【中山機械】

【解析】



$$\begin{vmatrix} \sigma_x - \sigma & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y - \sigma & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z - \sigma \end{vmatrix} = 0$$

$$\begin{vmatrix} 3-\sigma & 0 & 6 \\ 0 & 10-\sigma & 0 \\ 6 & 0 & 12-\sigma \end{vmatrix} = 0$$

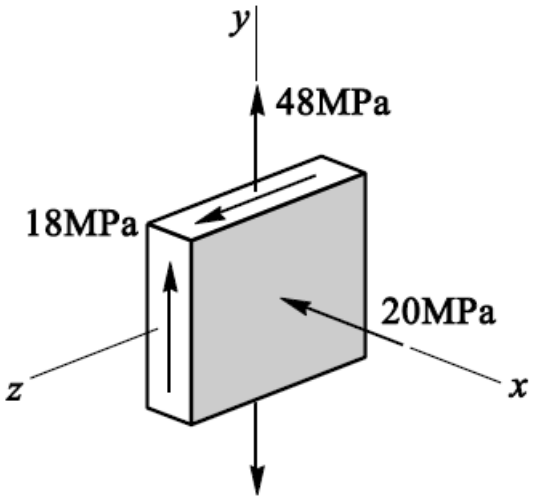
$$(10-\sigma)\sigma(\sigma-15) = 0$$

$\sigma = 0,10,15$

$$\tau_{\max} = \frac{1}{2}(\sigma_{\max} - \sigma_{\min}) = \frac{1}{2}(15-0) = 7.5$$

例 51

試求圖中應力狀態之最大剪應力。



【解析】

y - z 平面上之主應力

$$\sigma_{b,c}=\frac{\sigma_y+\sigma_z}{2}\pm\sqrt{\left(\frac{\sigma_y-\sigma_z}{2}\right)^2+\tau_{xy}^2}=\frac{48+0}{2}\pm\sqrt{\left(\frac{48-0}{2}\right)^2+18^2}=24\pm30$$

$\sigma_b=54\ \text{MPa}$，$\sigma_c=-6\ \text{MPa}$

因 x 面上無剪應力，故 $\sigma_x=-20\ \text{MPa}$ 為另一個主應力

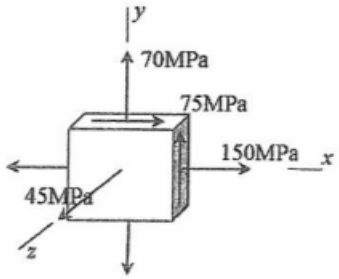
將三個主應力按大小次序排列：$\sigma_1=54\ \text{MPa}$，$\sigma_2=-6\ \text{MPa}$，$\sigma_3=-20\ \text{MPa}$

故 $\tau_{\max}=\frac{1}{2}(\sigma_1-\sigma_3)=\frac{1}{2}[54-(-20)]=37\ \text{MPa}$

例 52

、右圖爲構件某點在受力條件下之應力狀態，該點的最大剪應力爲

(A)85MPa　　(B)125MPa
(C)150MPa　　(D)165MPa
(E)以上皆非



【99 台大機械】

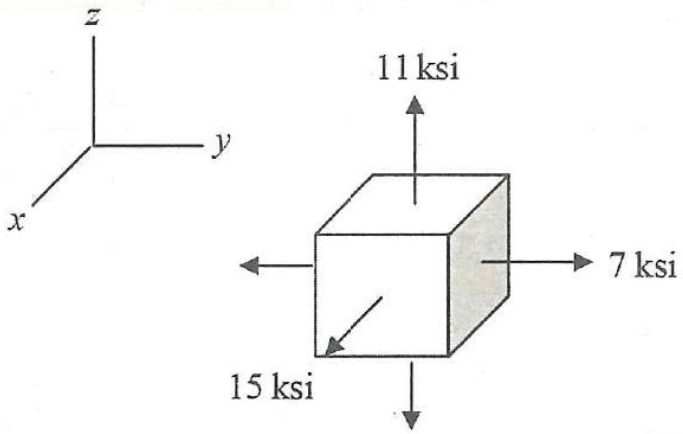
【解析】

$$\sigma_{1,2}=\frac{150+70}{2}\pm\sqrt{(\frac{150-70}{2})^2+75^2}=195\,,25(MPa)$$

$$\tau_{\max}=\frac{\sigma_{\max}-\sigma_{\min}}{2}=\frac{195-25}{2}=85(MPa)$$

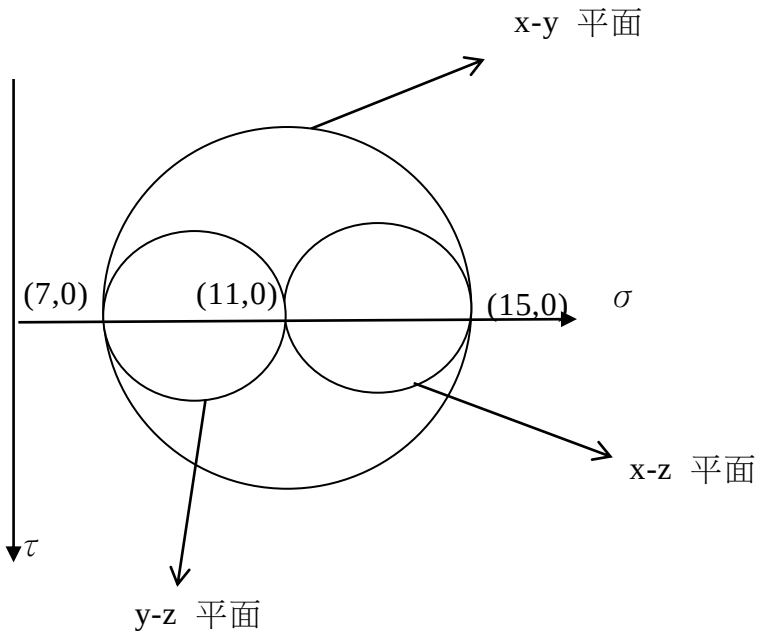
例 53

The principal stresses acting at a point in a body are shown in the figure. Draw the three Mohr's circles that describe this state of stress, and find the maximum in-plane shear stresses and associated average normal stresses for the x-y, y-z, and x-z planes. For each case, show the results on the element oriented in the appropriate direction. (25%)



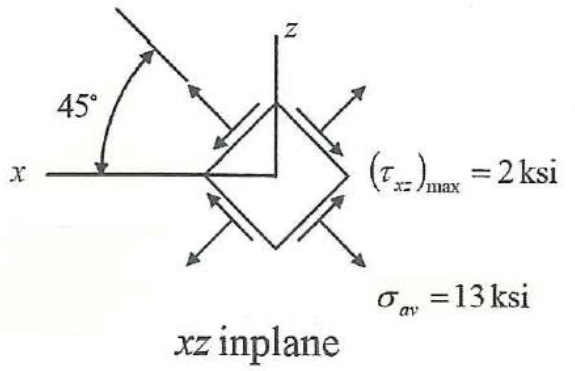
【中山機電】

【解析】



由莫爾圓x-z平面半徑 $\mathrm{R}=\dfrac{15-11}{2}=2(ksi)=(\tau_{xz})_{\max}$

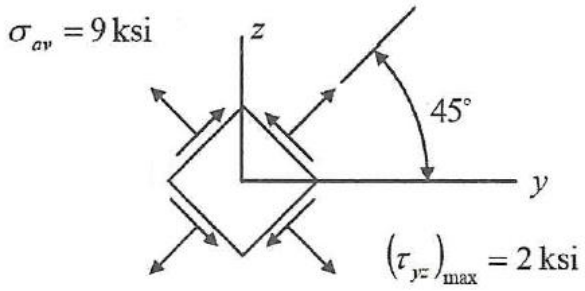
$\sigma_{\mathrm{av}}=11+\mathrm{R}=13(kpsi)$



xz inplane

由莫爾圓y-z平面半徑 $\mathrm{R} = \dfrac{11-17}{2} = 2(ksi)$

$\sigma_{\mathrm{av}} = 7 + \mathrm{R} = 9(kpsi)$



yz inplane

例 54

A 結構中某點 x 、 y 和 z 軸座標的應力方分量表示如下，

$$\begin{bmatrix} -10 & 0 & -8 \\ 0 & 2 & 0 \\ -8 & 0 & 2 \end{bmatrix} \text{ksi}$$

試求：

（a）最大主應力量值與方向

（b）最大剪應力

【解析】

$$\begin{vmatrix} \sigma_x - \sigma & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y - \sigma & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z - \sigma \end{vmatrix} = 0$$

$$\begin{vmatrix} (-10-\sigma_i) & 0 & -8 \\ 0 & (2-\sigma_i) & 0 \\ -8 & 0 & (2-\sigma_i) \end{vmatrix} = 0$$

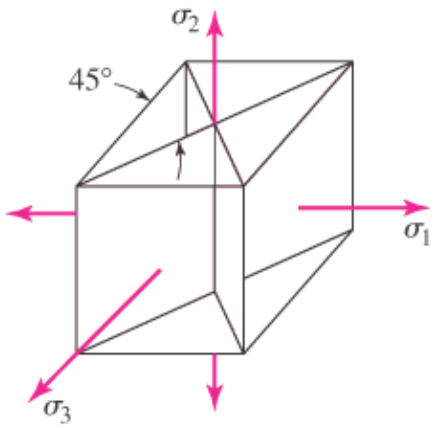
可擴展成：

$(2-\sigma_i)[(\sigma_i+10)(\sigma_i-2)-64]=0$

或 $\sigma_1 = 6\ ksi, \quad \sigma_2 = 2\ ksi, \quad \sigma_3 = -14\ ksi$ ◂

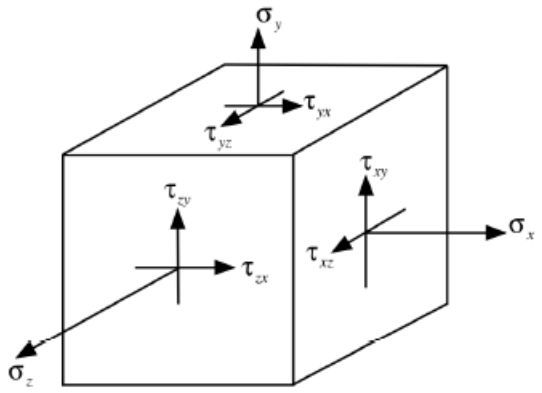
(b)

$\tau_{\max} = \frac{1}{2}(6+14) = 10\ ksi$ ◂



例 55

A triaxial stress element has a critical three dimensional stress. Calculate the first, second, and third stress invariants. Determine the principal normal stresses, the maximum shear stresses and draw the three Mohr circles



$\sigma_x = 60{,}000\,psi$
$\sigma_y = -30{,}000$
$\sigma_z = -15{,}000$
$\tau_{xy} = 9000$
$\tau_{yz} = -2000$
$\tau_{zx} = 3500$

【解析】

$$\begin{vmatrix} \sigma_x - \sigma & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y - \sigma & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z - \sigma \end{vmatrix} = 0$$

$$\begin{vmatrix} 60000 - \sigma & 9000 & 3500 \\ 9000 & -30000 - \sigma & -2000 \\ 3500 & -2000 & -15000 - \sigma \end{vmatrix} = 0$$

$$\sigma^3 - I_1\sigma^2 + I_2\sigma - I_3 = 0$$

$$I_1 = \sigma_x + \sigma_y + \sigma_z = 60{,}000 - 30{,}000 - 15{,}000 = 15{,}000$$

$$I_2 = \sigma_x\sigma_y + \sigma_y\sigma_z + \sigma_z\sigma_x - \tau_{xy}^2 - \tau_{yz}^2 - \tau_{zx}^2$$
$$= (60{,}000)(-30{,}000) + (-30{,}000)(-15{,}000) + (-15{,}000)(60{,}000)$$
$$- (9{,}000)^2 - (-2{,}000)^2 - (3{,}500)^2 = 2.347E^9$$

$$I_3 = \sigma_x\sigma_y\sigma_z + 2\tau_{xy}\tau_{yz}\tau_{zx} - \sigma_x\tau_{yz}^2 - \sigma_y\tau_{zx}^2 - \sigma_z\tau_{xy}^2$$
$$= 60{,}000(-30{,}000)(-15{,}000) + 2(9{,}000)(-2{,}000)(3{,}500)$$
$$- 60{,}000(-2{,}000)^2 - (-30{,}000)(3{,}500)^2$$
$$- (-15{,}000)(9{,}000)^2 = 2.8216E^{13}$$

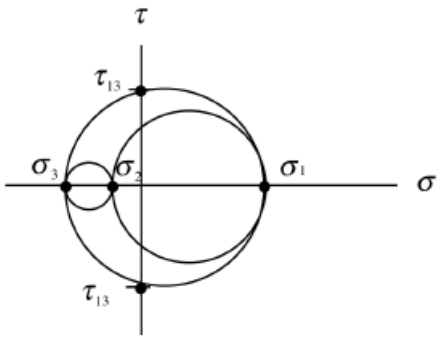
$$\sigma^3 - 15{,}000\sigma^2 - 2.347E^9\sigma - 2.8216E^{13} = 0$$

$$\sigma_1 = 61{,}033;\ \sigma_2 = -14{,}804;\ \sigma_3 = -31{,}230\,psi$$

$$\tau_{13} = \frac{|\sigma_1 - \sigma_3|}{2} = \frac{|61{,}033 - (-31{,}230)|}{2} = 46{,}131$$

$$\tau_{21} = \frac{|\sigma_2 - \sigma_1|}{2} = \frac{|-14{,}804 - 61{,}033|}{2} = 37{,}918$$

$$\tau_{32} = \frac{|\sigma_3 - \sigma_2|}{2} = \frac{|-31{,}230 - (-14{,}804)|}{2} = 8{,}213 psi$$



例 56

已知物體上某一點之應力狀態為σ_x = 72 MPa，σ_y = −32 MPa，σ_z = 0，τ_{xy} = 21 MPa，τ_{yz} = 0，τ_{xz} = 21 MPa，試求該點之主應力及最大剪應力。

【解析】

$$\begin{vmatrix} \sigma_x - \sigma & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y - \sigma & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z - \sigma \end{vmatrix} = 0$$

σ_x = 72 MPa，σ_y = −32 MPa，σ_z = 0，τ_{xy} = 21 MPa，τ_{yz} = 0，τ_{xz} = 21 MPa

帶入得

$$\sigma_p^3 - a\sigma_p^2 + b\sigma_p - c = 0$$

$$a = \sigma_x + \sigma_y + \sigma_z = 72 - 32 + 0 = 40$$

$$b = \sigma_x\sigma_y + \sigma_x\sigma_z + \sigma_y\sigma_z - \tau_{xy}^2 - \tau_{xz}^2 - \tau_{yz}^2$$

$$= (72)(-32) + (72)(0) + (-32)(0) - (21)^2 - (21)^2 - (0)^2 = -3186$$

$$c = \sigma_x\sigma_y\sigma_z + 2\tau_{xy}\tau_{xz}\tau_{yz} - \sigma_x\tau_{yz}^2 - \sigma_y\tau_{xz}^2 - \sigma_z\tau_{xy}^2 = 0 + 0 - 0 - (-32)(21)^2 - 0 = 14112$$

$$\sigma_p^3 - a\sigma_p^2 + b\sigma_p - c = 0 \text{ ， } \sigma_p^3 - 40\sigma_p^2 - 3186\sigma_p - 14112 = 0$$

σ_{p1} = 81.3 MPa，σ_{p2} = −4.75 MPa，σ_{p3} = −36.6 MPa

$$\tau_{\max} = \frac{1}{2}(\sigma_{p1} - \sigma_{p3}) = \frac{1}{2}[81.3 - (-36.6)] = 58.9 \text{ MPa}$$